

## 1. ARHAR WHOLE

Common name: अरहर / तुर

English name : Red gram

Botanical name: *Cajanus cajan*



### 1. Arhar whole shall be -

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety
- free from any obnoxious smell and fungus infestation

2. Variety .....(to be filled by assayer)

3. Actual Picture of the Lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| Sr.No.   | Tradable Parameters                     | Range-1                       | Range-2 | Range-3 |
|----------|-----------------------------------------|-------------------------------|---------|---------|
| <b>A</b> | <b>Essential</b>                        |                               |         |         |
| (i)      | Moisture(% by wt) Max                   | 10.0                          | 12.0    | 14.0    |
| (ii)     | Foreign matter (% by wt) Max            | 1.0                           | 1.5     | 2.0     |
| (iii)    | Other edible grains (% by wt) Max       | 1.0                           | 2.0     | 3.0     |
| (iv)     | Damaged grains (% by wt) Max            | 2.5                           | 3.5     | 5.0     |
| (v)      | Weevilled Grains<br>(% by count) Max    | 3.0                           | 5.0     | 10.0    |
| (vi)     | Admixture (% by wt) Max                 | 1.0                           | 2.0     | 3.0     |
| (vii)    | Immature & Shriveled<br>(% by wt) Max   | 1.0                           | 2.0     | 3.0     |
| <b>B</b> | <b>Optional (Processor/ Industrial)</b> |                               |         |         |
| (viii)   | Uric Acid                               | Not more than 100mg/kg        |         |         |
| (ix)     | Aflatoxin                               | Not more than 30 microgram/kg |         |         |

### 5. Sampling

- 5 % or minimum one container shall be randomly selected for sampling
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter: includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains: means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains: means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses
- Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 2. MOONG WHOLE

Common name: मूंग

English name : Green gram

**Botanical name:** *Vigna radiata* (L)



### 1. Moong whole shall be :

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety
- free from obnoxious smell and fungus infestation.

### 2. Variety..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr.No.    | Tradable Parameters                                             | Range-1                           | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                |                                   |         |         |
| (i)       | Moisture (% by wt) Max                                          | 10.0                              | 12.0    | 14.0    |
| (ii)      | Foreign matter (% by wt) Max                                    | 1.0                               | 1.5     | 2.0     |
| (iii)     | Other edible grains(% by wt) Max                                | 1.0                               | 2.0     | 3.0     |
| (iv)      | Damaged grains (% by wt) Max                                    | 2.5                               | 3.5     | 5.0     |
| (v)       | Weevilled Grains (% by count) Max                               | 3.0                               | 5.0     | 10.0    |
| (vi)      | Admixture (% by wt) Max                                         | 1.0                               | 2.0     | 3.0     |
| (vii)     | Immature & Shriveled (% by wt) Max                              | 1.0                               | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                          |                                   |         |         |
| (viii)    | Uric Acid                                                       | Not more than 100mg/kg            |         |         |
| (ix)      | Aflatoxin                                                       | Not more than 30 microgram/kg     |         |         |
| (x)       | Total of foreign matter, other edible grains and damaged grains | Not more than 9 percent by weight |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 3. MASOOR WHOLE

Common name: मसूर

English name : Lentil

**Botanical name:** *Lens culinaris*



**1. Masoor shall be:**

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

**2. Variety** .....(to be filled by assayer)

**3. Actual Picture of the lot**.....( to be uploaded by assayer)

**4. Criteria for Range designation :**

| Sr.No.    | Tradable Parameters                                             | Range-1                           | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                |                                   |         |         |
| (i)       | Moisture<br>(% by wt) Max                                       | 10.0                              | 12.0    | 14.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                                 | 1.0                               | 1.5     | 2.0     |
| (iii)     | Other edible grains<br>(% by wt) Max                            | 1.0                               | 2.0     | 3.0     |
| (iv)      | Damaged grains<br>(% by wt) Max                                 | 2.5                               | 3.5     | 5.0     |
| (v)       | Weevilled Grains<br>(% by count) Max                            | 3.0                               | 5.0     | 10.0    |
| (vi)      | Admixture (% by wt) Max                                         | 1.0                               | 2.0     | 3.0     |
| (vii)     | Immature & Shriveled<br>(% by wt) Max                           | 1.0                               | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/ Processor)</b>                         |                                   |         |         |
| (vii)     | Uric Acid                                                       | Not more than 100mg/kg            |         |         |
| (viii)    | Aflatoxin                                                       | Not more than 30 microgram/kg     |         |         |
| (ix)      | Total of foreign matter, other edible grains and damaged grains | Not more than 8 percent by weight |         |         |

**5. Sampling:-**

- 5% or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

**6. Definitions:**

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanically injured grains in case of whole pulses ;
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

#### 4. URAD WHOLE

Common name : उरद

English name : Black gram

**Botanical name:** *Vigna mungo*



**1. Urad shall be-**

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation,

**2. Variety .....** (to be filled by assayer)

**3. Actual Picture of the lot .....** ( to be uploaded by assayer)

**4. Criteria for Range designation :**

| Sr.No.    | Tradable Parameters                                                   | Range-1                           | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                      |                                   |         |         |
| (i)       | Moisture(% by wt) Max                                                 | 10.0                              | 12.0    | 14.0    |
| (ii)      | Foreign matter(% by wt) Max                                           | 1.0                               | 1.5     | 2.0     |
| (iii)     | Other edible grains(% by wt) Max                                      | 1.0                               | 2.0     | 3.0     |
| (iv)      | Damaged grains(% by wt) Max                                           | 2.5                               | 3.5     | 5.0     |
| (v)       | Weevilled Grains<br>(% by count) Max                                  | 3.0                               | 5.0     | 10.0    |
| (vi)      | Admixture (% by wt) Max                                               | 1.0                               | 2.0     | 3.0     |
| (vii)     | Immature & Shriveled<br>(% by wt) Max                                 | 1.0                               | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                |                                   |         |         |
| (x)       | Uric Acid                                                             | Not more than 100mg/kg            |         |         |
| (xi)      | Aflatoxin                                                             | Not more than 30 microgram/kg     |         |         |
| (xii)     | Total of foreign matter, other<br>edible grains and damaged<br>grains | Not more than 9 percent by weight |         |         |

**5. Sampling:-**

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

**6. Definitions:**

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 5. WHEAT

Common name : गेहूँ

English name : Wheat

**Botanical name:** *Triticum aestivum* (L).



### 1. Wheat shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation,

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                                                          | Range-1                          | Range-2 | Range-3 |
|-----------|------------------------------------------------------------------------------|----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                             |                                  |         |         |
| (i)       | Moisture (% by wt) Max                                                       | 11.0                             | 12.0    | 14.0    |
| (ii)      | Foreign matter (% by wt) Max                                                 | 1.5                              | 3.0     | 4.0     |
| (iii)     | Admixture/Other edible grains (% by wt) Max                                  | 5.0                              | 7.0     | 10.0    |
| (iv)      | Immature & Shriveled Grains (% by wt) Max                                    | 3.0                              | 5.0     | 7.0     |
| (v)       | Damaged/Discoloured grains (% by wt) Max                                     | 3.0                              | 5.0     | 7.0     |
| (vi)      | Weevilled Grains (% by count) Max                                            | 4.0                              | 6.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                       |                                  |         |         |
| (vii)     | Protein (per cent. on dry matter basis)                                      | Not less than 11.0 (durum wheat) |         |         |
| (viii)    | Minimum Test weight (weight of 100 litre volume expressed in Kg) /Hectoliter | Not less than 70                 |         |         |
| (ix)      | Uric Acid                                                                    | Not more than 100mg/kg           |         |         |
| (x)       | Aflatoxin                                                                    | Not more than 30 microgram/kg    |         |         |
| (xi)      | Deoxynivalenol (DON)                                                         | Not more than 1000 micro gm/ kg  |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 500 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged/Discoloured Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 6. MAIZE

Common name : मक्का

English name : Maize

**Botanical name:** *Zea mays*



### 1. Maize shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety,
- free from obnoxious smell and fungus infestation,

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                                             | Range-1                           | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                |                                   |         |         |
| (i)       | Moisture (% by wt) Max                                          | 12.0                              | 14.0    | 16.0    |
| (ii)      | Foreign matter (% by wt) Max                                    | 1.5                               | 2.0     | 3.0     |
| (iii)     | Admixture/Other edible grains (% by wt) Max                     | 1.0                               | 2.0     | 3.0     |
| (iv)      | Damaged/ immature & Shriveled grains (% by wt) Max              | 7.0                               | 10.0    | 15.0    |
| (v)       | Weevilled Grains (% by count) Max                               | 4.0                               | 6.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/ Processor)</b>                         |                                   |         |         |
| (vi)      | Uric Acid                                                       | Not more than 100mg/kg            |         |         |
| (vii)     | Aflatoxin                                                       | Not more than 30 microgram/kg     |         |         |
| (viii)    | Total of foreign matter, other edible grains and damaged grains | Not more than 9 percent by weight |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 500gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 7. CHANA

Common name: साबुत चना

English name : Bengal Gram

**Botanical name:** *Cicer arietinum*



### 1. Chana shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr.No.    | Tradable Parameters                                                   | Range-1                           | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                      |                                   |         |         |
| (i)       | Moisture<br>(% by wt) Max                                             | 10.0                              | 12.0    | 14.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                                       | 1.0                               | 2.0     | 3.0     |
| (iii)     | Admixture/Other edible<br>grains(% by wt) Max                         | 2.0                               | 3.0     | 4.0     |
| (iv)      | Damaged/ immature &<br>Shriveled grains<br>(% by wt) Max              | 4.0                               | 5.0     | 7.0     |
| (v)       | Weevilled Grains<br>(% by count) Max                                  | 3.0                               | 6.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                |                                   |         |         |
| (vi)      | Uric Acid                                                             | Not more than 100mg/kg            |         |         |
| (vii)     | Aflatoxin                                                             | Not more than 30 microgram/kg     |         |         |
| (viii)    | Total of foreign matter, other<br>edible grains and damaged<br>grains | Not more than 9 percent by weight |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 500gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 8. BAJRA

Common name: बाजरा

English name : Pearl millet

**Botanical name:** *Pennisetum americanum* (L)



### 1. Bajra shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr.No. | Tradable Parameters                                             | Range-1                            | Range-2 | Range-3 |
|--------|-----------------------------------------------------------------|------------------------------------|---------|---------|
| A.     | <b>Essential</b>                                                |                                    |         |         |
| (i)    | Moisture (% by wt) Max                                          | 12.0                               | 14.0    | 16.0    |
| (ii)   | Admixture (% by wt) Max                                         | 1.0                                | 2.0     | 3.0     |
| (iii)  | Immature & Shriveled grains(% by wt) Max                        | 4.0                                | 6.0     | 8.0     |
| (iv)   | Foreign matter (% by wt) Max                                    | 0.1                                | 0.6     | 1.0     |
| (v)    | Other edible grains (% by wt) Max                               | 3.0                                | 4.0     | 6.0     |
| (vi)   | Damaged/Discoloured grains (% by wt) Max                        | 5.0                                | 7.0     | 10.0    |
| (vii)  | Weevilled Grains (% by count) Max                               | 2.0                                | 4.0     | 6.0     |
| B.     | <b>Optional (Industrial/Processors)</b>                         |                                    |         |         |
| (viii) | Uric Acid                                                       | Not more than 100mg/kg             |         |         |
| (ix)   | Aflatoxin                                                       | Not more than 30 microgram/kg      |         |         |
| (x)    | Total of foreign matter, other edible grains and damaged grains | Not more than 10 percent by weight |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 500 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged/Discoloured Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 9. BARLEY

Common name : जौ

English name : Barley

**Botanical name:** *Hordeum vulgare*(L)



### 1. Barley shall be:

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from , abnormal flavours, obnoxious smell and fungus infestation

### 2. Variety ..... (to be filled by assayer)

### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                             | Range-1                            | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------|------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                |                                    |         |         |
| (i)       | Moisture (% by wt) Max                                          | 11.0                               | 12.0    | 14.0    |
| (ii)      | Foreign matter (% by wt) Max                                    | 1.5                                | 3.0     | 4.0     |
| (iii)     | Admixture/Other edible grains (% by wt) Max                     | 5.0                                | 7.0     | 10.0    |
| (iv)      | Immature & Shriveled Grains (% by wt) Max                       | 3.0                                | 5.0     | 7.0     |
| (v)       | Damaged/Discoloured grains (% by wt) Max                        | 3.0                                | 5.0     | 7.0     |
| (vi)      | Weevilled Grains (% by count) Max                               | 4.0                                | 6.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                          |                                    |         |         |
| (i)       | Uric Acid                                                       | Not more than 100mg/kg             |         |         |
| (ii)      | Aflatoxin                                                       | Not more than 30 microgram/kg      |         |         |
| (iii)     | Total of foreign matter, other edible grains and damaged grains | Not more than 12 percent by weight |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 500 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged/Discoloured Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 10. JOWAR

Common name: ज्वार

English name : Jowar

**Botanical name:** *Sorghum bicolor*



### 1. Jowar shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                             | Range-1                            | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------|------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                |                                    |         |         |
| (i)       | Moisture (% by wt) Max                                          | 12.0                               | 14.0    | 16.0    |
| (ii)      | Admixture (% by wt) Max                                         | 1.0                                | 2.0     | 3.0     |
| (iii)     | Immature & Shriveled grains(% by wt) Max                        | 4.0                                | 6.0     | 8.0     |
| (iv)      | Foreign matter (% by wt) Max                                    | 0.1                                | 0.6     | 1.0     |
| (v)       | Other edible grains (% by wt) Max                               | 3.0                                | 4.0     | 6.0     |
| (vi)      | Damaged/Discoloured grains (% by wt) Max                        | 5.0                                | 7.0     | 10.0    |
| (vii)     | Weevilled Grains (% by count) Max                               | 2.0                                | 4.0     | 6.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                          |                                    |         |         |
| (viii)    | Uric Acid                                                       | Not more than 100mg/kg             |         |         |
| (ix)      | Aflatoxin                                                       | Not more than 30 microgram/kg      |         |         |
| (x)       | Total of foreign matter, other edible grains and damaged grains | Not more than 10 percent by weight |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 500 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged/Discoloured Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 11. PADDY

Common name: धान

English name : Paddy

**Botanical name:** *Oryza sativa* (L)



### 1. Paddy shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                     | Range-1                       | Range-2 | Range-3 |
|-----------|---------------------------------------------------------|-------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                        |                               |         |         |
| (i)       | Moisture<br>(% by wt) Max                               | 17.0                          | 18.0    | 20.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                         | 2.0                           | 4.0     | 6.0     |
| (iii)     | Admixture<br>(% by wt) Max                              | 10.0                          | 15.0    | 20.0    |
| (iv)      | Immature & Shriveled grains<br>(% by wt) Max            | 3.0                           | 5.0     | 7.0     |
| (v)       | Damaged/ discolored &<br>Weevilled grains (% by wt) Max | 4.0                           | 6.0     | 8.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                  |                               |         |         |
| (vi)      | Uric Acid                                               | Not more than 100mg/kg        |         |         |
| (vii)     | Aflatoxin                                               | Not more than 30 microgram/kg |         |         |

### 5. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 500gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- (iii) Damaged/Discoloured Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iv) Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- (v) Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- (vi) Admixture means the presence of a variety of the same grain other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 12. RAJMA

Common name: राजमा

English name : Kidney beans

Botanical name: *Phaseolus vulgaris*



### 1. Rajma shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation

### 2. Variety ..... (to be filled by assayer)

### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                                   | Range-1                            | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------------|------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                      |                                    |         |         |
| (i)       | Moisture<br>(% by wt) Max                                             | 10.0                               | 12.0    | 14.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                                       | 1.0                                | 2.0     | 3.0     |
| (iii)     | Admixture/Other edible<br>grains(% by wt) Max                         | 2.0                                | 3.0     | 4.0     |
| (iv)      | Damaged/ immature &<br>Shriveled grains<br>(% by wt) Max              | 4.0                                | 5.0     | 7.0     |
| (v)       | Weevilled Grains<br>(% by count) Max                                  | 3.0                                | 6.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                |                                    |         |         |
| (vi)      | Uric Acid                                                             | Not more than 100mg/kg             |         |         |
| (vii)     | Aflatoxin                                                             | Not more than 30 microgram/kg      |         |         |
| (viii)    | Total of foreign matter, other<br>edible grains and damaged<br>grains | Not more than 12 percent by weight |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Admixture means the presence of a variety of the same grain other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 13. RAGI

Common name: रागी

English name: Finger millet

Botanical name: *Eleusine coracana*



#### 1. Ragi shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                      | Range-1                | Range-2 | Range-3 |
|-----------|------------------------------------------|------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                         |                        |         |         |
| (i)       | Moisture(% by wt) Max                    | 12.0                   | 14.0    | 16.0    |
| (ii)      | Admixture(% by wt) Max                   | 1.0                    | 2.0     | 3.0     |
| (iii)     | Immature & Shriveled grains(% by wt) Max | 4.0                    | 6.0     | 8.0     |
| (iv)      | Foreign matter (% by wt) Max             | 0.1                    | 0.6     | 1.0     |
| (v)       | Other edible grains (% by wt) Max        | 3.0                    | 4.0     | 6.0     |
| (vi)      | Damaged/Discoloured grains (% by wt) Max | 5.0                    | 7.0     | 10.0    |
| (vii)     | Weevilled Grains (% by count) Max        | 2.0                    | 4.0     | 6.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>   |                        |         |         |
| (viii)    | Uric Acid                                | Not more than 100mg/kg |         |         |

#### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample

#### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged/Discoloured Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 14. LOBIA

Common name: लोबिआ

English name : Cowpea

Botanical name : *Vigna sinensis*



### 1. Lobia shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                                   | Range-1                            | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------------|------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                      |                                    |         |         |
| (i)       | Moisture<br>(% by wt) Max                                             | 10.0                               | 12.0    | 14.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                                       | 1.0                                | 2.0     | 3.0     |
| (iii)     | Admixture/Other edible<br>grains(% by wt) Max                         | 2.0                                | 3.0     | 4.0     |
| (iv)      | Damaged/ immature &<br>Shriveled grains<br>(% by wt) Max              | 4.0                                | 5.0     | 7.0     |
| (v)       | Weevilled Grains<br>(% by count) Max                                  | 3.0                                | 6.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                |                                    |         |         |
| (vi)      | Uric Acid                                                             | Not more than 100mg/kg             |         |         |
| (vii)     | Aflatoxin                                                             | Not more than 30 microgram/kg      |         |         |
| (viii)    | Total of foreign matter,<br>other edible grains and<br>damaged grains | Not more than 12 percent by weight |         |         |

### 5. Sampling:-

- 5% or minimum one container shall be randomly selected for sampling.
- 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shriveled Grains means kernels or pieces of kernels that are not fully developed
- Admixture means the presence of a variety of the same grain other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 15. BASMATI RICE

Common name : बासमती चावल

English name : Basmati Rice

Botanical name : *Oryza sativa*



### 1. Basmati Rice shall be:

- clean, wholesome, long, slender, reasonably uniform in size, shape and colour with marked degree of natural fragrance characteristic to the variety;
- free from obnoxious smell and fungus infestation.

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                          | Range-1            | Range-2 | Range-3 |
|-----------|----------------------------------------------|--------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                             |                    |         |         |
| (i)       | Length                                       | 6.61mm and above   |         |         |
| (ii)      | Length- breadth ratio                        | 3.5 and above      |         |         |
| (iii)     | Moisture (% by wt) Max                       | 12.0               | 13.0    | 14.0    |
| (iv)      | Foreign matter (% by wt) Max                 | 0.40               | 0.50    | 0.60    |
| (v)       | Broken/Damaged/Chalky grains (% by wt) Max   | 8.5                | 13.0    | 17.0    |
| (vi)      | Red /Green grains/Paddy Grains (% by wt) Max | 5.0                | 7.0     | 9.0     |
| (vii)     | Other varieties of rice (% by count) Max     | 7.0                | 10.0    | 15.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>       |                    |         |         |
| (viii)    | Average cooked rice length                   | 12.0 mm and above  |         |         |
| (ix)      | Average volume expansion ratio               | More than 3.5      |         |         |
| (x)       | Average pre-cooked milled rice breadth       | Not more than 2 mm |         |         |
| (xi)      | Elongation ratio after cooking               | Not less than 1.7  |         |         |
| (xii)     | Alkali Spreading Value                       | 4 to 7             |         |         |
| (xiii)    | Amylose content (%)                          | 20 to 25           |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other variety of rice means any the presence of a variety of the same rice grain other than the variety under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Broken grains means pieces of sound kernels that are less than three fourths of the size of full kernels.
- Chalky Grains are pieces of kernels of which at least half the portion is opaque, milky white in colour and brittle in nature.
- Red grains shall be the kernels, whole or broken which have their surface coated with red brand. Green grains shall be the kernels, whole or broken which are greenish in color.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 16. White Peas

Common name: सफ़ेद मटर /Mochai (TAMIL)/ Vella pattaani (Malyalam)/Thella Bataaneelu (Telegu)  
English name: White Peas  
Botanical name: Lathyrus sativus



### 1. White Peas shall be-

- intact, firm, clean, reasonably uniform size, & shape, color characteristics to the variety
- free from visible foreign matter, abnormal external moisture, foreign smell, smoke, burn

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                   | Range-1                            | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------------|------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                      |                                    |         |         |
| (i)       | Moisture<br>(% by Weight) Max                                         | 12.0                               | 12.0    | 14.0    |
| (ii)      | Foreign matter<br>(% by Weight) Max                                   | 1.0                                | 1.5     | 2.0     |
| (iii)     | Other edible grains<br>(% by Weight) Max                              | 1.0                                | 2.0     | 3.0     |
| (iv)      | Damaged grains(% by<br>Weight) Max                                    | 3.0                                | 4.0     | 5.0     |
| (v)       | Weevilled grains<br>(% by Weight) Max                                 | 7.0                                | 9.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                |                                    |         |         |
| (vi)      | Uric Acid                                                             | Not more than 100mg/kg             |         |         |
| (vii)     | Aflatoxin                                                             | Not more than 30 microgram/kg      |         |         |
| (viii)    | Total of foreign matter, other<br>edible grains and damaged<br>grains | Not more than 12 percent by weight |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- 250 g shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter:- shall include sand, dirt, stones, glass and metallic pieces, lumps of earth, clay and mud, husk, chaff, straw, weed seeds and other inedible grains.
- Damaged and Discolored seeds: -Kernels or pieces of kernels that are sprouted or internally damaged as a result of heat, moisture, weather or microbes, and mechanical injured grains in case of whole pulses; discoloured, or discolouration materially affecting the quality.
- Other edible grain/seeds:- means any edible grains (including oil seeds) other than the one which is under consideration;
- Weevilled grains:-means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 17. CASTOR SEED

Common name : अरंडी

English name : Castor Seed

**Botanical name:** *Ricinus communis* (L)



### 1. Castor Seed shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation .

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                               | Range-1                           | Range-2 | Range-3 |
|-----------|---------------------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                  |                                   |         |         |
| (i)       | Oil Content<br>(% by wt) Min                      | 47.0                              | 45.0    | 40.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                   | 1.5                               | 2.5     | 3.5     |
| (iii)     | Damaged and discoloured<br>seeds<br>(% by wt) Max | 4.0                               | 6.0     | 7.0     |
| (iv)      | Broken Seeds<br>(% by wt) Max                     | 1.0                               | 2.0     | 3.0     |
| (v)       | Moisture<br>(% by wt) Max                         | 5.0                               | 6.0     | 7.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>            |                                   |         |         |
| (vi)      | Admixture                                         | Not more than 2 percent by weight |         |         |
| (vii)     | Fotri (Husk)                                      | Not more than 5 percent by weight |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Damaged and Discoloured seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Broken seeds are pieces of sound kernels that are less than three fourths of the size of full kernels.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 18. MUSTARD SEED

Common name: सरसों

English name : Mustard Seed

**Botanical name:** *Brassica nigra* (L)



### 1. Mustard Seed shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                         | Range-1                             | Range-2 | Range-3 |
|-----------|---------------------------------------------|-------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                            |                                     |         |         |
| (i)       | Oil content (% by wt) Min                   | 42.0                                | 38.0    | 36.0    |
| (ii)      | Moisture (% by wt) Max                      | 8.0                                 | 9.0     | 10.0    |
| (iii)     | Foreign matter (% by wt) Max                | 2.0                                 | 3.0     | 4.0     |
| (iv)      | Damaged and discoloured seeds (% by wt) Max | 2.0                                 | 3.0     | 5.0     |
| (v)       | Other Edible Seeds (% by wt) Max            | 0.5                                 | 0.8     | 1.0     |
| (vi)      | Argemone seeds                              | Absent                              |         |         |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>      |                                     |         |         |
| (vii)     | Allyl iso thiocynate (m/m) on dry basis     | Not less than 0.7 percent by weight |         |         |

### 5. Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged and discoloured seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iii) are pieces of sound kernels that are less than three fourths of the size of full kernels.
- (iv) Other Edible seeds means any edible seed other than the one which is under consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 19.SOYABEAN

Common name: सोयाबीन

English name : Soyabean

**Botanical name:** *Glycine max (L)*



### 1. Soyabean Seed shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                    | Range-1                              | Range-2 | Range-3 |
|-----------|----------------------------------------|--------------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                       |                                      |         |         |
| (i)       | Oil content<br>(% by wt) Min           | 20.0                                 | 18.0    | 13.0    |
| (ii)      | Moisture<br>(% by wt) Max              | 7.0                                  | 9.0     | 12.0    |
| (iii)     | Foreign matter<br>(% by wt) Max        | 1.5                                  | 2.0     | 3.0     |
| (iv)      | Damaged seeds<br>(% by wt) Max         | 3.0                                  | 5.0     | 7.0     |
| (v)       | Other edible seeds<br>(% by wt) Max    | 1.0                                  | 3.0     | 6.0     |
| (vi)      | Weevilled Seeds<br>(% by count) Max    | 3.0                                  | 4.0     | 5.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                                      |         |         |
| (vii)     | Green Seed                             | Not more than 7.0 percent by weight  |         |         |
| (viii)    | Protein                                | Not less than 34.0 percent by weight |         |         |

### 5. Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250 g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 g shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Other Edible seeds means any edible seed other than the one which is under consideration;
- (iii) Damaged seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iv) Weevilled seeds are partially or wholly bored by insects injurious to seed but do not include germ eaten seeds and egg spotted seeds.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 20. PEANUT KERNEL

Common name: मूँगफली /Nilakadalai

(Tamil)/Nilakadala (Malyalam)/ Verushanaga (Telugu)

English name: Peanut (deshelled), Groundnut kernel (deshelled)

Botanical name: Arachis hypogaea



### 1. Peanut shall be–

- intact, firm, clean, reasonably uniform size & shape, color characteristics to the variety
- free from visible foreign matter, abnormal external moisture, foreign smell, smoke, burn

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                       | Range-1                            | Range-2 | Range-3 |
|-------|-------------------------------------------------------------------------------------------|------------------------------------|---------|---------|
| A.    | <b>Essential</b>                                                                          |                                    |         |         |
| (i)   | <b>Moisture</b><br>(% by Weight) Max                                                      | 5.0                                | 6.0     | 7.0     |
| (ii)  | <b>Damaged kernel including slightly damaged, discoloured kernel</b><br>(% by Weight) Max | 3.0                                | 4.0     | 5.0     |
| (iii) | <b>Immature &amp; Shriveled</b><br>(% by Weight) Max                                      | 2.0                                | 3.0     | 4.0     |
| (iv)  | <b>Foreign matter</b><br>(% by Weight) Max                                                | 0.5                                | 1.0     | 2.0     |
| B.    | <b>Optional (Industrial/Processor)</b>                                                    |                                    |         |         |
| (v)   | Aflatoxins                                                                                | Not more than 30 parts per billion |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 200 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- 200 g shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter: means stones, dirt, clay or any other edible/non-edible seeds such as mahua, castor, neem or argemone.
- Immature and shriveled kernel: are those kernels which are imperfectly developed and /or shrunken.
- Damaged kernel: are those kernels that are damaged mechanically or by mould, weevil or any other insect attack or those showing internal discoloration materially affecting the quality.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 21. SUNFLOWER SEED

Common name: सूर्यमुखी बीज

English name : Sunflower

Botanical name: *Helianthus annus* (L)



### 1. Sunflower seed shall be-

- i. clean, wholesome, nearly reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No | Tradable Parameters                    | Range-1                           | Range-2 | Range-3 |
|--------|----------------------------------------|-----------------------------------|---------|---------|
| A.     | <b>Essential</b>                       |                                   |         |         |
| i.     | Oil Content<br>(% by wt) Min           | 28.0                              | 25.0    | 22.0    |
| ii.    | Moisture<br>(% by wt) Max              | 4.0                               | 4.5     | 5.0     |
| iii.   | Foreign matter<br>(% by wt) Max        | 2.0                               | 4.0     | 6.0     |
| iv.    | Damaged seeds<br>(% by wt) Max         | 4.0                               | 5.0     | 6.0     |
| B.     | <b>Optional (Industrial/Processor)</b> |                                   |         |         |
| viii.  | Admixture                              | Not more than 2 percent by weight |         |         |

### 5. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.

## 22. KUSUM SEED

Common name: कुसुम

English name: Kusum

Botanical name: *Schleichera oleosa*



### 1. Kusum seed shall be-

- i. clean, wholesome, nearly reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

### 2. Variety.....(to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No  | Tradable Parameters             | Range-1                           | Range-2 | Range-3 |
|-------|---------------------------------|-----------------------------------|---------|---------|
| A.    | Essential                       |                                   |         |         |
| (i)   | Oil Content<br>(% by wt) Min    | 32.0                              | 30.0    | 28.0    |
| (ii)  | Moisture<br>(% by wt) Max       | 6.0                               | 6.5     | 7.0     |
| (iii) | Foreign Matter<br>(% by wt) Max | 1.5                               | 2.5     | 4.0     |
| (iv)  | Damaged seeds (% by wt) Max     | 3.0                               | 5.0     | 7.0     |
| B.    | Optional (Industrial/Processor) |                                   |         |         |
| (v)   | Acid value of the extracted oil | Not more than 8 percent by weight |         |         |

### 5. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 23. SESAME SEED

Common name: तिल

English name: Sesame seed

Botanical name: *Sesamum indicum*



### 1. Sesame seed shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No  | Tradable Parameters                                                        | Range-1                 | Range-2 | Range-3 |
|-------|----------------------------------------------------------------------------|-------------------------|---------|---------|
| A.    | <b>Essential</b>                                                           |                         |         |         |
| (i)   | Oil Content<br>(% by wt) Min                                               | 45.0                    | 40.0    | 35.0    |
| (ii)  | Moisture<br>(% by wt) Max                                                  | 5.0                     | 6.0     | 7.0     |
| (iii) | Foreign matter<br>(% by wt) Max                                            | 1.0                     | 1.5     | 2.0     |
| (iv)  | Damaged seed<br>(% by wt) Max                                              | 1.0                     | 2.0     | 3.0     |
| (v)   | Admixture of other varieties<br>(%by wt) Max                               | 10.0                    | 15.0    | 20.0    |
| B.    | <b>Optional (Industrial/Processor)</b>                                     |                         |         |         |
| (vi)  | Total of Immature , shriveled, dead , damaged, discoloured seeds by weight | Not more than 5 percent |         |         |

### 5. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iii) Admixture means the presence of a variety of the same seed other than the variety in consideration

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 24. NIGER SEED

Common name: सरहगया /Uchellu (Tamil) Valisalu

(Telegu)/ खुरसणीबी(Marathi)

English name : Niger Seed

Botanical name: Guizotia abyssinica



### 1. Niger Seed shall be-

- intact, firm, clean, reasonably uniform size & shape, color characteristics to the variety
  - free from visible foreign matter, abnormal external moisture, foreign smell
- Variety ..... (to be filled by assayer)
  - Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No  | Tradable Parameter                                                                              | Range-1                           | Range-2 | Range-3 |
|-------|-------------------------------------------------------------------------------------------------|-----------------------------------|---------|---------|
| A.    | <b>Essential</b>                                                                                |                                   |         |         |
| (i)   | <b>Oil Content</b><br>(% by Weight) Min                                                         | 30.0                              | 25.0    | 20.0    |
| (ii)  | <b>Moisture</b><br>(% by Weight) Max                                                            | 7.0                               | 8.0     | 9.0     |
| (iii) | <b>Extraneous Matter</b><br>(% by Weight) Max                                                   | 1.0                               | 2.0     | 3.5     |
| (iv)  | <b>Broken seeds</b><br>(Damaged, shriveled, discoloured and immature seed)<br>(% by Weight) Max | 3.0                               | 4.0     | 5.0     |
| (v)   | <b>Insect damaged seeds</b><br>(% by Weight) Max                                                | 0.5                               | 0.75    | 1.0     |
| B.    | <b>Optional (Industrial/Processor)</b>                                                          |                                   |         |         |
| (vi)  | Total of Immature , shriveled, dead , damaged, discolored and slightly damaged Seeds            | Not more than 5 percent by weight |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 250 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- 250 g shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Extraneous /Foreign matter:- Includes inorganic and organic matter. The inorganic matter shall include sand, gravel, dirt, pebbles, stones, glass and metallic pieces, lumps of earth, clay and mud. Organic matter shall include husk, chaff, straw, weed seeds and other inedible grains.
- Broken seeds: -Kernels or pieces of kernels that are sprouted or internally damaged as a result of heat, moisture, weather or microbes, discoloured, and immature seed.
- Insect damaged seeds: shall be those seeds which are wholly or partly bored or eaten by the weevils.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 25. SAL SEED

Common name: साल बीज /Venkungiliyam (Tamil)/ Karimauthu with (Malayalam)/ Guggilam (Telegu)/राळाबी (Marathi)  
English name: Sal seed  
Botanical name: Shorea robusta



### 1. Sal Seed shall be-

- i. intact, firm clean, reasonably uniform size & shape, color characteristics to the variety
  - ii. free from visible foreign matter, abnormal external moisture, foreign smell
2. Variety .....(to be filled by assayer)
3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable parameter                                                                              | Range-1                           | Range-2 | Range-3 |
|-------|-------------------------------------------------------------------------------------------------|-----------------------------------|---------|---------|
| A.    | <b>Essential</b>                                                                                |                                   |         |         |
| (i)   | <b>Oil content</b><br>(% by Weight) Min                                                         | 14.0                              | 10.0    | 8.0     |
| (ii)  | <b>Moisture</b><br>(% by Weight) Max                                                            | 11.0                              | 11.0    | 11.0    |
| (iii) | <b>Extraneous matter</b><br>(% by Weight) Max                                                   | 1.5                               | 3.0     | 4.0     |
| (iv)  | <b>Broken seeds</b><br>(Damaged, shriveled, discoloured and immature seed)<br>(% by Weight) Max | 9.0                               | 11.0    | 12.0    |
| (v)   | <b>Insect damaged seed</b><br>(% by Weight) Max                                                 | 3.0                               | 5.0     | 6.0     |
| B.    | <b>Optional (Industrial/Processor)</b>                                                          |                                   |         |         |
| (vi)  | <b>Admixture</b>                                                                                | Not more than 5 percent by weight |         |         |

### 5. Sampling:

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 200 g shall be drawn from each selected container and shall be mixed homogenously will be called as primary sample
- (iii) 200 g shall be drawn from primary sample will be called as Laboratory sample

### 6. Definitions:

- (i) Extraneous /Foreign matter:- Includes inorganic and organic matter. The inorganic matter shall include sand, gravel, dirt, pebbles, stones, glass and metallic pieces, lumps of earth, clay and mud. Organic matter shall include husk, chaff, straw, weed seeds and other inedible grains.
- (ii) Broken seeds: -Kernels or pieces of kernels that are sprouted or internally damaged as a result of heat, moisture, weather or microbes, discoloured, and immature seed.
- (iii) Insect damaged seeds: shall be those seeds which are wholly or partly bored or eaten by the weevils.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 26. COTTON SEED

Common name: कपास बीज / Paruthi Vidai (Tamil)/

Parathi kuru (Malyalam)/ Patti Vittanam (Telugu)

English name : Cotton Seed

Botanical name: Gossypium spp.



Actual Picture for illustration

### 1. Cotton seed shall be-

- i. intact, firm, clean, of same variety and
- ii. free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot..... (to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                    | Range-1                            | Range-2 | Range-3 |
|-------|----------------------------------------|------------------------------------|---------|---------|
| A.    | <b>Essential</b>                       |                                    |         |         |
| (i)   | Oil Content<br>(% by Weight) Min       | 15.0                               | 15.0    | 10.0    |
| (ii)  | Moisture<br>(% by Weight) Max          | 10.0                               | 11.0    | 12.0    |
| (iii) | Foreign matter<br>(% by Weight) Max    | 1.0                                | 2.0     | 4.0     |
| (iv)  | Damaged seed<br>(% by Weight) Max      | 3.0                                | 4.0     | 5.0     |
| B.    | <b>Optional (Industrial/Processor)</b> |                                    |         |         |
| (v)   | Linters                                | Not more than 10 percent by weight |         |         |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250 g shall be drawn from each selected container shall be mixed homogenously called as primary sample.
- (iii) 250 g shall be drawn from primary sample will be called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter:-All foreign material other than cottonseed material, where cottonseed material includes cotton fiber attached to the seed coat.
- (ii) Damaged seeds: -Includes smutty seed, hot seed, musty seed, sour seed, rotted seed and mouldy seed.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 27. PONGAM SEED

Common name: पोंगम बीज / Pongam vidai (Tamil)/  
Pongu With (Malayalam)/ Kaanuga Vittanam  
(Telugu)

English name: Pongam seeds

Botanical name: Pongamia pinnata



Actual Picture for illustration

### 1. Pongam Seed shall be-

- i. intact, firm, clean, reasonably uniform size, & shape, color characteristics to the variety
  - ii. free from visible foreign matter, abnormal external moisture, foreign smell
2. Variety .....(to be filled by assayer)
  3. Actual Picture of the lot.....( to be uploaded by assayer)
  4. **Criteria for Range designation:**

| S.No.     | Tradable Parameter                     | Range-1                           | Range-2 | Range-3 |
|-----------|----------------------------------------|-----------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                       |                                   |         |         |
| (i)       | Oil content<br>(% by Weight) Min       | 30                                | 25.0    | 20.0    |
| (ii)      | Moisture<br>(% by Weight) Max          | 11.0                              | 11.0    | 11.0    |
| (iii)     | Foreign matter<br>(% by Weight) Max    | 1.5                               | 3.0     | 4.0     |
| (iv)      | Broken seeds<br>(% by Weight) Max      | 9.0                               | 11.0    | 12.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                                   |         |         |
| (v)       | Admixture                              | Not more than 5 percent by weight |         |         |

### 5. Sampling:-

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- (iii) 250 g shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter:- Includes inorganic and organic matter. The inorganic matter shall include sand, gravel, dirt, pebbles, stones, glass and metallic pieces, lumps of earth, clay and mud. Organic matter shall include husk, chaff, straw, weed seeds and other inedible grains.
- (ii) Broken seed:- Pieces of sound kernels that are less than three fourths of the size of the full kernels

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 28. APPLE

Common name: सेब

English name : Apple

Botanical name: *Malus pumila* (L)

Varieties name: Apple Kashmiri, Kinnor, Royal Delicious, Royal Gold, Apple Golden, American, Pittu etc



### 1. Apples shall be-

- intact, firm, clean, of reasonably uniform size & shape, color (Red/Yellowish/Pinkish Green/Greenish Red) characteristics to the variety
- free from bruises, soft patches, insect/mechanical injury, visible disorders, fungal infestation,

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable parameters                                                                                                        | Range-1      | Range-2          | Range-3         |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|------------------|-----------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |              |                  |                 |
| (i)       | Size (Diameter (mm) of the equatorial section)<br>For all varieties                                                        | Up to 90     | Up to 70         | Up to 50        |
| (ii)      | Defects (% by count) (Max)<br>(i) Dry Healed Hail<br>(2-5 sq.cm of total surface area)                                     | 10           | 15               | 20              |
|           | (ii) Russetting<br>(2-5 sq.cm of total surface area)                                                                       | 2            | 4                | 6               |
|           | (iii) Soft Patch<br>(2-5 sq.cm of total surface area)                                                                      | 2            | 4                | 6               |
| (iii)     | Range Acceptance                                                                                                           | 90 %         | 85 %             | 80 %            |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |              |                  |                 |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between<br>40-70 | Less than<br>40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 29. PEAR

Common name: **नाशपाती**

English name : Pear

Botanical name: *Pyrus communis*

Varieties name: Naashpaati, Pear Kashmir, William, Kashmir Nakh, Vicar of Wakefield, Beuree D Amanlis, Goshbagu, Babugosa, Beurre Hardy etc.



### 1. Pears shall be–

- intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- free from, visible foreign matter, abnormal external moisture, foreign smell.

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                     | Range-1                          | Range-2                 | Range-3                          |
|-----------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|----------------------------------|
| <b>A.</b> | <b>Essential</b>                                                                                                        |                                  |                         |                                  |
| (i)       | Size (mm)<br><b>Pear Shimla/Babugosa</b><br><b>Pear Kashmiri</b><br><b>Pear Nakh</b>                                    | Above 55<br>Above 45<br>Above 65 | 45-55<br>40-45<br>65-50 | Below 45<br>Below 40<br>Below 50 |
| (ii)      | <b>Defects</b><br><b>% by count (Max):-</b><br>(i) Colour<br>(0.5 to 1 sq. cm. of total surface area)                   | 2%                               | 5%                      | 10%                              |
|           | (ii) Skin defects (bruising, brown spots) (0.5 to 2 sq. cm. of total surface area)                                      | 5%                               | 10%                     | 15%                              |
| (iii)     | <b>Range Acceptance</b>                                                                                                 | 95 %                             | 90 %                    | 85 %                             |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                  |                                  |                         |                                  |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70                     | Between 40-70           | Less than 40                     |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 30. ORANGE

Common name: संतरा

English name: Mandarin

Botanical name: *Citrus reticulata*

Varieties name: Kinnow mandarin, Darjeeling Mandarin, Sumithra mandarin



#### 1. Mandarins shall be –

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell ,

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                     | Range-1      | Range-2       | Range-3             |
|-----------|-------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------------|
| <b>A.</b> | <b>Essential</b>                                                                                                        |              |               |                     |
| (i)       | <b>Size in mm</b><br>(diameter of the equatorial section)                                                               | 66 to 80mm   | 55 to 65mm    | Above 80 & Below 55 |
| (ii)      | <b>Defects % by count (Max):-</b><br>(i) Discoloration (0.5 to 5 sq cm of total surface area)                           | 2%           | 5%            | 10%                 |
|           | (ii) Slight bruising/minor cut/ (0.5 to 2 sq cm of total surface area)                                                  | 2%           | 5%            | 10%                 |
| (iii)     | <b>Range Acceptance</b>                                                                                                 | 95 %         | 90 %          | 85%                 |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                  |              |               |                     |
| (iv)      | Total soluble solids (TSS) (minimum) in degree Brix ( Min)                                                              | 8            | 6.5           |                     |
| (v)       | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40        |

#### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 31. SAPOTA

Common name: चीकू

English name: Sapodilla

Botanical name: *Manilkara zapota*

Varieties name: Cricket ball, Kalipatti, Calcutta round, DHS-1, DHS-2, Baramasi, Baharu, Gandhevi Barada etc.



#### 1. Sapota shall be-

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- ii. Sapota shall not be very soft, shriveled or water soaked.
- iii. Free from, visible foreign matter, abnormal external moisture, foreign smell ,

#### 2. Variety .....(to be filled by assayer)

#### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S. No.        | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|---------------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.            | <b>Essential</b>                                                                                                           |              |               |              |
| (i)           | <b>Size</b><br>(diameter of the equatorial section))                                                                       | Above 50mm   | Above 40mm    | Below 40mm   |
| (ii)<br>(iii) | <b>Defects</b><br><b>% by count (Max):-</b><br>Skin defects (scratches scars, scrapes and blemishes)                       | 10 %         | 15 %          | 20%          |
| (iv)          | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85%          |
| B.            | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (v)           | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

#### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 32. MUSK MELON

Common name: खरबूजा

English name: Musk melon

Botanical name: Cucumis *melo*

Varieties name: Arka rajhans, Hara Madhu, Pusa Asaarbati, Arka Jest, Durgapura Madhu, Punjab sunahari, Lanow safed, Annamalai, Harbhari etc.



### 1. Musk Melons shall be-

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell ,

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                                                                                        | Range-1           | Range-2          | Range-3      |
|-----------|----------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |                   |                  |              |
| (i)       | Weights (gms)                                                                                                              | Above 500 to 1500 | Above 300 to 500 | Below 300    |
| (ii)      | <b>Defects % by count (Max):-</b><br>Superficial Crack & Bruises<br>(1-3sq. cm. of total surface area)                     | 5%                | 10%              | 15%          |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 95 %              | 90 %             | 85%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |                   |                  |              |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70      | Between 40-70    | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement



### 33. GRAPES

Common name: अंगूर

English name: Grapes

Botanical name: *Vitis vinifera*

Varieties name: Anab-e-Shah, Bangalore Blue, Bhokri, Gulabi,

Kali Sahebi, Perlette, Thompson Seedless, Sharad Seedless, Arkavati, Arka Hans etc



#### 1. Bunches and berries of Table grapes shall be-

- intact, clean, firmly attached to the stalk, of reasonably uniform size, color characteristics to the variety
- free from insect infestation, water sacs, shrivel age, foreign smell, mould

#### 2. Variety .....(to be filled by assayer)

#### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1                                         | Range-2                                                                        | Range-3                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| A.     | Essential                                                                                                                  |                                                 |                                                                                |                                                                                |
| (i)    | <b>Defects:-<br/>(% by count) Max</b>                                                                                      | 4%                                              | 6%                                                                             | 8%                                                                             |
|        | (i)Discolouration<br>(per bunch)                                                                                           | Up to 200<br>(evenly spaced<br>along the stalk) | Up to 150<br>(less evenly<br>spaced along the<br>stalk than in the<br>Range-1) | Up to 100<br>(less evenly<br>spaced along the<br>stalk than in the<br>Range-2) |
|        | (ii)Bunch size/weight<br>(gm)                                                                                              |                                                 |                                                                                |                                                                                |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %                                            | 90 %                                                                           | 85%                                                                            |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |                                                 |                                                                                |                                                                                |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the<br>variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70                                    | Between 40-70                                                                  | Less than 40                                                                   |
| (iv)   | Total soluble solids<br>(TSS) % (minimum)                                                                                  | 16 degree Brix                                  |                                                                                |                                                                                |

#### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 200 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- 200 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 34. LITCHI

Common name: लीची

English name: Litchi

Botanical name: *Litchi chinensis*

Varieties name: Shahi, Swarna Roopa, China, Kasba, Elachi, Purbi, Early Seedless, Rose Scented etc



#### 1. Litchi shall be:-

- i. intact, firm, clean, firmly attached to the stalk, of reasonably uniform size, color(Red/Orange) characteristics to the variety
- ii. free from soft brown patches on the skin, insect infestation, other visible defects

#### 2. Variety .....(to be filled by assayer)

#### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                        | Range-1                                                                                                                    | Range-2                                                                                                                    | Range-3                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |                                                                                                                            |                                                                                                                            |                                                                                                                            |
| (i)       | <b>Defects (% by count) (Max):-</b><br>(i) Discolouration<br>(1-3 sq cm of total surface area)                             | 10%                                                                                                                        | 15%                                                                                                                        | 20%                                                                                                                        |
|           | (ii) Shriveled                                                                                                             | 2%                                                                                                                         | 5%                                                                                                                         | 7%                                                                                                                         |
| (ii)      | <b>Size/weight of the bunch</b>                                                                                            | Not less than 500 gm<br><br>(Litchis must be firm, firmly attached to the stalk, loose berries shall not be more than 2 %) | Not less than 400 gm<br><br>(Litchis must be firm, firmly attached to the stalk, loose berries shall not be more than 4 %) | Not less than 300 gm<br><br>(Litchis must be firm, firmly attached to the stalk, loose berries shall not be more than 6 %) |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 95 %                                                                                                                       | 90 %                                                                                                                       | 85%                                                                                                                        |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |                                                                                                                            |                                                                                                                            |                                                                                                                            |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70                                                                                                               | Between 40-70                                                                                                              | Less than 40                                                                                                               |
| (v)       | Size equatorial diameter (minimum)                                                                                         | 33 mm                                                                                                                      | 28 mm                                                                                                                      | 23 mm                                                                                                                      |

#### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 35. POMEGRANATE

Common name: अनार

English name: Pomegranate

Botanical name: *Punica granatum*

Varieties name: Bhagava, Ganesh, Arakta, Mrudula, Muskat, Jyoti, Ruby, Dholka etc.



### 1. Pomegranates shall be :

- i. intact, firm, clean, firmly attached to the stalk of reasonably uniform size, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell ,

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                   | Range-1           | Range-2          | Range-3          |
|--------|-------------------------------------------------------------------------------------------------------|-------------------|------------------|------------------|
| A.     | <b>Essential</b>                                                                                      |                   |                  |                  |
| (i)    | <b>Size (mm)</b>                                                                                      | <b>Upto 90 mm</b> | <b>Upto 75</b>   | <b>Upto 65</b>   |
| (ii)   | <b>Defects<br/>% by count (Max):-</b><br><br>Slight bruising<br>(2-5sq. cm. of total<br>surface area) | 2%                | 5%               | 10%              |
| (iii)  | <b>Range Acceptance</b>                                                                               | 95 %              | 90 %             | 85%              |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                |                   |                  |                  |
| (iv)   | <b>Weight in Grams<br/>(minimum)</b>                                                                  | More than<br>300  | More than<br>200 | Less than<br>200 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

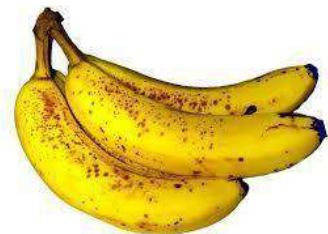
## 36. BANANA

Common name: केला

English name: Banana

Botanical name: *Musa acuminata*

Varieties name: Dwarf Cavendish, Robusta, Rasthali, Safed Velchi Musa, Karpuravalli, Monthan etc.



### 1 Bananas shall be –

- intact, firm, clean, firmly attached to the stalk of reasonably uniform size, color (Green/Yellow/Golden Yellow) characteristics to the variety
- free from chitti, insect infestation, cuts, mechanical injury ,
- Bananas shall not be over ripened /under ripened

2 Variety .....(to be filled by assayer)

3 Actual Picture of the lot..... ( to be uploaded by assayer)

### 4 Criteria for Range designation:

| Sl.No.    | Tradable Parameters                                  | Range-1                                                                                       | Range-2 | Range-3   |
|-----------|------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------|-----------|
| <b>A.</b> | <b>Essential</b>                                     |                                                                                               |         |           |
| (i)       | <b>Defects (% by count) (Max)</b>                    |                                                                                               |         |           |
|           | (i) Cut/ Bruising (10 to 20mm of total surface area) | 5                                                                                             | 10      | Above 10  |
|           | (ii) Chitti (5-10mm of surface area)                 | 5.0                                                                                           | 8.0     | Above 8.0 |
|           | (iii) Black spot (5-10mm of surface area)            | 2.0                                                                                           | 4.0.    | Above 4.0 |
| (ii)      | <b>Range Acceptance</b>                              | 95 %                                                                                          | 90 %    | 85%       |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>               |                                                                                               |         |           |
| (iii)     | <b>Sizing characteristics</b>                        | The minimum length should not be less than 14.0 cm and the minimum grade not less than 2.7 cm |         |           |

### 5 Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Lungars shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- 03 bunches shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 37.PLUM

Common name: आलूबुखारा

English name: Plum

Botanical name: *Prunus domestica*

Varieties name: Kala Amritsari, Satluj purple, Jamuni Merruti, Titron, Katari chak, Alu Bukhara, Howe, Alpha, Alucha Black etc.



### 1 Plums shall be-

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell ,  
Mold/ Fungal infested.

2 Variety .....(to be filled by assayer)

3 Actual Picture of the lot..... ( to be uploaded by assayer)

### 4 Criteria for Range designation:

| S. No.       | Tradable Parameters                                                                                                        | Range-1         | Range-2         | Range-3         |
|--------------|----------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|
| <b>A.</b>    | <b>Essential</b>                                                                                                           |                 |                 |                 |
| <b>(i)</b>   | Size (mm)                                                                                                                  | <b>Above 40</b> | <b>30 to 39</b> | <b>Below 30</b> |
| <b>(ii)</b>  | <b>Defects % by count (Max):-</b><br>Slight bruising/ hail patches<br>( 0.5 -1 sq. cm of total surface area)               | 3%              | 5%              | 7%              |
| <b>(iii)</b> | <b>Range Acceptance</b>                                                                                                    | 95 %            | 90 %            | 85%             |
| <b>B.</b>    | <b>Optional (Industrial/Processor)</b>                                                                                     |                 |                 |                 |
| <b>(iv)</b>  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70    | Between 40-70   | Less than 40    |

### 5 Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## BB. PEACH

Common name: **आम**

English name: Peach

Botanical name: *Prunus persica*

Varieties name: TA-170, Shan-e-Punjab Flordasun, Sharbati etc.



### 1. Peach shall be-

- i. intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell ,

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.       | Tradable Parameters                                                                                                        | Range-1         | Range-2         | Range-3         |
|--------------|----------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|
| <b>A.</b>    | <b>Essential</b>                                                                                                           |                 |                 |                 |
| <b>(i)</b>   | Size (mm)                                                                                                                  | <b>Above 45</b> | <b>35 to 45</b> | <b>Below 35</b> |
| <b>(ii)</b>  | <b>Defects<br/>% by count (Max):-</b><br>Slight bruising/ soft patches<br>( 0.5 -1 sq. cm of total surface area)           | 3%              | 5%              | 7%              |
| <b>(iii)</b> | <b>Range Acceptance</b>                                                                                                    | 95 %            | 90 %            | 85%             |
| <b>B.</b>    | <b>Optional (Industrial/Processor)</b>                                                                                     |                 |                 |                 |
| <b>(iv)</b>  | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70    | Between 40-70   | Less than 40    |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement



### 39. MANGO

Common name: आम

English name: Mango

Botanical name: *Mangifera indica*

Varieties name: Dusherry, Alphonso, Totapuri, Baneshan, Safeda, Malda, Malda, kesar etc



#### 1. Mangoes shall be-

- i. Fully developed, matured, clean, of reasonably uniform size & color (Yellow/Reddish, Yellow/Reddish Green) characteristics to the variety
  - ii. free from, insect infestation, internal spongy tissues, visible foreign matter, foreign smell ,
2. Variety ..... (to be filled by assayer)
3. Actual Picture of the lot ..... ( to be uploaded by assayer)
4. **Criteria for Range designation:**

| S.No.     | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |              |               |              |
| (i)       | <b>Size by Weight (in gm)</b><br>For all varieties                                                                         | Up to 500    | Up to 200     | Up to 100    |
| (ii)      | <b>Defects (% by count) (Max)</b><br><br>Cut skin/Bruising (10 to 30 mm of total surface area)                             | 10%          | 15%           | 20%          |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 90 %         | 85 %          | 80%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

#### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample,

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 40. SWEET ORANGE

Common name: **मौसंबी**

English name: Sweet Orange

Botanical name: *Citrus aurantium*

Varieties name: Mosambi, Malta (Common), Malta (Blood Red), Sathgudi etc.



### 1. Mosambi shall be-

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety.
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell.

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                                                         | Range-1                      | Range-2       | Range-3      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                            |                              |               |              |
| (i)       | <b>Size</b><br>(Diameter)                                                                                                   | Upto 80mm                    | Upto 65 mm    | Upto 55 mm   |
| (ii)      | <b>Defects</b><br><b>% by count (Max):-</b><br>Scratches scars, scrapes and blemishes (0.5-2 sq. cm. of total surface area) | <b>3%</b>                    | 5%            | 7%           |
| (iii)     | <b>Range Acceptance</b>                                                                                                     | 95 %                         | 90 %          | 85%          |
| <b>B</b>  | <b>Optional (Industrial/Processor)</b>                                                                                      |                              |               |              |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour)        | More than 70                 | Between 40-70 | Less than 40 |
| (v)       | <b>Total soluble solids</b>                                                                                                 | Not less than 10 degree brix |               |              |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 41.CUSTARD APPLE

Common name: शरीफा

English name: Custard Apple

Botanical name: *Annona reticulata*

Varieties name: Balanagar, Arka Sahan, Barbados Seedling, Kakarlapahad, British Guinea, Mahboobnagar, Saharanpur local, Washington etc.



### 1. Custard apple shall be-

- i. intact, firm, clean, & color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1            | Range-2            | Range-3            |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|
| A.     | <b>Essential</b>                                                                                                           |                    |                    |                    |
| (i)    | <b>Size (Diameter)</b>                                                                                                     | <b>Above 60 mm</b> | <b>50 to 60 mm</b> | <b>Below 50 mm</b> |
| (ii)   | <b>Defects % by count (Max):-</b><br>Blemishes, scars, sunspots (0.5-2 sq. cm. of total surface area)                      | 3%                 | 5%                 | 10%                |
| (iii)  | <b>Range Acceptance</b>                                                                                                    | 95 %               | 90 %               | 85 %               |
|        | <b>Optional (Industrial/Processor)</b>                                                                                     |                    |                    |                    |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70       | Between 40-70      | Less than 40       |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 42. WATERMELONS

Common name: तरबूज

English name: Watermelon

Botanical name: *Citrullus lanatus*

Varieties name: Sugar Baby and Arka Manik, PKM 1, Arka Jyoti and Pusa Bedana



### 1. Watermelons shall be –

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                                                 | Range-1                      | Range-2       | Range-3      |
|-------|---------------------------------------------------------------------------------------------------------------------|------------------------------|---------------|--------------|
| A.    | Essential                                                                                                           |                              |               |              |
| (i)   | Weight (kgs)                                                                                                        | above 5 to 10                | 2 to 5        | Below 2      |
| (ii)  | Defects<br>% by count (Max):-<br><br>Colour (paler part of any region including bruises)                            | 10%                          | 15%           | 20%          |
| (iii) | Range Acceptance                                                                                                    | 95 %                         | 90 %          | 85 %         |
| B.    | Optional (Industrial/Processor)                                                                                     |                              |               |              |
| (iv)  | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70                 | Between 40-70 | Less than 40 |
| (v)   | Total Soluble Solids                                                                                                | Not less than 10 degree brix |               |              |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 3 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 3 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

### 43. LEMON

Common name: नींबू

English name: Lemon

Botanical name: *Citrus limon*

Varieties name: Kargzi lime, Assam lemon, Italian lemon, Pant lemon, Galgal, Eureka lemon, Sevilla, Malta lemon, Vikram, Prumalini, PKM, and Sai Sharbati etc.



#### 1. Lemons shall be-

- i. intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell, Mold infestation

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                                      | Range-1       | Range-2       | Range-3      |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                                         |               |               |              |
| (i)    | <b>Size</b><br>(Diameter)                                                                                                                | Above 30 mm   | 20 to 30 mm   | Below 20 mm  |
| (ii)   | <b>Defects % by count (Max):-</b><br>(i) Colour<br><br>(ii) Scratches scars, scrapes and blemishes (0.5-1 sq. cm. of total surface area) | 3 %<br><br>3% | 5%<br><br>5%  | 7%<br><br>7% |
| (iii)  | <b>Range Acceptance</b>                                                                                                                  | 95 %          | 90 %          | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                                   |               |               |              |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour)               | More than 70  | Between 40-70 | Less than 40 |

#### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 44.GUAVA

Common name: अमरुद/ Goiyya (Tamil)/Perakka (Malyalam)/  
Jaama (Telugu)/ JAMFAL (Gujarati) /गु (Marathi)

English name:Guava

Botanical name: *Psidium guajava*



Variety Name: Allahabad safeda, Lucknow 49, Anakapalli, Banarasi, Chittidar, Hafshi, Sardar, Smooth Green, Safed Jam, Arka Mridula,-49, Lucknow Safeda, Apple Colour, Chittidar, Red Fleshed, Allahabad Surkha, Mirzapuri Seedless, CISH-G-1, CISH-G-2, CISH-G-3

### 1. Guava shall be–

- i. Wholesome, firm, fresh, clean, of same variety and
- ii. Free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameter                                                                                                         | Range-1      | Range-2                   | Range-3                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------|---------------------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |              |                           |                           |
| (i)       | Size mm                                                                                                                    | 50 to 90     | Below 50 and more than 90 | Below 50 and more than 90 |
| (ii)      | <b>Defects % by count (Max):-</b><br><br>Skin defects (Bruising/scratches/ Sunburn)                                        | 5%           | 10%                       | 15%                       |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %                      | 85 %                      |
|           | <b>Optional (Industrial/Processor)</b>                                                                                     |              |                           |                           |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70             | Less than 40              |

### 5. Sampling:

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 45. JACKFRUIT

Common name: कटहल /Palapazham (Tamil)/chakka (Malayalam)/ Panasa (Telugu)/फणस (Gujarati & Marathi)

English name: Jackfruit

Botanical name: *Artocarpus heterophyllus*



### 1. Jackfruit shall be –

- i. Wholesome, firm, fresh, clean, of same variety and
- ii. Free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.        | Tradable Parameter                                                                                                               | Range-1      | Range-2           | Range-3      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------|--------------|
| <b>A.</b>    | <b>Essential</b>                                                                                                                 |              |                   |              |
| <b>(i)</b>   | <b>Defects<br/>% by count (Max):-</b>                                                                                            |              |                   |              |
|              | Irregular Shape<br>maturity development                                                                                          | 2%           | 4%                | 6%           |
|              | Skin defects                                                                                                                     | 3%           | 5%                | 10%          |
| <b>(ii)</b>  | <b>Range Acceptance</b>                                                                                                          | 95 %         | 90 %              | 85 %         |
| <b>B.</b>    | <b>Optional (Industrial/Processor)</b>                                                                                           |              |                   |              |
| <b>(iii)</b> | <b>Color (%)</b><br>(Characteristic true to the<br>variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety<br>colour) | More than 70 | Between 40-<br>70 | Less than 40 |

### 5. Sampling:

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 3 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 3 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 46. KINNOW

Common name: किन्नु Naarinja (Telugu)

English name: Mandarin

Botanical name: *Citrus reticulata*



### 1. KINNOW shall be –

- Wholesome, firm, fresh, clean, of rind moderately thick, of reasonably uniform size, & shape, color characteristics to the variety
- Free from, visible foreign matter, abnormal external moisture, foreign smell

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                                                                                          | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                                                                                             |              |               |              |
| (i)    | <b>Defects<br/>% by count (Max):-</b><br>(i) Discolouration (1-4 sq cm of total surface area)<br><br>(ii) Skin defects due to rubbing (Bruising/scratches) (1-4 sq cm of total surface area) | 2%           | 4%            | 6%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                                                                                      | 95 %         | 90 %          | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                                                                                       |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour)                                                                         | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 47. RAW MANGO (For Pickles)

Common name: कच्चा आम/Maangai (Tamil)/ Pacha mango(Malyalam)/Pachi Mamidi (Telugu)/KARY (Gujarati)  
English name: Raw Mango  
Botanical name: Mangifera indica



**1. Raw mangoes shall be-**

- i. Firm, Wholesome, fresh, clean, of same variety and
- ii. Free from visible foreign matter, abnormal external moisture, foreign smell

**2. Variety .....**(to be filled by assayer)

**3. Actual Picture of the lot.....** ( to be uploaded by assayer)

**4. Criteria for Range designation:**

| S.No. | Tradable Parameter                                                                                                   | Range-1           | Range-2       | Range-3            |
|-------|----------------------------------------------------------------------------------------------------------------------|-------------------|---------------|--------------------|
| A.    | <b>Essential</b>                                                                                                     |                   |               |                    |
| (i)   | <b>Hardness</b>                                                                                                      | <b>Very hard</b>  | <b>Hard</b>   | <b>Slight hard</b> |
| (ii)  | <b>Color</b>                                                                                                         | <b>Dark Green</b> | <b>Green</b>  | <b>Light Green</b> |
| (iii) | <b>Defects (Max):-</b><br>Cut,<br>Bruising/scratches (5mm to 15)                                                     | 5%                | 10%           | Above 10%          |
| (iv)  | <b>Range Acceptance</b>                                                                                              | 95 %              | 90 %          | 85 %               |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                               |                   |               |                    |
| (v)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour) | More than 70      | Between 40-70 | Less than 40       |

**5. Sampling:**

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 48. ONION

Common name: **प्याज**

English name :Onion

Botanical name: *Allium cepa* (L)

Varieties name: Bhima Raj, Bhima Red, Bhima Super, Bhima Super, Bhima Shakti, Bhima Shweta, PAU, Ludhiana, Pusa Red, N-2-4-1, Pusa Madhavi, NHRDF, Nasik, Early Grano, Arka Niketan, Arka Kalyan etc



### 1. Onion shall be-

- i. dry, intact, clean, of reasonably uniform size, & shape, color characteristic to the variety,
- ii. free from sprouting, seed stem, double split, staining, insect infestation, soft root.

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sl.No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                           |              |               |              |
| (i)    | <b>Defects<br/>(% by count) (Max)</b>                                                                                      |              |               |              |
|        | (i) Cut<br>(5-15mm long)                                                                                                   | 5.0          | 10.0          | 15.0         |
|        | (ii) Double split                                                                                                          | 2.0          | 5.0           | Above 8.0    |
|        | (iii) Sprouted                                                                                                             | 2.0          | 5.0           | Above 8.0    |
|        | (iv) Rooting                                                                                                               | 2.0          | 5.0           | Above 8.0    |
| (ii)   | <b>Size</b><br>(maximum diameter of the equatorial section)                                                                | Up to 65 mm  | Up to 45mm    | Up to 25 mm  |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling.

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- (iii) 500 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 49. POTATO

Common name: **आलू**

English name: Potato

Botanical name: *Solanum tuberosum*

Varieties name: Kufri Sindhuri, Kufri Chandramukhi, Kufri Jyoti, Kufri Lauvkar, Kufri Badshah, Kufri Bahar, Kufri Lalima, Kufri Jawahar, Kufri Suttlej, Kufri Ashoka, Kufri Pukhraj, Kufri Chipsona-1, Kufri Chipsona-2 Kufri Anand



### 1. Potato shall be-

- firm, fresh, of reasonably uniform size, shape & colour characteristic to the variety
- free from rotten spot, greening, chitti, cracks, cuts, major wholes, mechanical injury

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sl.No.    | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |              |               |              |
| (i)       | <b>Defects (% by count) (Max)</b>                                                                                          |              |               |              |
|           | (i) Cracks/cut (5 mm to 20mm)                                                                                              | 5%           | 10%           | Above 10%    |
|           | (ii) Greening (1-3 sq cm of total surface area)                                                                            | 2.0          | 3.0           | Above 3.0    |
|           | (iii) Holes                                                                                                                | 1.0          | 2.0           | Above 2      |
|           | (iv) Mechanical injury                                                                                                     | 2.0          | 3.0           | Above 3.0    |
| (ii)      | <b>Size (mm)</b><br>(maximum diameter of the equatorial section)                                                           | Up to 70     | Up to 55      | Up to 25     |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 90%          | 85%           | 80%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling
- 1Kg shall be drawn from each selected container and shall be mixed homogeneously, called as primary sample
- 500g. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 50. TOMATO

Common name: टमाटर

English name : Tomato

**Botanical name:** *Solanum lycopersicum*

Varieties name :Arka Saurabh, Arka Vikas, Arka Ahuti, Arka Ashish, Arka Abha , Arka Alok, HS101, HS102, HS110, Hisar Arun, Hisar Lalima, Hisar Lalit, Hisar Anmol, KS.2, Narendra Tomato 1, Narendra Tomato 2, Pusa Red Plum, Pusa Early Dwarf, Pusa Ruby, Co-1, CO 2, CO 3, S-12, Punjab Chhuhara, PKM 1, Pusa Ruby, Paiyur-1, Shakthi, SL 120, Pusa Gaurav, S 12, Pant Bahar, Pant T3, Solan Gola and Arka Meghali etc



### 1. Tomatoes shall be-

- intact, firm, clean, of reasonably uniform size& shape, color characteristics to the variety
- free from insect damage, mechanical injury, rotten spot, over ripening

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sl.No.    | Tradable Parameters                                                                                                  | Range-1      | Range-2       | Range-3      |
|-----------|----------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                     |              |               |              |
| (i)       | <b>Defects (% by count) (Max)</b>                                                                                    |              |               |              |
|           | (i) Dry crack/cut (5-10mm)                                                                                           | 2.0          | 3.0           | Above 3.0    |
|           | (ii) Sunburn (for 1-3 sq.cm of surface area)                                                                         | 2.0          | 3.0           | Above 3.0    |
|           | (iii) Mechanical injury                                                                                              | 2.0          | 3.0           | Above 3.0    |
|           | (iv) Discolouration (Green /Green Top/ Blotchy) (for 1-4 sq.cm of surface area)                                      | 3.0          | 5.0           | Above 5.0    |
| (ii)      | <b>Size</b> (maximum diameter of the equatorial section)                                                             | Upto 65mm    | Upto 55mm     | Upto 45 mm   |
| (iii)     | <b>Range Acceptance</b>                                                                                              | 95%          | 90%           | 85%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                               |              |               |              |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling.

- 5 % or minimum one container shall be randomly selected for sampling.
- 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- 500 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 51. PEAS

Common name: मटर

English name :Shelling peas

Botanical name: *Pisum sativum* (L)

Varieties name : Arkel, Jawahar Matar-4, Bonneville, P-8, Lincoln, Jawahar Matar-1, VL-3, Pant Uphar etc.



### 1. The Peas pods shall be –

- i. intact, clean and sound, turgid, fresh in appearance
- ii. free from black/brown spots on pods, visible insect damage, rotten tips,

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sl.No.    | Tradable Parameters                                                                                                        | Range-1                       | Range-2                         | Range-3                         |
|-----------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------|---------------------------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |                               |                                 |                                 |
| (i)       | <b>Defects (Max)</b><br>(i) Under developed (% by count)                                                                   | 3.0                           | 5.0                             | Above 5.0                       |
|           | (ii) Black/ brown spots                                                                                                    | 1 sq cm of total surface area | 1-2 sq cm of total surface area | 2-3 sq cm of total surface area |
|           | (iii) Dry skin                                                                                                             | Covering 10% surface area     | Covering 20% surface area       | Covering above 20% surface area |
| (ii)      | Size (Length in mm)<br>(maximum diameter of the equatorial section)                                                        | Upto 65mm                     | Upto 55mm                       | Upto 45 mm                      |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 95%                           | 90%                             | 85%                             |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |                               |                                 |                                 |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70                  | Between 40-70                   | Less than 40                    |

### 5. Sampling.

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample
- (iii) 200 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 52.BOTTLE GOURD

Common name: लौकी

English name: Bottle Gourd

Botanical name: *Lagenaria siceraria*

Varieties name: PKM1, CO.1, Pusa Navven, Arka Bahar, Summer prolific Round, Summer prolific long, Pusa Manjiri, Pusa Meghdoot, Samrat etc.



### 1. Bottle gourd shall be-

- intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
- free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                     | Range-1      | Range-2       | Range-3       |
|--------|-------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|
| A.     | <b>Essential</b>                                                                                                        |              |               |               |
| (i)    | Weight (min.)                                                                                                           | Upto 750 gms | 500- 350 gms  | Below 349 gms |
| (ii)   | <b>Defects % by count (Max):-</b><br>(i) shape/color & stains<br>(1-4 sq cm of the surface area)                        | 5%           | 10%           | 15%           |
|        | (ii) rubbing and handling (1-4 sq cm of the surface area)                                                               | 5%           | 10%           | 15%           |
| (iii)  | <b>Range Acceptance</b>                                                                                                 | 95 %         | 90 %          | 85 %          |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                  |              |               |               |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40  |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



### 53. BITTER GOURD

Common name: करेला

English name: Bitter Gourd

Botanical name: *Momordica charantia*

Varieties name: Co 1, MDU 1, COBgoH 1 (Hybrid), Arka Harit, Priya and Preethi etc



#### 1. Bitter gourds shall be-

- i. intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                                                   | Range-1          | Range-2             | Range-3      |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|--------------|
| A.     | Essential                                                                                                                                             |                  |                     |              |
| (i)    | Length( mm)                                                                                                                                           | Above 150 to 200 | Above 100 mm to 150 | 60 to 100    |
| (ii)   | <b>Defects<br/>% by count (Max):-</b><br>Skin, Color defects (scratches, scars, scraps and blemishes slight yellow color) (1-4 sq cm of surface area) | 3%               | 5%                  | 7%           |
| (iii)  | <b>Range Acceptance</b>                                                                                                                               | 95 %             | 90 %                | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                                                |                  |                     |              |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour)                                  | More than 70     | Between 40-70       | Less than 40 |

#### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 54. CUCUMBER

Common name: खीरा

English name: Cucumber

Botanical name: *Cucumis sativus*

Varieties name: Amira, Lemon, Greensleeves, Armenian, Salad bush, Pickle bush etc.



### 1. Cucumber shall be–

- i. intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell
2. Variety .....(to be filled by assayer)
3. Actual Picture of the lot..... ( to be uploaded by assayer)
4. **Criteria for Range designation:**

| S.No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|-------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.    | <b>Essential</b>                                                                                                           |              |               |              |
| (i)   | <b>Defects<br/>% by count (Max):-</b>                                                                                      |              |               |              |
|       | (i) color stains &                                                                                                         | 3%           | 5%            | 7%           |
|       | (ii) Skin defects<br>(Bruising/scratches/ cut) (1-4 sq cm of total surface area)                                           | 3%           | 5%            | 7%           |
| (ii)  | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii) | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 55. BRINJAL

Common name: बैंगन

English name: Brinjal

Botanical name: *Solanum melongena*

Varieties name: Pusa Purple Long, Pusa Purple Cluster, Pusa Kranti, Pusa Barsati, Manjri Gota, Vaishali, Arka Navneet, Arka Sheel, Arka Kusmukar, Arka Nidhi ( BWR – 12), Arka Keshav ( BWR – 21 ), Arka Neelkanth ( BWR – 54 ), Pusa Ankur, Eggplant- cloud nine, Hari, Andaj, Bijli, Bharta



### 1. Brinjal shall be -

- iii. intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
  - free from visible foreign matter, abnormal external moisture, foreign smell
  - Variety.....(to be filled by assayer)
  - Actual Picture of the lot ..... ( to be uploaded by assayer)

### 2. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                           |              |               |              |
| (i)    | <b>Defects</b><br><b>% by count (Max):-</b><br>(i) Bruising/<br>Scars/<br>Scratches<br>(1-3 sq cm of total surface area)   | 3%           | 5%            | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale<br>(1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 3. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 15 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 15 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 56. CABBAGE

Common name: पत्तागोभी

English name: Cabbage

Botanical name: *Brassica oleracea var. capitata*

Varieties name: Golden Acre, Pride of India, Copenhagen Market, Pusa Mukta, Pusa Synthetic, Midseason Market, September Early, Pusa Drumhead, Early Drum Head, Late Large Drum Head, K-1



### 1 Cabbage shall be-

- i. fresh, firm, clean, well trimmed, of reasonably uniform size ,variety
  - ii. free from visible foreign matter, abnormal external moisture, foreign smell
- 2 Variety .....(to be filled by assayer)
- 3 Actual Picture of the lot.....( to be uploaded by assayer)
- 4 **Criteria for Range designation:**

| S. No. | Tradable Parameters                                                                                                        | Range-1               | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                           |                       |               |              |
| (i)    | <b>Defects<br/>% by count (Max):-</b>                                                                                      |                       |               |              |
|        | (i) Burst Head                                                                                                             | 3%                    | 5%            | 7%           |
|        | (ii) Mechanical/Physical Injury                                                                                            | 3%                    | 5%            | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %                  | 90 %          | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |                       |               |              |
| (iii)  | <b>Weight</b>                                                                                                              | Not less than 200 gms |               |              |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70          | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 57.CAULIFLOWER

Common name: फूलगोभी

English name: Cauliflower/ BROCCOLI

Botanical name: *Brassica oleracea var. botrytis*/ L. var. italic



Varieties name: Pusa Deepali, Early Kunwari, Punjab Giant-26, Punjab Giant-35, Pant Shubhra, Pusa Snowball-1, Sonwball-16, Pusa Early Synthetic, Pant Gobhi-2, Pant Gobhi-3, Dania Kalimpong, KTS-I, Solan Green, Lucky, Fiesta, Pushpa, Aishwarya, Palam Vichitra etc.

### 1. Cauliflower shall be-

- i. fresh, firm, clean, well trimmed, of reasonably uniform size , & shape, color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                           |              |               |              |
| (i)    | <b>Defects % by count (Max):-</b>                                                                                          |              |               |              |
|        | (i) Bruising<br>(1-3 sq cm of total surface area)                                                                          | 3 %          | 5 %           | 7 %          |
|        | (ii) Colour<br>(1-5 sq cm of total surface area)                                                                           | 3 %          | 5 %           | 7 %          |
|        | (iii) Wooliness<br>(1-10 sq cm of total surface area)                                                                      | 3%           | 5%            | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 58. GREEN CHILLIES

Common name: हरी मिर्च

English name: Green Chilly

Botanical name: *Capsicum annum L*

Varieties name: Agnirekha, Volcano, Tapan, Savitri, Bela, NS991, Balwant, Green wonder, CCS-1, G-4, Kiran and Jwala etc.



### 1. Green Chillies shall be

- i. fresh, firm, clean, of color characteristics to the variety
- ii. Free from visible foreign matter, abnormal external moisture, foreign smell, holes, rotten, insect infestation

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|-------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.    | <b>Essential</b>                                                                                                           |              |               |              |
| (i)   | <b>Defects</b><br><b>% by count (Max):-</b><br>(i) Bruising<br>(5-10 mm of total surface area)                             | 3%           | 5%            | 7%           |
|       | (ii) Colour<br>(Including sun spots)<br>(05-10 mm of total surface area)                                                   | 3%           | 5%            | 7%           |
| (ii)  | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii) | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5% or minimum one container shall be randomly selected for sampling
- (ii) 100g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (iii) 100g. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 59. CARROT

Common name: गाजर

English name: Carrot

Botanical name: *Daucus carota* subsp. *Sativus*

Varieties name: Pusa Kesar, Pusa Meghali, Selection 223, No. 29, Nantes Half Long, Early Nantes, Chantenay, Imperator, Zeno, Pusa Yamdagini etc.



### 1. Carrots shall be-

- i. fresh, firm, clean, of color characteristics to the variety
- ii. free from root splitting, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                     | Range-1           | Range-2                 | Range-3                          |
|--------|-------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------|----------------------------------|
| A.     | <b>Essential</b>                                                                                                        |                   |                         |                                  |
| (i)    | <b>Size (mm)</b>                                                                                                        | <b>150 to 200</b> | <b>Above 200 to 275</b> | <b>Below 150 &amp; above 275</b> |
| (ii)   | <b>Defects % by count (Max):-</b><br>(i) Bruising/ Scars/ Scratches<br>(1-10 sq cm of total surface area)               | 5%                | 10%                     | 15%                              |
|        | (ii) discoloration<br>(1-4 sq cm of total surface area)                                                                 | 5%                | 10%                     | 15%                              |
|        | (iii) Cracks/Fissures                                                                                                   | 3%                | 5%                      | 7%                               |
| (iii)  | <b>Range Acceptance</b>                                                                                                 | 95 %              | 90 %                    | 85 %                             |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                  |                   |                         |                                  |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70      | Between 40-70           | Less than 40                     |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 15 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 15 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 60. SWEET POTATO

Common name: शकरकंद

English name: Sweet Potato

Botanical name: *Ipomoea batatas*

Varieties name: Co 3, Co CIP 1, Sree Nandini, Sree Vardhini, Kiran, Sree Bhadra, Sree Rethna, Gouri and Sankar. Certain local cultivars like SP local, SP 4, SP 13, SP 18 and Musiri Thandel



### 1. Sweet potatoes shall be –

- i. fresh, firm of color characteristics to the variety
- ii. free from black rot, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | Essential                                                                                                                  |              |               |              |
| (i)    | <b>Defects<br/>% by count (Max):-</b>                                                                                      |              |               |              |
|        | (i) Discoloration<br>(1-4 sq cm of total surface area)                                                                     | 3%           | 5%            | 7%           |
|        | (ii) Skin defects<br>(Bruising/scratches/cut)<br>(1-4 sq cm of total surface area)                                         | 3%           | 5%            | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B      | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale<br>(1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 05 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 61. SPINACH

Common name: पालक

English name: Spinach

Botanical name: *Spinacia oleracea*



### 1. Spinach shall be –

- i. intact, firm, clean, properly drained if washed and
- ii. free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                  | Range-1                                          | Range-2                                          | Range-3                                           |
|--------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------|
| A.     | Essential                                                                                                            |                                                  |                                                  |                                                   |
| (i)    | <b>Defects % by weight (Max):-</b><br>(i) Discolouration                                                             | 3%                                               | 5%                                               | 7%                                                |
|        | (ii) Leaf Stem                                                                                                       | Spinach Leaf stem must not exceed 5 cm in length | Spinach Leaf stem must not exceed 7 cm in length | Spinach Leaf stem must not exceed 10 cm in length |
| (ii)   | <b>Range Acceptance</b>                                                                                              | 95 %                                             | 90 %                                             | 85 %                                              |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                               |                                                  |                                                  |                                                   |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour) | More than 70                                     | Between 40-70                                    | Less than 40                                      |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 500 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 62. MUSTARD LEAF

Common name: सरसों साग

English name: Mustard Leaf

Botanical name: *Brassica nigra*



### 1. Mustard Leaf shall be –

- i. intact, firm, clean, properly drained if washed and
- ii. free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1                                          | Range-2                                          | Range-3                                           |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------|
| A.     | <b>Essential</b>                                                                                                           |                                                  |                                                  |                                                   |
| (i)    | <b>Defects<br/>% by weight (Max):-</b><br>(i) Discolouration                                                               | 3%                                               | 5%                                               | 7%                                                |
|        | (ii) Leaf Stem                                                                                                             | Mustard Leaf stem must not exceed 5 cm in length | Mustard Leaf stem must not exceed 7 cm in length | Mustard Leaf stem must not exceed 10 cm in length |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %                                             | 90 %                                             | 85 %                                              |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |                                                  |                                                  |                                                   |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70                                     | Between 40-70                                    | Less than 40                                      |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 500 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 63.CORIANDER LEAVES

Common name: धनिया पत्ता

English name: Coriander leaves

Botanical name: *Coriandrum sativum*

Varieties name: CO-1, CO-2, CO-3, CS-2, CS-287, Karan



### 1. Coriander Leaves shall be –

- intact, firm, clean, properly drained if washed and
- free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                           |              |               |              |
| (i)    | <b>Defects</b><br><b>% by weight (Max):-</b><br>(i) Discolouration                                                         | 3%           | 5%            | 7%           |
|        | (ii) Split leaves                                                                                                          | 3%           | 5%            | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- 100 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 64.GARLIC

Common name: लहसुन

English name: Garlic

Botanical name: *Allium sativum*

Varieties name: Bhima Omkar, Bhima Purple, Agrifound White (G-41) Yamuna Safed(G-1), Yamuna Safed-2 (G-50), Yamuna Safed-3 (G-282), Yamuna Safed-4 (G-323) Yamuna Safed-5 (G-189), Godavari, Shweta, Phule Baswant, GG 4, Ooty 1, VL Garlic 1, VL Lahsun 2, Agrifound Parvati, Agrifound Parvati-2 (G-408)



### 1. Garlic bulbs shall be-

- i. intact, firm, clean, of reasonably uniform size & shape, color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell and

Garlic may be Fresh or Semidry or Dry.

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                    | Range-1             | Range-2 | Range-3 |
|-------|----------------------------------------|---------------------|---------|---------|
| A.    | <b>Essential</b>                       |                     |         |         |
| (i)   | <b>Defects<br/>% by count (Max):-</b>  |                     |         |         |
|       | (i) Irregular shape                    | 3%                  | 5%      | 7%      |
|       | (ii) Slight tears in outer skin        | 2%                  | 4%      | 6%      |
| (ii)  | <b>Range Acceptance</b>                | 95 %                | 90 %    | 85 %    |
| B.    | <b>Optional (Industrial/Processor)</b> |                     |         |         |
| (iii) | Garlic Bulb diameter                   |                     |         |         |
|       | (i) with cloves                        | Not less than 25 mm |         |         |
|       | (ii) single clove                      | Not less than 15 mm |         |         |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 100 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 100 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 65. Bhindi/Okra

Common name: भिंडी

English name: Lady Finger

Botanical name: *Abelmoschus esculentus*

Varieties name: Pusa Makhmali, Punjab no. 13, Punjab Padmini, Pusa Sawani, Parbhani Kranti, Arka Anamika etc.



### 1. Lady finger shall be-

- i. whole with peduncle and intact tip, firm, fresh, clean, of color characteristics to the variety
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                                                               | Range-1      | Range-2          | Range-3              |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|----------------------|
| A.    | Essential                                                                                                                         |              |                  |                      |
| (i)   | Size (mm)                                                                                                                         | 30 to 110    | Above 110 to 135 | Below 30 & above 135 |
| (ii)  | <b>Defects</b><br><b>% by count (Max):-</b><br>(i) scars, blemishes, scratches, bruises, scraps (1-4 sq cm of total surface area) | 3%           | 5%               | 7%                   |
|       | (ii) Broken/without stalk                                                                                                         | 2%           | 4%               | 6%                   |
|       | (iii) Hardness                                                                                                                    | 10%          | 15%              | 20%                  |
| (iii) | <b>Range Acceptance</b>                                                                                                           | 95 %         | 90 %             | 85 %                 |
| B.    | Optional (Industrial/Processor)                                                                                                   |              |                  |                      |
| (iv)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour)              | More than 70 | Between 40-70    | Less than 40         |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 100 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 66.GINGER

Common name: अदरक

English name: Ginger

Botanical name: *Zingiber officinale*

Varieties name: Maran, Assam, Himachal, Kuruppampadi, Wynad Local, Suprabha, Suruchi, Suravi, Himgiri, Varada, Mahima, Rejatha etc.



### 1. Ginger shall be-

- Wholesome, firm, fresh, clean, same variety and
  - Free from visible foreign matter, abnormal external moisture, foreign smell
- Variety..... (to be filled by assayer)
  - Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1               | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|--------------|
| A.     | <b>Essential</b>                                                                                                           |                       |               |              |
| (i)    | <b>Defects<br/>% by weight (Max):-</b><br><br>Skin defects<br>(Bruising/scratches/cut) (1-3 sq cm of total surface area)   | 3%                    | 5 %           | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %                  | 90 %          | 85 %         |
|        | <b>Optional (Industrial/Processor)</b>                                                                                     |                       |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70          | Between 40-70 | Less than 40 |
| (iv)   | <b>Weight</b>                                                                                                              | Not less than 20 grms |               |              |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 100 g shall be drawn from each selected container and shall be mixed homogeneously called as primary sample.
- 100 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 67. BEETROOT

Common name: चुकंदर

English name: Beetroot

Botanical name: [Beta vulgaris](#)

Varieties name: Detroit Dark Red, Crimson Globe



### 1 Beet root shall be:

- i. Fresh, firm deep bright red-purple flesh, clean, of reasonably uniform size, same variety and
  - ii. free from, visible foreign matter, abnormal external moisture, foreign smell
2. Variety .....(to be filled by assayer)
  3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3      |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.     | Essential                                                                                                                  |              |               |              |
| (i)    | <b>Defects<br/>% by count (Max):-</b>                                                                                      |              |               |              |
|        | (i) Discolouration<br>(1-4 sq cm of total surface area)                                                                    | 2%           | 4%            | 6%           |
|        | (ii) Skin defects<br>(Bruising/scratches/cut) (1-4 sq cm of total surface area)                                            | 3%           | 5%            | 7%           |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85 %         |
| B.     | Optional (Industrial/ Processor)                                                                                           |              |               |              |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 500 gms shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 500 gms shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 68. RIBBED CELERY

Common Name: अजमोद

English name: Ribbed Celery

Botanical name: *Apium graveolens*



### 1 Ribbed celery shall be :

- i. intact, firm, clean, properly drained if washed and
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell

2 Variety .....(to be filled by assayer)

3 Actual Picture of the lot.....( to be uploaded by assayer)

### 4 Criteria for Range designation :

| S. No.    | Tradable Parameters                                                                                                        | Range-1         | Range-2           | Range-3         |
|-----------|----------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|-----------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |                 |                   |                 |
| (i)       | Defects<br>% by weight (Max):-<br>(i) Discolouration<br><br>(ii) Crushed & Split leaves                                    | 3%<br><br>2%    | 5%<br><br>4%      | 7%<br><br>6%    |
| (ii)      | Range Acceptance                                                                                                           | 95 %            | 90 %              | 85 %            |
| <b>B.</b> | <b>Optional (Industrial/ Processor)</b>                                                                                    |                 |                   |                 |
| (iii)     | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than<br>70 | Between 40-<br>70 | Less than<br>40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 100 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 100 g shall be drawn from primary sample called as Laboratory sample

## 69. CLUSTER BEANS

Common name: गँवार फली/ Kothavarangai

(Tamil)/Kothamara (Malayam)/Chikkudu (Telugu)/ Valore Ke Dane (Gujarati)

English name: Cluster Beans

Botanical name: Cyamopsis tetragonoloba

Varieties name: Durgajay, Agaita Guara-112, etc



### 1. Cluster Beans shall be-

- intact, firm, clean, properly drained if washed and
- free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                  | Range-1      | Range-2       | Range-3      |
|-------|----------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.    | <b>Essential</b>                                                                                                     |              |               |              |
| (i)   | <b>Defects<br/>% by weight (Max):-</b><br>(i) Discolouration                                                         | 3%           | 5%            | 7%           |
|       | (ii) Crushed & Split                                                                                                 | 2%           | 4%            | 6%           |
| (ii)  | <b>Range Acceptance</b>                                                                                              | 95 %         | 90 %          | 85 %         |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                               |              |               |              |
| (iii) | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual (*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 200 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- 200 g shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 70. RIDGE GOURD

Common name: तुरई/ Peerkangai (Tamil)/Peechinga (Malayalam)/

Beerakaya (Telugu)/दोङ्का (Marathi)

English name: Ridge Gourd

Botanical name: *Luffa acutangula*



### 1. Ridge Gourd shall be-

- i. Wholesome, firm, fresh, clean, of reasonably uniform size, of same variety and
  - ii. Free from visible foreign matter, abnormal external moisture, foreign smell
2. Variety .....(to be filled by assayer)
3. Actual Picture of the lot.....( to be uploaded by assayer)
4. **Criteria for Range designation :**

| S.No. | Tradable Parameters                                                                                                              | Range-1         | Range-2           | Range-3         |
|-------|----------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|-----------------|
| A.    | Essential                                                                                                                        |                 |                   |                 |
| (i)   | <b>Defects<br/>% by count (Max):-</b>                                                                                            |                 |                   |                 |
|       | (i) Discoloration                                                                                                                | 3%              | 5%                | 10%             |
|       | (ii) Skin defects<br>(Bruising/scratches/cut) (1-7<br>sq cm of total<br>surface area)                                            | 3%              | 5%                | 10%             |
| (ii)  | <b>Range Acceptance</b>                                                                                                          | 95 %            | 90 %              | 85 %            |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                           |                 |                   |                 |
| (iii) | <b>Color (%)</b><br>(Characteristic true to the<br>variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety<br>colour) | More than<br>70 | Between 40-<br>70 | Less than<br>40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 71. RADISH

Common name: मूली / Mullangi (Tamil, Malayalam and Telugu)  
English name: Radish  
Botanical name: Raphanus sativus  
Variety Name: Pusa Des, Pusa Chetki, Pusa Reshmi, Nadauni etc



### 1. Radish shall be-

- i. Wholesome, firm, fresh, clean, of reasonably uniform size, of same variety and
- ii. Free from visible foreign matter, abnormal external moisture, foreign smell

### 2. Variety..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S. No. | Tradable Parameters                                                                                                        | Range-1         | Range-2         | Range-3         |
|--------|----------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|
| A.     | <b>Essential</b>                                                                                                           |                 |                 |                 |
| (i)    | <b>Size (mm)</b>                                                                                                           | <b>Upto 350</b> | <b>Upto 250</b> | <b>Upto 150</b> |
| (ii)   | <b>Defects<br/>% by count (Max):-</b>                                                                                      |                 |                 |                 |
|        | (i) Mechanical injury                                                                                                      | 2%              | 5%              | 10%             |
|        | (ii) Skin defects (Bruising/scratches/cut) (1-10 sq cm of total surface area)                                              | 2%              | 5%              | 7%              |
| (iii)  | <b>Range Acceptance</b>                                                                                                    | 95 %            | 90 %            | 85 %            |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |                 |                 |                 |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70    | Between 40-70   | Less than 40    |

### 5. Sampling:

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 72. Sweet Corn

Common name: स्वीट कॉर्न/ Makkacholam (Tamil)/  
Madhura cholam (Malyalam) / Theepi Mokkaionna  
(Telugu)  
English name: Sweet Corn  
Botanical name: Zea mays



### 1. Sweet Corn shall be-

- Wholesome, firm, fresh, clean, of reasonably uniform size, of same variety and
- Free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S. No. | Tradable Parameters                                                                                                       | Range-1      | Range-2           | Range-3         |
|--------|---------------------------------------------------------------------------------------------------------------------------|--------------|-------------------|-----------------|
| A.     | Essential                                                                                                                 |              |                   |                 |
| (i)    | Size (mm)                                                                                                                 | Upto 250     | Upto 200          | Upto 150        |
| (ii)   | Defects<br>% by count (Max):-<br>(i) Broken                                                                               | 2%           | 4%                | 6%              |
| (iii)  | Range Acceptance                                                                                                          | 95 %         | 90 %              | 85 %            |
| B.     | Optional (Industrial/Processor)                                                                                           |              |                   |                 |
| (iv)   | Color (%)<br>(Characteristic true to the<br>variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety<br>colour) | More than 70 | Between 40-<br>70 | Less than<br>40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 73. Capsicum (Green)

Common name: शिमला मिर्च / Kodaimiliga (Tamil)/Kuda mulaga

(Malyalam)/ Vanta Mirapakaya (Telugu)

English name: Capsicum/ Bell-Pepper/Green pepper

Botanical name: *Capsicum annuum*



**1. Capsicum shall be –**

- i. Wholesome, firm, fresh, clean, of same variety and
- ii. Free from, visible foreign matter, abnormal external moisture, foreign smell

**2. Variety .....**(to be filled by assayer)

**3. Actual Picture of the lot.....**( to be uploaded by assayer)

**4. Criteria for Range designation :**

| S. No. | Tradable Parameters                                                                                                        | Range-1         | Range-2          | Range-3         |
|--------|----------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|-----------------|
| A.     | <b>Essential</b>                                                                                                           |                 |                  |                 |
| (i)    | <b>Defects<br/>% by count (Max):-</b><br>(i) Discoloration                                                                 | 2%              | 4%               | 6%              |
|        | (ii) Skin defects due to rubbing<br>(Bruising/scratches)<br>(1-4 sq cm of total surface area)                              | 5%              | 7%               | 10%             |
| (ii)   | <b>Range Acceptance</b>                                                                                                    | 95 %            | 90 %             | 85 %            |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |                 |                  |                 |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than<br>70 | Between<br>40-70 | Less than<br>40 |

**5. Sampling:**

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 74. Jimikand (Suran)

Common name: सुरन /Boomi Sakkarailangu (Tamil)/ Chaena

(Malyalam)/ Kandha (Telugu)

English name: Elephant foot yam

Botanical name: *Amorphophallus paeoniifolius*



**1. Jimikand shall be –**

- i. Wholesome, firm, fresh, clean, of same variety and
- ii. Free from, visible foreign matter, abnormal external moisture, foreign smell

**2. Variety .....** (to be filled by assayer)

**3. Actual Picture of the lot .....** ( to be uploaded by assayer)

**4. Criteria for Range designation :**

| S.No.       | Tradable Parameter                         | Range-1 | Range-2 | Range-3 |
|-------------|--------------------------------------------|---------|---------|---------|
| <b>A.</b>   | <b>Essential</b>                           |         |         |         |
| <b>(i)</b>  | <b>Defects<br/>% by count (Max):-</b>      |         |         |         |
|             | (i) Irregular Shape & maturity development | 2%      | 4%      | 6%      |
|             | (ii) Skin defects (Cracks/Cuts/Scratches)  | 3%      | 5%      | 10%     |
| <b>(ii)</b> | <b>Range Acceptance</b>                    | 95 %    | 90 %    | 85 %    |

**5. Sampling:**

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 3 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 3 nos. shall be drawn from primary sample called as Laboratory sample

## 75. CUMIN

Common name : जीरा

English name : Cumin

**Botanical name:** *Cuminum cyminum* (L)



### 1. Cumin shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                          | Range-1           | Range-2 | Range-3 |
|-----------|----------------------------------------------|-------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                             |                   |         |         |
| (i)       | Moisture<br>(% by wt) Max                    | 9.0               | 10.0    | 10.0    |
| (ii)      | Foreign matter<br>(% by wt) Max              | 2.0               | 3.0     | 5.0     |
| (iii)     | Broken/Damaged /discoloured<br>(% by wt) Max | 3.0               | 4.0     | 5.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>       |                   |         |         |
| (iv)      | Non volatile ether extract % (m/m)           | Not less than 8.0 |         |         |
| (v)       | Volatile oil % (ml/100 gm) (min.)            | Not less than 1.0 |         |         |

### 5. Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 100gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 100 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions :

- ⓪ Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- ⓪ Damaged and discoloured seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- ⓪ Broken seeds are pieces of sound kernels that are less than three fourths of the size of full seeds.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 76. RED CHILLIES

Common name: लाल मिर्च

English name : Red Chillies

**Botanical name:** *Capsium annum* (L)



### 1. Red Chillies shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                      | Range-1           | Range-2 | Range-3 |
|-----------|------------------------------------------|-------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                         |                   |         |         |
| (i)       | Moisture (% by wt) Max                   | 9.0               | 10.0    | 11.0    |
| (ii)      | Foreign matter (% by wt) Max             | 1.0               | 1.5     | 2.0     |
| (iii)     | Broken/Damaged/Discoloured (% by wt) Max | 4.0               | 5.0     | 6.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>   |                   |         |         |
| (iv)      | Capsaicinoids content % (m/m)            | Not less than 0.3 |         |         |

### 5. Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged and discoloured are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iii) Broken fruits are pieces of sound fruits that are less than three fourths of the size of full kernels.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 77. TURMERIC

Common name : हल्दी

English name : Turmeric

**Botanical name:** *Curcuma longa* L



### 1. Turmeric shall be-

- i. clean, dried, wholesome, having characteristic odour and flavor, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation,

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                    | Range-1           | Range-2 | Range-3 |
|-----------|----------------------------------------|-------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                       |                   |         |         |
| (i)       | Moisture (% by wt) Max                 | 10.0              | 11.0    | 12.0    |
| (ii)      | Foreign matter<br>(% by wt) Max        | 1.0               | 1.5     | 2.0     |
| (iii)     | Defective Rhizomes<br>(% by wt) Max    | 5.0               | 7.0     | 10.0    |
| (iv)      | Damaged matter<br>(% by wt) Max        | 1.0               | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                   |         |         |
| (v)       | Curcuminoid content % (m/m)            | Not less than 2.0 |         |         |

### 5. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogeneously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Defective rhizomes” means Shriveled fingers and or bulbs, internally damaged, hollow or porous rhizomes, rhizomes scorched by boiling and other types of damaged rhizomes.
- (ii) Foreign matter includes dust, dirt, stones, lumps of earth, chaff, stems or straw, food grains including oil seeds of any other variety or any other impurity.
- (iii) Damaged fruits are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanically injured fingers or bulbs.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 78. AJWAIN

Common name : अजवाइन

English name : Bishops Seed

Botanical name : *Trachyspermum ammi*.L Sprague



### 1. Ajwain seeds shall be :-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                                    | Range-1           | Range-2 | Range-3 |
|-----------|--------------------------------------------------------|-------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                       |                   |         |         |
| (i)       | Moisture<br>(% by wt) Max                              | 10.0              | 11.0    | 11.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                        | 1.5               | 2.0     | 3.0     |
| (iii)     | Weevilled/Damaged/dicoloured<br>seeds<br>(% by wt) Max | 1.5               | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                 |                   |         |         |
| (iv)      | Volatile oil % (ml/100 gm)                             | Not less than 1.5 |         |         |

### 5. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 100gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 100 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged/Discoloured seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iii) Weevilled seeds are partially or wholly bored by insects injurious to seed but do not include germ eaten seeds and egg spotted seeds.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## CORIANDER WHOLE

Common name: साबतू धनिया

English name: Coriander whole

Botanical name: *Coriandrum sativum*



### 1 Coriander( Whole) shall be :-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
  - ii. free from obnoxious smell and fungus infestation.
- 2 Variety ..... (to be filled by assayer)
  - 3 Actual Picture of the lot..... ( to be uploaded by assayer)
  - 4 **Criteria for Range designation:**

| S.No      | Tradable Parameters                                    | Range-1           | Range-2     | Range-3 |
|-----------|--------------------------------------------------------|-------------------|-------------|---------|
| <b>A.</b> | <b>Essential</b>                                       |                   |             |         |
| (i)       | Visible Color                                          | Green             | Light Brown | Brown   |
| (ii)      | Moisture<br>(% by weight) Max                          | 8.0               | 9.0         | 11.0    |
| (iii)     | Foreign matter<br>(% by weight) Max                    | 1.0               | 2.0         | 3.0     |
| (iv)      | Damaged /Coriander Split<br>seeds<br>(% by weight) Max | 8.0               | 12.0        | 15.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                 |                   |             |         |
| (v)       | Total Ash % (m/m)                                      | Not more than 7.0 |             |         |
| (vi)      | Acid insoluble ash % (m/m)                             | Not more than 1.5 |             |         |
| (vii)     | Volatile oil % (ml/100 gm)                             | Not less than 0.1 |             |         |

### 5 Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6 Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged/Discoloured seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iii) Coriander Split is pieces of seeds that are one fourth or less than the full size of the whole seed.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 80. DRIED RAW MANGO SLICES

Common name: आमचूर / Mangavathal (Tamil)/Unakka Mango pool (Malyalam)/ Endina Mamidi mukkalu (Telegu) English name: Dried Raw Mango Slices  
Botanical name: *Mangifera indica*



### 1. Dried Mango Slices shall be–

- intact, firm, clean, of same variety and
- free from visible foreign matter, abnormal external moisture, foreign smell ,

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                  | Range-1       | Range-2        | Range-3        |
|-----------|------------------------------------------------------|---------------|----------------|----------------|
| <b>A.</b> | <b>Essential</b>                                     |               |                |                |
| (i)       | Colour                                               | Natural White | White to Brown | Brown to Black |
| (ii)      | Moisture<br>(% by Weight) Max                        | 12.0          | 12.0           | 12.0           |
| (iii)     | Damaged & discoloured<br>Slices<br>(% by Weight) Max | 2.0           | 4.0            | 5.0            |
| (iv)      | Extraneous substance<br>(% by Weight) Max            | 1.0           | 2.0            | 4.0            |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>               |               |                |                |
| (v)       | Seed Coatings<br>(% by Weight) Max                   | 1.0           | 3.0            | 6.0            |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 100 g shall be drawn from each selected container and shall be mixed homogenously will be called as primary sample
- 100 g shall be drawn from primary sample will be called as Laboratory sample

### 6. Definitions:

- Damaged & discoloured slices mean the slices that are eaten by weevils or other insects and includes slices internally damaged by fungus, moisture or heating
- Seed coatings are exterior covering of the seed.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 81. GUAR SEED

Common name : गुआर

English name : Guar Seed

**Botanical name:** *Cyamopsis tetragonoloba*



### 1. Guar seed shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
  - ii. free from obnoxious smell and fungus infestation.
2. Variety ..... (to be filled by assayer)
  3. Actual Picture of the lot..... ( to be uploaded by assayer)
  4. **Criteria for Range designation:**

| S.No  | Tradable Parameters                       | Range-1           | Range-2 | Range-3 |
|-------|-------------------------------------------|-------------------|---------|---------|
| A.    | <b>Essential</b>                          |                   |         |         |
| (i)   | Moisture<br>(% by wt) Max                 | 10.0              | 11.0    | 12.0    |
| (ii)  | Foreign matter<br>(% by weight) Max       | 3.0               | 4.0     | 5.0     |
| (iii) | Admixture/discolored<br>(% by weight) Max | 10.0              | 15.0    | 20.0    |
| B.    | <b>Optional (Industrial/Processor)</b>    |                   |         |         |
| (iv)  | Protein percentage by weight              | Not more than 9 % |         |         |
| (v)   | Gum percent by weight                     | Not less than 67  |         |         |

### 2. Sampling:-

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 3. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged seeds are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iii) Admixture means the presence of a variety of the same seed other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 82. MAHUA FLOWER

Common name : महुआ फूल

English name : Mahua Flower

**Botanical name:** *Madhuca longifolia*



### 1. Mahua flower shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                                        | Range-1 | Range-2 | Range-3 |
|-----------|------------------------------------------------------------|---------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                           |         |         |         |
| (i)       | Moisture content<br>(% by wt) Max                          | 3.0     | 5.0     | 7.0     |
| (ii)      | Foreign matter (% by wt) Max                               | 12.0    | 14.0    | 16.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                     |         |         |         |
| (iii)     | Total reducing sugar plus<br>sucrose per cent (Min)        | 60      | 50      | 40      |
| (iv)      | Crude fibre per cent by mass on<br>dry basis (Max.)        | 3       | 4       | 5       |
| (v)       | Acid insoluble ash per cent by<br>mass on dry basis (Max.) | 0.5     | 1.0     | 1.5     |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions :

- Foreign matter includes sand, stones, pebbles, lumps of earth, clay, mud and vegetable matter of plant other than Mahua flower.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

### 83. TAMARIND

Common name: इमली

English name : Tamarind

**Botanical name:** *Tamarindus indica* (L)



#### 1 Tamarind shall be-

- i. clean, wholesome, reasonably in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

2 Variety ..... (to be filled by assayer)

3 Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4 Criteria for Range designation:

| S. No.    | Tradable Parameters                           | Range-1 | Range-2 | Range-3 |
|-----------|-----------------------------------------------|---------|---------|---------|
| <b>A.</b> | <b>Essential</b>                              |         |         |         |
| (i)       | Foreign matter<br>(% by wt) Max               | 1.0     | 3.0     | 5.0     |
| (ii)      | Moisture content<br>(% by wt) Max             | 18.0    | 20.0    | 25.0    |
| (iii)     | Damaged Pods<br>(% by wt) Max                 | 3.0     | 5.0     | 8.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>        |         |         |         |
| (iv)      | Total Ash, per cent By<br>mass on dry basis   | 4.0     | 5.0     | 6.0     |
| (v)       | Crude fibre, per cent<br>by mass on dry basis | 7.0     | 8.0     | 9.0     |

#### 5 Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogeneously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

#### 6 Definitions :

- (i) Foreign matter includes sand, stones, and pebbles, lumps of earth, clay, mud and vegetable matter of plant other than tamarind.
- (ii) Damaged pods means insect damaged pods and those which are devoid of pulp.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 84. JAGGERY

Common name: गुड़  
English name: Jaggery



### 1. Jaggery shall be-

- of clean, firm consistency, non sticky, reasonable size & shape(*Lumps/Bheli/Chaku/Luddoo/Balti/Solid Cake/Cubes/bricks etc.*), colour characteristic to the variety,
- free from abnormal flavours, obnoxious smell, fungus infestation, fermentation and dirt .

2. Variety.....(to be filled by assayer)

3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                           | Range-1                                                       | Range-2                   | Range-3                         |
|-----------|-----------------------------------------------|---------------------------------------------------------------|---------------------------|---------------------------------|
| <b>A.</b> | <b>Essential</b>                              |                                                               |                           |                                 |
| (i)       | Colour                                        | Light Golden/Golden /Yellow White                             | Dark Golden/Yellow Golden | Light Brown/Brown/ Dark Reddish |
| (ii)      | Moisture content (% by wt) Max                | 8.0                                                           | 12.0                      | 16.0                            |
| (iii)     | Foreign matter (% by wt) Max                  | 1.0                                                           | 1.5                       | 2.0                             |
| (iv)      | Broken Part (% by weight) Max                 | 2.0                                                           | 5.0                       | 10.0                            |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>        |                                                               |                           |                                 |
| (i)       | Total Sugar expressed as invert sugar % (m/m) | Not less than 90 percent and sucrose not less than 60 percent |                           |                                 |
| (ii)      | Total ash %                                   | Not more than 6 percent                                       |                           |                                 |
| (iii)     | Ash insoluble in HCl %                        | Not more than 0.5                                             |                           |                                 |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 3 Nos or 250 gm whichever is applicable shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 3 Nos or 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 85. GROUNDNUT WITH PODS

Common name: छिलके के साथ मूंगफली  
English name : Groundnut  
Botanical name: *Arachis hypogaea* (L)



### 1. Groundnut shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. free from obnoxious smell and fungus infestation.

### 2. Variety.....(to be filled by assayer)

### 3. Actual Picture of the lot .....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                    | Range-1          | Range-2 | Range-3 |
|-----------|----------------------------------------|------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                       |                  |         |         |
| (i)       | Moisture (% by wt)                     | 8.0              | 10.0    | 12.0    |
| (ii)      | Foreign matter (% by wt)               | 1.0              | 2.0     | 3.0     |
| (iii)     | Damaged pods (% by wt)                 | 2.0              | 3.0     | 5.0     |
| (iv)      | Pods of other varieties (% by wt)      | 3.0              | 5.0     | 7.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                  |         |         |
| (v)       | Shelling (% by wt.)                    | Not less than 62 |         |         |

### 5. Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 250 gm shall be drawn from each selected container and shall be mixed homogeneously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Pods of other varieties means pods other than the one which is under consideration;
- (iii) Damaged pods are internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- (iv) Shelling means the weight of kernels found in 100 grams of pods.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 86. COCONUT

Common name: खोपरा / Theengai (Tamil)/Thenga (Malayalam)/

Kobbari (Telugu)

English name: Coconut

Botanical name: *Cocos nucifera*



### 1. Coconut (Copra/ dried kernal of coconut)shall be–

- i intact, firm, clean, of same variety and
- ii free from visible foreign matter, abnormal external moisture, foreign smell, mold, smoke, burn

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameter                     | Range-1 | Range-2 | Range-3 |
|-----------|----------------------------------------|---------|---------|---------|
| <b>A.</b> | <b>Essential</b>                       |         |         |         |
| (i)       | Oil content (% by Weight) Min          | 60.0    | 58.0    | 55.0    |
| (ii)      | Moisture (% by Weight) Max             | 7.0     | 7.0     | 7.0     |
| (iii)     | Wrinkled kernels (% by Weight) Max     | 10.0    | 10.0    | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |         |         |         |
| (iv)      | Size Diameter in (mm)                  | 85      | 75      | 60      |

### 5. Sampling:

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

### 6. Definitions:

- (i) Wrinkled kernels include balls that are shrunk out of normal shape or are not fully matured or developed or have a rubbery structure and uneven surface. Such kernels are often discoloured.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 87. REETHA

Common name: रीठा /Soppuka, pasakka (Malyalam/  
Kumkudu(Telgu)  
English name: Soapnut  
Botanical name: Sapindus mucorossi



### 1 Reetha nuts shall be –

- intact, firm, clean, of color characteristics to the variety and
- free from, visible foreign matter, abnormal external moisture, foreign smell ,

### 2 Variety .....(to be filled by assayer)

### 3 Actual Picture of the lot.....( to be uploaded by assayer)

### 4 Criteria for Range designation:

| Sr. No. | Tradable parameters                                                                                                        | Range-1         | Range-2          | Range-3      |
|---------|----------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|--------------|
| A.      | <b>Essential</b>                                                                                                           |                 |                  |              |
| (i)     | Foreign matter<br>(% by Weight) Max                                                                                        | 0.5             | 1.0              | 2.0          |
| (ii)    | Immature & shriveled<br>nuts(% by Weight) Max                                                                              | 2.0             | 3.0              | 5.0          |
| (iii)   | Damaged and discolored<br>nuts<br>(% by Weight) Max                                                                        | 2.0             | 3.0              | 5.0          |
| (iv)    | Moisture<br>(% by Weight) Max                                                                                              | 9.0             | 10.0             | 12.0         |
| (v)     | Pods of other variety<br>(% by Weight) Max                                                                                 | 4.0             | 6.0              | 8.0          |
| B.      | <b>Optional (Industrial/Processor)</b>                                                                                     |                 |                  |              |
| (vi)    | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than<br>70 | Between<br>40-70 | Less than 40 |

### 5 Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 200 g shall be drawn from each selected container and shall be mixed homogenously will be called as primary sample
- 200 g shall be drawn from primary sample will be called as Laboratory sample

### 6 Definitions

- Foreign matter:-Includes inorganic and organic matter. The inorganic matter shall include sand, gravel, dirt, pebbles, stones, glass and metallic pieces, lumps of earth, clay and mud. Organic matter shall include husk, chaff, straw, weed seeds and other inedible grains.
- Damaged and Discolored pods: -Kernels or pieces of kernels that are sprouted or internally damaged as a result of heat, moisture, weather or microbes; discoloured, or discolouration materially affecting the quality.
- Immature and shriveled pods: are those pods which are imperfectly developed and /or shrunken.
- Pods of other varieties: means the pods of other than the principal variety.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement



## 88. Bamboo

Common name: बांस /Muungil (Tamil)/Mula (Malyalam)/

Veduru (Telugu)/ Baa/Baash (Aasmi)

English name: **Bamboo**

Botanical name: *Bambusa vulgaris* as per the variety\*



### 1. Bamboo shall be –

- i. clean, dry, mature and of same variety;
- ii. cut after same year monsoon
- iii. free from excessive dryness, tattering, excessive moisture, damage caused by frost/pests/diseases.

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                 | Range-1                                                                              | Range-2                        |
|-----------|-----------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------|
| <b>A.</b> | <b>Essential</b>                                    |                                                                                      |                                |
| (i)       | Girth at thin end                                   | not less than 8 cm                                                                   | not less than 06 cm            |
| (ii)      | Standard Lengths<br>(In meter)                      | 1, 2, 2.15, 2.5, 3.10,<br>3.70, 4.60, 5.50, 6.40,<br>7.30 ,8.20,9.10,10.0            | 0.9 - 1.5,1.5 to 2.14          |
| (iii)     | Girth Class<br>(In cm)<br>(Measured at<br>fat end ) | 12-15<br>15-20<br>20-25<br>25-30<br>30-35<br>35-40<br>40-45<br>45-50<br>More than 50 | 09-12<br><br>More than 12      |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>              |                                                                                      |                                |
| (iv)      | Colour                                              | Dull green to Dark<br>Green, Yellow                                                  | Dull green to Green,<br>Yellow |

7. **Sampling:** 5% or 5 No (whichever is less) of the Lot shall be randomly selected and that will be called as Laboratory Sample.

8. **Definitions:** Variety- that differs from others of the same general kind or of the group to which it belongs.

## 89. COTTON

Common name: कपास

English name : Cotton

**Botanical name:** *Gossypium hirsutum* (L)



### 1. Cotton Seed shall be-

- Wholesome, clean, dried; and
- Free from insect & fungus infestation, mould growth, rodent hair and excreta.

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No  | Tradable Parameters               |                    | Range-1            | Range-2          | Range-3          |
|-------|-----------------------------------|--------------------|--------------------|------------------|------------------|
| A.    | Essential                         |                    |                    |                  |                  |
| (i)   | Staple length (in mm)             | Extra long         | Above 32.5         |                  |                  |
|       |                                   | Long staple        | Above 27.5 to 32.5 |                  |                  |
|       |                                   | Medium long staple | Above 25.0 to 27.5 |                  |                  |
|       |                                   | Medium staple      | 20.5 to 25.0       |                  |                  |
|       |                                   | Short Staple       | Below 20.5         |                  |                  |
| (ii)  | Trash (% by wt) Max               |                    | 3.0                | 5.0              | 7.0              |
| (iii) | Bundle Strength (gm/tex) (mm)     |                    | Above 28.0         | 26.0 and above   | 24.0 and above   |
| (iv)  | Micronaire                        |                    | 2.8 to 3.0         | Above 3.0 to 3.5 | Above 3.5 to 4.0 |
| (v)   | Moisture (% by Weight)            |                    | Not more than 10.0 |                  |                  |
| B.    | Optional (Industrial/Processor)   |                    |                    |                  |                  |
| (i)   | Percentage of Matured fibers (mm) |                    | Above 81.0         | Above 75.0-81.0  | Above 70.0-75.0  |

### 5. Sampling:-

- The lint cotton of one variety having reasonably uniform characteristic offered for sale at one time shall constitute a lot.
- From each of the bale so selected the representative adequate material may be drawn.

### 6. Definitions:

- Micronaire (MIC) is a measure of the air permeability of compressed cotton fibers. It is often used as an indication of fiber fineness and maturity.
- Trash: often referred to as non-lint, commonly comprises fragments of leaves, bark and grass, as well as particles of sand and dust.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 90. RAW CASHEWNUT

Common name: काजू (कच्चा) /Mundhri

(Tamil)/Kasuvandi (Malayalam)/ Jeedi Pappu (Telugu)

English name : Raw Cashewnut

Botanical name: Anacardium occidentale



Actual Picture above for illustration

### 1. Raw Cashew nuts shall be –

- Intact, firm, clean, of same variety and
- Free from, visible foreign matter, abnormal external moisture, foreign smell ,

2. Variety Name Variety.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable parameters                                                                                                           | Range-1      | Range-2          | Range-3      |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                              |              |                  |              |
| (i)       | Moisture<br>(% by Weight) Max                                                                                                 | 12.0         | 12.0             | 12.0         |
| (ii)      | Number of nuts per kg.<br>(% by Weight) Max                                                                                   | 160          | 180              | 210          |
| (iii)     | Void nuts<br>(% by Weight) Max                                                                                                | 1.0          | 2.0              | 3.0          |
| (iv)      | Immature nuts<br>(% by Weight) Max                                                                                            | 2.0          | 3.0              | 5.0          |
| (v)       | Damaged and<br>discolored nuts<br>(% by Weight) Max                                                                           | 2.0          | 3.0              | 5.0          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                        |              |                  |              |
| (vi)      | <b>Color (%)</b><br>(Characteristic true to the<br>variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety<br>colour) | More than 70 | Between<br>40-70 | Less than 40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 100 g shall be drawn from each selected container and shall be mixed homogenously will be called as primary sample
- 100 g shall be drawn from primary sample will be called as Laboratory sample

### 6. Definitions:

- Extraneous matter: includes stones, dust, dirt, parts of dried fruits and leaf, etc.
- Void nuts: in which there are no kernels.
- Immature nuts: the nuts, the kernels of which are not properly developed and/or shriveled.
- Damaged and discolored nuts: the nuts containing discolored kernels, decayed, diseased and broken nuts or mechanical injury, materially affecting the quality.

## 91. MOTH

Common names: मोठ /मटकी (मराठी)/ Prani, Chellu(Malayalam)

English name: Moth

Botanical name: *Vigna aconitifolia* Jacq Marechal



### 1. Moth shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
  - free from obnoxious smell and fungus infestation.
- Variety .....(to be filled by assayer)
  - Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                     | Range-1                       | Range-2 | Range-3 |
|-----------|-----------------------------------------|-------------------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                        |                               |         |         |
| (i)       | Moisture<br>(% by wt) Max               | 10.0                          | 12.0    | 14.0    |
| (ii)      | Foreign matter<br>(% by wt) Max         | 0.1                           | 0.60    | 1.0     |
| (iii)     | Other edible grains<br>(% by wt) Max    | 1.0                           | 3.0     | 6.0     |
| (iv)      | Damaged grains<br>(% by wt) Max         | 1.0                           | 3.0     | 5.0     |
| (v)       | Weevilled Grains<br>(% by count) Max    | 2.5                           | 5.0     | 10.0    |
| (vi)      | Admixture<br>(% by wt) Max              | 1.0                           | 2.0     | 3.0     |
| (vii)     | Immature & Shriveled<br>(% by wt) Max   | 1.0                           | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Processor/ Industrial)</b> |                               |         |         |
| (viii)    | Added Color                             | Absent                        |         |         |
| (ix)      | Uric Acid                               | Not more than 100mg/kg        |         |         |
| (x)       | Aflatoxin                               | Not more than 30 microgram/kg |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 250g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.
- Immature & Shriveled grains means kernels or pieces of grains that are not fully developed.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 92. LINSEED

Common Name: - अलसी / Avise Ginjalu  
(Telgu)/Tisi,Mosina (Bangla)/ Chananvithu  
(Malyalam)/जवस (मराठी)/

English Name: - **Linseed**

Botanical name:-*Linum usitatisimum* Linn.



### 1. Linseed shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
  - free from obnoxious smell and fungus infestation.
- Variety .....(to be filled by assayer)
  - Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                     | Range-1 | Range-2 | Range-3 |
|-----------|-----------------------------------------|---------|---------|---------|
| <b>A.</b> | <b>Essential</b>                        |         |         |         |
| (i)       | Oil Content<br>(% by wt) Min            | 39.0    | 33.0    | 24.0    |
| (ii)      | Moisture<br>(% by wt) Max               | 7.0     | 8.0     | 9.0     |
| (iii)     | Foreign matter<br>(% by wt) Max         | 2.0     | 4.0     | 6.0     |
| (iv)      | Other oil seed<br>(% by wt) Max         | 1.0     | 2.0     | 3.0     |
| (v)       | Damaged seed<br>(% by wt) Max           | 2.0     | 4.0     | 8.0     |
| <b>B.</b> | <b>Optional (Processor/ Industrial)</b> |         |         |         |
| (vi)      | Immature, shriveled and<br>dead seeds   | 1.0     | 3.0     | 6.0     |

### 5. Sampling:-

- 5 % or minimum one container shall be randomly selected for sampling.
- 250g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Seed means any edible seed (including oil seeds) other than the one which is under consideration;
- Damaged Seed means grains that are sprouted or internally damaged as a result of heat, microbe, and moisture or weather and mechanical injured seed.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 93. RAW JUTE

Common Name: - रॉ जूट / Janapa Nara (Telgu), Pat

(Bangla)/ Chanam (Malyalam)/ताग

(मराठी)/ English Name: - **Raw Jute**

Botanical name: - *Corchorus* spp.



#### 1. Raw Jute shall be-

- Dried, of minimum reed length of 150 cm, or the effective reed length should not be less than 100 cm
  - free from HUNKA, mud and other foreign materials.
- Variety.....(to be filled by assayer)
  - Actual Picture of the lot.....( to be uploaded by assayer)
  - Criteria for Range designation :**

| Tradable Parameter          | Range-1                                                                                         | Range-2                                                                                             | Range-3                                                                                             | Range-4                                                                                     | Range-5                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Strength (Marks)</b>     | Need strength to break the fiber and sharp audible sound at the time of breakage (Excellent 30) | Need less strength to break the fiber and sound will be available at the time of breakage (Good 23) | Need less strength to break the fiber and sound will be available at the time of breakage (Good 23) | Need less strength to break the fiber and feeble sound at the time of breakage (Average 13) | Easily break the fiber and no sound at the time of breakage (Poor 04) |
| <b>Defects (Marks)</b>      | Free from major defects but 10% minor defects may be allowed (25)                               | 90% free from major defects but 20% minor defects may be allowed (17)                               | 80% free from major defects and 30% minor defects may be allowed (09)                               | 80% free from major defects and 30% minor defects may be allowed (09)                       | 70% free from major defects (05)                                      |
| <b>Root Content (Marks)</b> | < 5% length wise (20)                                                                           | > 5% - 8% length wise (15)                                                                          | > 8% - 10% length wise (08)                                                                         | > 8% - 10% length wise (08)                                                                 | > 10% length wise (03)                                                |
| <b>Fineness (Marks)</b>     | Very Fine (15)                                                                                  | Very Fine (15)                                                                                      | Fine (10)                                                                                           | Coarse (05)                                                                                 | Coarse (05)                                                           |
| <b>Colour (Marks)</b>       | Light creamy to reddish yellow with luster (10)                                                 | Light creamy to reddish yellow with luster (10)                                                     | Light creamy to reddish yellow with luster (10)                                                     | Reddish / brownish with some light grey (05)                                                | Light grey to dark grey (03)                                          |
| <b>Total Score</b>          | Above 80                                                                                        | Above 60                                                                                            | Above 40                                                                                            | Above 20                                                                                    | Less than 20                                                          |

#### 5. Sampling:-

- 5% or minimum one Bunch shall be randomly selected for sampling.
- 1 kg shall be drawn from each selected Bunch and shall be mixed homogenously, called as primary sample.
- 1 kg shall be drawn from primary sample called as Laboratory sample.

#### 6. Note

- Natural dust may be allowed in Range 3 to 5 with proportionate discount.
- Root content will include hard baky croppy ends.
- Scores on the table may be taken as guidance for determining the discount.
- HUNA: Defects caused by non-removal of dried up base and hard bark from the fibers



## 94. SAFFRON

Common Name: - केसर/ केसर(मराठी),Kunkuma Puvvu (Telgu)/  
Japharan(Jafran) (Bangla)/Kunkumapoovu(Malyalam)  
English Name: - **Saffron** Botanical name:-*Crocus sativus*  
*Linnaeus*



### 1. Saffron shall be-

- clean, of dark red colour, pleasant aroma characteristic to the variety;
- free from obnoxious smell and fungus infestation.

### 2. Variety ..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| Sr. No.   | Tradable Parameters                                                                      | Range-1            | Range-2 | Range-3 |
|-----------|------------------------------------------------------------------------------------------|--------------------|---------|---------|
| <b>A.</b> | <b>Essential</b>                                                                         |                    |         |         |
| (i)       | Moisture<br>(% by wt) Max                                                                | 8.0                | 10.0    | 12.0    |
| (ii)      | Foreign matter<br>(% by wt) Max                                                          | 0.5                | 1.0     | 1.5     |
| (iii)     | Floral waste<br>(% by wt) Max                                                            | 7.0                | 8.0     | 10.0    |
| <b>B.</b> | <b>Optional (Processor/ Industrial)</b>                                                  |                    |         |         |
| (iv)      | Added color                                                                              | Absent             |         |         |
| (v)       | Total Ash on dry basis (% by wt)<br>(Max.)                                               | 8                  |         |         |
| (vi)      | Ash Insoluble in dilute HCl on dry<br>basis (% by wt) (Max.)                             | Not more than 1.5% |         |         |
| (vii)     | Solubility in cold water on dry wt.<br>basis (% by wt)                                   | Not less than 65%  |         |         |
| (viii)    | Crude fibre on dry basis (% by<br>wt)                                                    | Not more than 6%   |         |         |
| (ix)      | Total Nitrogen on dry basis                                                              | Not more than 2 %  |         |         |
| (x)       | Coloring strength expressed as<br>direct reading of absorbance of<br>440 nm of dry basis | Not less than 80 % |         |         |

### 5. Sampling:-

- 5% or minimum one container shall be randomly selected for sampling.
- 1 g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 1 g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- "Foreign matter":-It includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Floral Waste:- Yellow filaments that are unattached and separated pollens, stamens, parts of ovaries and other parts of flowers of *Crocus sativus* Linnaeus.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 95. DRUMSTICK

Common Name:- ड्रमस्टिक / शेवगा (मराठी)/ Guggala  
(Kannada)/Munaga Kaya (Telugu)/Sajna data  
(Bangla)/Muringakkaya (Malyalam), English Name:  
- Drumstick  
Botanical name:- *Guilandina moringa* L.



### 1. Drumstick shall be-

- clean, wholesome, of colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No  | Tradable Parameters                                 | Range-1  | Range-2   | Range-3 |
|-------|-----------------------------------------------------|----------|-----------|---------|
| A.    | Essential                                           |          |           |         |
| (i)   | Appearance(Size, Shape & Color)                     | Superior | Very Good | Good    |
| (ii)  | Defects<br>Scars/cut/ Bruising<br>(% by weight) Max | 5        | 10        | 15      |
| (iii) | Range Acceptance (%)                                | 95       | 90        | 85      |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 200g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 200g shall be drawn from primary sample called as Laboratory sample.

## 96. RAISINS

Common Name: - किसमिस / बेदाने(मराठी)/ Endu Draksha  
(Telgu)/ Kismis (Bangla)/ Unakkamuntiri (Malyalam)  
English Name: -Raisins  
Botanical name:- *Vitis vinifera* L



### 1. Raisins shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

### 2. Variety..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                                              | Range-1 | Range-2 | Range-3  |
|-----------|------------------------------------------------------------------|---------|---------|----------|
| <b>A.</b> | <b>Essential</b>                                                 |         |         |          |
| (i)       | Moisture<br>(% by wt) Max                                        | 13.0    | 14.0    | 15.0     |
| (ii)      | Damaged Raisins<br>(% by wt) Max                                 | 1.0     | 1.5     | 2.0      |
| (iii)     | Pieces of stem/kg.<br>(Max.)                                     | Absent  | 2       | 4        |
| <b>B.</b> | <b>Optional (Processor/ Industrial)</b>                          |         |         |          |
| (iv)      | Cap-Stem/500 gm.<br>(Max)                                        | 40      | 50      | Above 50 |
| (v)       | Immature raisin percent<br>by weight<br>(Max.)                   | 2       | 4       | 6        |
| (vi)      | Sugared Raisins (Dried<br>Grapes) percent by<br>weight<br>(Max.) | 5       | 10      | 15       |

### 5. Sampling:-

- 5% or minimum one container shall be randomly selected for sampling.
- 500g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- “Damaged Raisins” means Raisins that are internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Cap-stem - Small woody stem exceeding 3 mm in length which attaches the grape to the branch of the bunch and whether or not attached to a raisin.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 97. MAHUA SEED

Common Name: - महुआ बीज / महुआ बीज (मराठी)/Eppa Puvvu  
(Telugu)/ Mohua beej (Bangla)/Ilippa (Malyalam)/ English  
Name: - Mahua Seed  
Botanical name:-*Madhuca longifolia*



### 1 Mahua Seed shall be-

- i. clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
  - ii. free from obnoxious smell and fungus infestation.
- 2 Variety .....(to be filled by assayer)
- 3 Actual Picture of the lot.....( to be uploaded by assayer)
- 4 **Criteria for Range designation :**

| S.No      | Tradable Parameters                       | Range-1        | Range-2    | Range-3 |
|-----------|-------------------------------------------|----------------|------------|---------|
| <b>A.</b> | <b>Essential</b>                          |                |            |         |
| (i)       | Oil Content<br>(% by wt) Min              | 40.0           | 38.0       | 35.0    |
| (ii)      | Moisture<br>(% by wt) Max                 | 7.0            | 8.0        | 9.0     |
| (iii)     | Foreign matter<br>(% by wt) Max           | 2.0            | 4.0        | 6.0     |
| (iv)      | Broken & damaged<br>seed<br>(% by wt) Max | 10.0           | 15.0       | 20.0    |
| <b>B.</b> | <b>Optional (Processor/ Industrial)</b>   |                |            |         |
| (i)       | Color                                     | Reddish yellow | Dark brown |         |

### 5. Sampling:-

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) "Foreign matter":-It includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) "Broken":- shall include those seeds which are less than three-fourth but more than one-fourth of a whole seed.
- (iii) "Damaged Seed" means grains that are damaged as a result of heat, microbe, moisture or weather and mechanical injury.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 98. PINEAPPLE

Common Name: - अनानास / अननस (मराठी)/ Anasa Pandu  
(Telgu)/ Anaras (Bangla)/ Kaitachakka (Malyalam)  
English Name: -Pineapple  
Botanical name:-*Ananas comosus*



### 1. Pineapple shall be-

- clean, wholesome of colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

### 2. Variety ..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No  | Tradable Parameters                   | Range-1 | Range-2 | Range-3       |
|-------|---------------------------------------|---------|---------|---------------|
| A.    | Essential                             |         |         |               |
| (i)   | Weight in Kg (Min)                    | 1.5     | 1.0     | Less than 1.0 |
| (ii)  | Defects (%)<br>Scars/cut/<br>Bruising | 2.0     | 4.0     | Above 4.0     |
| (iii) | Range<br>Acceptance (%)               | 90      | 85      | 80            |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 5 Nos. shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- 5 Nos. shall be drawn from primary sample called as Laboratory sample

## 99. TEZPATTA (BAY LEAVES)

Common Names: - तेजपत्ता /तमालपत्र (मराठी)/ Biryani Aku (Telgu), Tezpatta (Bangla), Tamalapatram (Malyalam),  
English Name: - Bay leaves  
Botanical name: **Laurus nobilis**



### 1. Tezpat shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No  | Tradable Parameters               | Range-1     | Range-2      | Range-3    |
|-------|-----------------------------------|-------------|--------------|------------|
| A.    | <b>Essential</b>                  |             |              |            |
| (i)   | Appearance                        | Superior    | Very Good    | Good       |
| (ii)  | Defects<br>(% by weight)<br>(Max) | 1.0         | 10.0         | above 10.0 |
| (iii) | Range<br>Acceptance<br>(%)        | 95          | 90           | 85         |
| (iv)  | Colour                            | Light green | Slight brown | Brown      |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 50g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 50g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Damage means leaves that are internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 100. TENDER COCONUT

Common name: हरा नारियल

English name: Coconut

Botanical name: *Cocos nucifera*



### 1. Coconut shall be–

- i. intact, firm, clean, of same variety and
- ii. free from visible defects, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S. No. | Tradable Parameters                                                                           | Range-1    | Range-2                     | Range-3                     |
|--------|-----------------------------------------------------------------------------------------------|------------|-----------------------------|-----------------------------|
| A.     | Essential                                                                                     |            |                             |                             |
| (i)    | Weight (kgs)                                                                                  | 1.0 to 2.0 | Below 1.0 and more than 2.0 | Below 1.0 and more than 2.0 |
| (ii)   | Defects<br>% by count (Max):-<br><br>Colour(paler/brown part of any region including bruises) | 10%        | 15%                         | Above 15%                   |
| (iii)  | Range Acceptance                                                                              | 95 %       | 90 %                        | 85 %                        |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample.

## 101. FENUGREEK (WHOLE)

Common names: साबुत मेंथी

English name: Fenugreek

Botanical name: *Trigonella foenum-graecum*



### 1. Fenugreek shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No   | Tradable Parameters                         | Range-1 | Range-2 | Range-3 |
|--------|---------------------------------------------|---------|---------|---------|
| A.     | Essential                                   |         |         |         |
| (i)    | Moisture<br>(% by wt) Max                   | 9.0     | 9.5     | 10.0    |
| (ii)   | Foreign matter<br>(% by wt) Max             | 0.7     | 1.3     | 1.9     |
| (iii)  | Other edible seeds<br>(% by wt) Max         | 1.0     | 1.5     | 2.0     |
| (iv)   | Damaged/Discoloured seeds<br>(% by wt) Max  | 1.0     | 1.5     | 2.0     |
| (v)    | Weevilled seeds<br>(% by count) Max         | 2       | 4       | 6       |
| B.     | Optional (Processor/ Industrial)            |         |         |         |
| (vi)   | Total Ash % (m/m) (max.)                    | 5.0     | 7.0     |         |
| (vii)  | Acid insoluble ash % (m/m)<br>(max.)        | 1.5     | 2.0     |         |
| (viii) | Cold water soluble extract %<br>(m/m) (Min) | 30.0    |         |         |
| (ix)   | Crude Fiber % (m/m) (Max.)                  | 15.0    | 18.0    |         |

### 5. Sampling:-

- 5 % or minimum one container shall be randomly selected for sampling.
- 100g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 100g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains” means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Discoloured seeds mean seeds or pieces of seeds that have changed colour as a result of deteriorative changes.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement



## 102. FENUGREEK LEAVES (Methi)

Common names: हरी मेंथी

English name: Fenugreek Leaves

Botanical name: *Trigonella foenum-graecum*



### 1. Methi shall be –

- i. intact, firm, clean, properly drained if washed and
  - ii. free from visible foreign matter, abnormal external moisture, foreign smell
2. Variety .....(to be filled by assayer)
  3. Actual Picture of the lot.....( to be uploaded by assayer)
  4. **Criteria for Range designation :**

| S. No. | Tradable Parameters                                                                | Range-1                                                      | Range-2                                                      | Rang-3                                                       |
|--------|------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| A.     | <b>Essential</b>                                                                   |                                                              |                                                              |                                                              |
| (i)    | <b>Defects<br/>% by weight (Max):-</b><br>(i) Discolouration<br><br>(ii) Leaf Stem | 3%<br><br>Fenugreek Leaf stem must not exceed 2 cm in length | 5%<br><br>Fenugreek Leaf stem must not exceed 4 cm in length | 7%<br><br>Fenugreek Leaf stem must not exceed 6 cm in length |
| (ii)   | <b>Range Acceptance</b>                                                            | 95 %                                                         | 90 %                                                         | 85 %                                                         |

### 5. **Sampling:** -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 500 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 500 g shall be drawn from primary sample called as Laboratory sample

## 103. FENNEL SEED

Common names: सौंफ

English name: Fennel seed

Botanical name: *Foeniculum vulgare* Mill



### 1. Fennel seed shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                                                       | Range-1 | Range-2 | Rang-3 |
|-----------|---------------------------------------------------------------------------|---------|---------|--------|
| <b>A.</b> | <b>Essential</b>                                                          |         |         |        |
| (i)       | Moisture<br>(% by wt) Max                                                 | 10.0    | 11.0    | 12.0   |
| (ii)      | Foreign matter<br>(% by wt) Max                                           | 1.5     | 3.5     | 4.5    |
| (iii)     | Other edible seeds<br>(% by wt) Max                                       | 1.0     | 1.5     | 2.0    |
| (iv)      | Damaged/Broken seeds<br>(% by wt) Max                                     | 1.5     | 3.0     | 4.0    |
| (v)       | Immature/<br>Shriveled/dicoloured/<br>blackened seeds<br>(% by count) Max | 2.5     | 4.0     | 6.0    |
| <b>B.</b> | <b>Optional Parameters (Industrial/Processor)</b>                         |         |         |        |
| (vi)      | Volatile Oil % (ml/100gm)                                                 | 2.0     | 1.5     | 1.0    |
| (vii)     | Crude Fiber (% by weight)                                                 | 11.0    | 13.0    | 15.0   |

### 5. Sampling:-

- 5 % or minimum one container shall be randomly selected for sampling.
- 100g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 100g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains” means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Discoloured seeds mean seeds or pieces of seeds that have changed colour as a result of deteriorative changes.
- Immature grains means kernels or pieces of kernels that are not fully developed.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 104. ISABGOL

Common names: इसबगोल

English name: Isabgol

Botanical name: *Plantago ovate* Forsk



### 1. Isabgol shall be-

- i. clean, white to off white in colour, odour less and have thin shaped structures, 2 to 3mm long and 0.5 to 1mm wide.
- ii. free from fungus infestation.

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No      | Tradable Parameters                                  | Range-1 | Range-2 | Rang-3 |
|-----------|------------------------------------------------------|---------|---------|--------|
| <b>A.</b> | <b>Essential</b>                                     |         |         |        |
| (i)       | Moisture<br>(% by wt) Max                            | 10.0    | 11.0    | 12.0   |
| (ii)      | Extraneous matter<br>(% by wt) Max                   | 1.0     | 1.5     | 2.0    |
| <b>B.</b> | <b>Optional (Processor/ Industrial)</b>              |         |         |        |
| (iii)     | Total Ash(% by weight )<br>on dry basis              | 2.0     | 2.5     | 3.0    |
| (iv)      | Acid insoluble Ash<br>(% by weight )<br>on dry basis | 0.1     | 0.2     | 0.3    |

### 5. Sampling:-

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 100g shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 100g shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Extraneous matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 105. Pointed GOURD

Common name: परवल

English name: Pointed Gourd

Botanical name: *Trichosanthes dioica*



### 1. Pointed Gourd shall be-

- i. Wholesome, tender, firm, fresh, clean and blemish free surface, of reasonably uniform size, of same variety and
- ii. Free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                               | Range-1      | Range-2       | Range-3      |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| A.    | Essential                                                                                                                         |              |               |              |
| (i)   | <b>Defects<br/>% by count (Max):-</b>                                                                                             |              |               |              |
|       | (i) Discoloration                                                                                                                 | 5%           | 10%           | 15%          |
|       | (ii) Skin defects<br>(Bruising/scratches/cut/blemishes) (1-3 sq cm of total surface area)                                         | 5%           | 10%           | 15%          |
| (ii)  | <b>Range Acceptance</b>                                                                                                           | 95 %         | 90 %          | 85 %         |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                            |              |               |              |
| (iii) | <b>Size as Diameter</b>                                                                                                           | 25 to 45 mm  |               |              |
| (iv)  | <b>Color –Green (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 100g Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (iii) 200g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 106. Pumpkin

Common name: कद्दू

English name: Pumpkin

Botanical name: *Cucurbita pepo* L.



### 1. Pumpkin shall be –

- Wholesome, firm (thick core), reasonably uniform size & shape, fresh, clean, of same variety and
- Free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameter                                                                                                                             | Range-1      | Range-2           | Range-3      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------|--------------|
| A.    | Essential                                                                                                                                      |              |                   |              |
| (i)   | Weight / Piece<br>( Kg)                                                                                                                        | 1- 20        |                   |              |
| (ii)  | Defects<br>% by count (Max):-<br><br>Skin defects<br>(cracks/cut/rotten<br>spots) (1-9 sq cm of<br>total surface area)                         | 10%          | 15%               | 20%          |
| (iii) | Range Acceptance                                                                                                                               | 90 %         | 85 %              | 80 %         |
| B.    | Optional (Industrial/Processor)                                                                                                                |              |                   |              |
| (iv)  | Color of flesh –<br>Yellow (%)<br>(Characteristic true to the<br>variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety<br>colour) | More than 70 | Between 40-<br>70 | Less than 40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 107. Large Cardamom

Common name: बड़ी इलायची

English name : Large Cardamom

**Botanical name:** Elettaria cardamomum (L)



### 1. Large Cardamom shall be-

- (i) Dried, clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- (ii) free from wild cardamom, obnoxious smell and fungus infestation.

### 2. Variety..... (to be filled by assayer)

### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No   | Tradable Parameters                                         | Range-1           | Range-2    | Range-3    |
|--------|-------------------------------------------------------------|-------------------|------------|------------|
| A.     | Essential                                                   |                   |            |            |
| (i)    | Moisture(% by wt) Max                                       | Up to 11 %        | up to 12 % | up to 13 % |
| (ii)   | Foreign matter(% by wt) Max                                 | 1.0               | 5.0        |            |
| (iii)  | Immature, Shriveled & Insect damaged capsules (% by wt) Max | 3.0               | 5.0        |            |
| (iv)   | Empty and malformed capsules (% by wt.)                     | 5.0               | 8.0        |            |
| (v)    | Mass in gm/L (Min)                                          | 350               | 300        |            |
| B.     | Optional (Industrial/Processor)                             |                   |            |            |
| (vi)   | Tail cut                                                    | Yes               |            |            |
| (vii)  | Colour                                                      | Pink              | Brown      | Brown      |
| (viii) | Acid insoluble ash % (m/m) (max.)                           | 2.0               |            |            |
| (ix)   | Total Ash % (m/m)(max.)                                     | 8.0               |            |            |
| (x)    | Volatile oil % (ml/100 gm) (min.)                           | Not less than 0.5 |            |            |

### 5. Sampling:

- (i) 5% or minimum one container shall be randomly selected for sampling.
- (ii) 25gm (100 g in case of single container) shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 100 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Empty and Malformed capsules are capsules without seed or are scantily filled with seeds.
- (iii) Immature and shriveled and insect damaged capsules which are not fully developed or are damaged by insects,

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 108. Colocasia vegetable

Common name: अरबी

English name: **Colocasia vegetable**

Botanical name: *Colocasia esculenta*



### 1. Colocasia vegetable shall be –

- i. fresh, firm, of color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                    | Range-1        | Range-2     | Range-3  |
|-------|----------------------------------------------------------------------------------------|----------------|-------------|----------|
| A.    | <b>Essential</b>                                                                       |                |             |          |
| (i)   | <b>Defects<br/>% by count (Max):-</b>                                                  |                |             |          |
|       | (i) Extraneous matter (% by weight) Max                                                | 5%             | 10%         | 12%      |
|       | (ii) Skin defects (cracks/scratches/cut/branching)<br>(2-5sq cm of total surface area) | 5%             | 10%         | 15%      |
| (ii)  | <b>Range Acceptance</b>                                                                | 95 %           | 90 %        | 85 %     |
| B     | <b>Optional (Industrial/Processor)</b>                                                 |                |             |          |
| (iii) | <b>Diameter</b>                                                                        | More than 25mm | 15 to 25 mm | up to 15 |
| (iv)  | <b>Length</b>                                                                          | 30-45          |             |          |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

### 7. Definitions :

- (i) Extraneous matter includes husk, straws, weed seeds, sand, gravel, dirt, pebbles, lumps of earth, clay, mud and animal filth etc.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 109. Betel Leaves

Common name: पान पत्ता

English name: **Betal Leaves**

Botanical name: *Piper betel* leaves.



### 1. Betal Leaves shall be –

- i. Mature & stalk intact, color characteristics true to variety, fresh, clean
- ii. Free from visible foreign matter, foreign smell and taste, pest, larvae and mites,

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot..... (to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                                                        | Range-1                             | Range-2                                   | Range-3                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------|-------------------------------------------|
| A.    | Essential                                                                                                                  |                                     |                                           |                                           |
| (i)   | <b>Defects % by weight (Max):-</b><br>(iii) Cracks/holes/tears/bruises                                                     | 5%                                  | 10%                                       | 15%                                       |
| (ii)  | <b>Range Acceptance</b>                                                                                                    | 95 %                                | 90 %                                      | 85 %                                      |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                     |                                     |                                           |                                           |
| (iii) | <b>Size</b>                                                                                                                | Length: upto 7cm<br>Width: upto 4cm | Length: 7cm to 15cm<br>Width: 4cm to 10cm | Length: 15 cm above<br>Width: 10 cm above |
| (iv)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70                        | Between 40-70                             | Less than 40                              |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 10 No. shall be drawn from each selected container and shall be mixed homogeneously called as primary sample.
- (iii) 10 No. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 110 PAPAYA

Common name: **पपीता**

English name: **Papaya**

Botanical name: ***Carica papaya***



### 1. Papaya shall be-

- i. intact, firm, clean, ripened to proper state, of reasonably uniform size & shape, color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell.

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                                                                                                                                        | Range-1      | Range-2       | Range-3      |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Essential</b>                                                                                                                                           |              |               |              |
| (i)       | <b>Defects % by count (Max):-</b><br>(Cuts/cracks/bruises/black spot/soft patches)<br>(1-3sq. cm. of total surface area)                                   | 5%           | 7%            | 10%          |
| (ii)      | <b>Range Acceptance</b>                                                                                                                                    | 95 %         | 90 %          | 85%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                                                     |              |               |              |
| (iii)     | <b>Color(%):-</b><br><b>Partly Green /Yellow</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |
| (iv)      | <b>Weight in kg</b>                                                                                                                                        | 0.5 to 2.0   | More than 2.0 |              |

### 5. Sampling: -

- (iv) 5 % or minimum one container shall be randomly selected for sampling
- (v) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (vi) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 111. IVY GOURD

Common name: कुंदा

English name: Ivy/Scarlet Gourd

Botanical name: *Coccinia grandis*



### 1 Ivy Gourd shall be-

- i. Wholesome, tender, firm, fresh, clean and blemish free surface, of reasonably uniform size, of same variety and
- ii. Free from visible foreign matter, abnormal external moisture, foreign smell

2 Variety .....(to be filled by assayer)

3 Actual Picture of the lot..... ( to be uploaded by assayer)

### 4 Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                                     | Range-1         | Range-2       | Range-3         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|-----------------|
| A.    | Essential                                                                                                                               |                 |               |                 |
| (i)   | <b>Defects<br/>% by count (Max):-</b>                                                                                                   |                 |               |                 |
|       | (i) Discoloration                                                                                                                       | 5%              | 10%           | 15%             |
|       | (ii) Skin defects<br>(Bruising/scratches/cut/<br>blemishes) (1-3 sq cm<br>of total surface area)                                        | 5%              | 10%           | 15%             |
| (ii)  | <b>Range Acceptance</b>                                                                                                                 | 95 %            | 90 %          | 85 %            |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                                  |                 |               |                 |
| (iii) | <b>Color –Green (%)</b><br>(Characteristic true to the<br>variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety<br>colour] | More than<br>70 | Between 40-70 | Less than<br>40 |

### 5 Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 15 nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 15 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 112. Areca nut (Betel Nut)

Common name: सुपारी

English name: **Areca nut (Betel Nut)**

Botanical name: *Areca catechu*

Varieties name: Moti special, Shrivardhan special, Jamnagar special, Jeeni special. Meetupalayam special, pooja special, koka special etc.



### 1. Areca nut shall be-

- i. Whole fully husked, of light color, reasonably matured, reasonably uniform size & shape, color characteristics to the variety
- ii. Free from blemish, cracks, fissure, shrinkage
- iii. shall not be worm eaten and hollow inside.

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3             |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |              |               |                     |
| (i)       | <b>Size</b><br>(diameter of the equatorial section))                                                                       | 20-30 mm     | 15-19mm       | Not less than 12 mm |
| (ii)      | <b>Defects</b><br><b>% by count (Max):-</b><br>Skin defects (scratches scars, scrapes and blemishes)                       | 10 %         | 15 %          | 20%                 |
| (iii)     | <b>Range Acceptance</b>                                                                                                    | 95 %         | 90 %          | 85%                 |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |                     |
| (iv)      | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40        |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 100gm shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 100 gm shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 113 Tuberose

Common name: रजनीगंधा

English name: Tuberose

Botanical name: Polianthes tuberosa Linn.



1. Tuberose shall be –
  - (i) not in dry condition, Clean free from any visible foreign matter, pest, insect cut and damage caused by them or diseases or mechanical damage due to handling, loading, unloading and transportation affecting the general appearance of the produce, reasonably uniform size & shape, color
  - (ii) free from artificial colouring
2. Variety .....(to be filled by assayer)
3. Actual Picture of the Lot.....
4. **Criteria for Range designation:**

| Sr.No.   | Tradable Parameters                       | Range-1                                                   | Range-2                                                    | Range-3                                                         |
|----------|-------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------|
| <b>A</b> | <b>Essential</b>                          |                                                           |                                                            |                                                                 |
| (i)      | Color<br>(characteristics to the variety) | Bright White                                              | slight brighter<br>white to milky<br>white                 | Reasonable<br>white to milky<br>white                           |
| (ii)     | Blooming and Shape                        | Perfect<br>bloomed and<br>conical shape<br>More than 95 % | Slightly<br>bloomed and<br>conical shape<br>More than 90 % | Reasonable<br>bloomed and<br>conical shape<br>More than 85<br>% |
| (iii)    | <b>Range Acceptance</b>                   | 95 %                                                      | 90 %                                                       | 85%                                                             |
| <b>B</b> | <b>Optional (Industrial/Processor)</b>    |                                                           |                                                            |                                                                 |
| (iv)     | Stem Length in cm                         | Above 45                                                  | 30 to 45                                                   | Below 30                                                        |

### 5. Sampling:

- (i) 5 % or minimum one bunch shall be randomly selected for sampling from each bunch.
- (ii) 5 Nos spikes shall be drawn from each selected bunch and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. spikes shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 114 Marigold

Common name: गेंदाफल

English name: Marigold

Botanical name: Tagetes erecta, Tagetes tenuifolia, Tagetes patula, Tagetes minuta



1. Marigold shall be -
  - (i) Clean free from any visible foreign matter, pest, insect cut and damage caused by them or diseases or mechanical damage due to handling, loading, unloading and transportation affecting the general appearance of the produce, reasonably uniform size & shape, color
  - (ii) free from artificial colouring
2. Variety .....(to be filled by assayer)
3. Actual Picture of the Lot.....
4. **Criteria for Range designation:**

| Sr.No.   | Tradable Parameters                       | Range-1                                                        | Range-2                                                    | Range-3                                                           |
|----------|-------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------|
| <b>A</b> | <b>Essential</b>                          |                                                                |                                                            |                                                                   |
| (i)      | Color<br>(characteristics to the variety) | Superior Bright<br>Yellow to<br>Golden yellow                  | slight brighter<br>reddish with<br>yellow<br>sparkling     | Reasonable<br>brighter brown<br>and white                         |
| (ii)     | Shape and Blooming                        | Perfect<br>bloomed and<br>round disc<br>shape<br>More than 95% | Slight bloomed<br>and round disc<br>shape<br>More than 90% | Reasonable<br>bloomed and<br>round disc<br>shape<br>More than 85% |
| (iii)    | <b>Range Acceptance</b>                   | 95 %                                                           | 90 %                                                       | 85%                                                               |
| <b>B</b> | <b>Optional (Industrial/Processor)</b>    |                                                                |                                                            |                                                                   |
| (iv)     | Moisture %                                | Upto 40%                                                       |                                                            |                                                                   |

### 5. Sampling:

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 100 gm shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 100 gm shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 115. Buckwheat

Common name : Kuttu / कुट्टू

English name : Buckwheat

**Botanical name:** *Fagopyrum esculentum*



### 1. Buckwheat shall be-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety,
- free from obnoxious smell and fungus infestation,

### 2. Variety ..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No      | Tradable Parameters                                 | Range-1                       | Range-2 | Range-3 |
|-----------|-----------------------------------------------------|-------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                      |                               |         |         |
| (i)       | Moisture (% by wt) Max                              | 12.0                          | 13.0    | 14.0    |
| (ii)      | Foreign matter (% by wt) Max                        | 1.5                           | 3.0     | 4.0     |
| (iii)     | Admixture/Other edible grains (% by wt) Max         | 3.0                           | 5.0     | 7.0     |
| (iv)      | Damaged/ immature & Shrivelled grains (% by wt) Max | 3.0                           | 5.0     | 7.0     |
| (v)       | Weevilled Grains (% by count) Max                   | 3.0                           | 5.0     | 7.0     |
| <b>B.</b> | <b>Optional (Industrial/ Processor)</b>             |                               |         |         |
| (vi)      | Uric Acid                                           | Not more than 100mg/kg        |         |         |
| (vii)     | Aflatoxin                                           | Not more than 30 microgram/kg |         |         |

### 5. Sampling:

- 5% or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 500 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injury.
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 116. SNAKE GOURD

Common name: चिचिण्डा

English name: Snake Gourd

Botanical name: *Trichosanthes cucumerina*

### 1. Snake Gourd shall be-

- i. Wholesome, firm, fresh, clean, of reasonably uniform size, of same variety and
- ii. Free from visible foreign matter, abnormal external moisture, foreign smell

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                                         | Range-1      | Range-2      | Range-3                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|-----------------------------------------|
| A.    | Primary                                                                                                                                     |              |              |                                         |
| (i)   | Length                                                                                                                                      | 300-750 MM   |              | All sizes not conforming to Range 1 & 2 |
|       | Diameter                                                                                                                                    | 40-60 MM     |              |                                         |
|       | Skin defects Damage<br>(Bruising/scratches/cut) (1-7 sq cm of total surface area)                                                           | 3%           | 5%           | 10%                                     |
| (ii)  | Range Acceptance                                                                                                                            | 95 %         | 90 %         | 85 %                                    |
| B.    | Optional (Industrial/Processor)                                                                                                             |              |              |                                         |
| (iii) | Color (%) Green with white strips<br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | More than 40 |                                         |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogeneously, called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 117. STRAWBERRIES

Common name: स्ट्रॉबेरी

English name: **Strawberries**

Botanical name: *Fragaria × ananassa*



### 1. Strawberries shall be-

- i. intact, firm, clean, of reasonably uniform size, & shape, color characteristics to the variety
- ii. Reasonably free from visible foreign matter, abnormal external moisture, foreign smell, Mould infestation
- iii. With calyx intact and short green un-withered stalk attached

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... (to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                                                            | Range-1       | Range-2       | Range-3       |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|
| <b>A.</b> | <b>Primary</b>                                                                                                                                                 |               |               |               |
| (i)       | <b>Size</b><br>(Diameter) mm                                                                                                                                   | Above 25      | Above 18      |               |
| (ii)      | <b>Traces of Soil</b>                                                                                                                                          | Absent        | Sight traces  |               |
| (iii)     | <b>Defects by count (Max):-</b><br>(i) White patches (up to 10% of surface area of fruit)<br><br>(ii) Scratches & bruises (up to 10% of surface area of fruit) | NIL<br><br>3% | 10%<br><br>5% | 20%<br><br>7% |
| (iv)      | <b>Range Acceptance</b>                                                                                                                                        | 95 %          | 90 %          | 85 %          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                                                         |               |               |               |
| (v)       | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual                                                                             | More than 70  | More than 40  |               |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 118. Amla

Common name: आंवला

English name: **Indian Goosberry**

Botanical name: *Phyllanthus emblica*



### 1. Amla shall be-

- i. clean, intact, firm, of reasonably uniform size, & shape, color characteristics to the variety
- ii. free from visible foreign matter, abnormal external moisture, foreign smell, mould infestation

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... (to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                     | Range-1      | Range-2      | Range-3 |
|--------|-------------------------------------------------------------------------------------------------------------------------|--------------|--------------|---------|
| A.     | <b>Primary</b>                                                                                                          |              |              |         |
| (i)    | <b>Size</b><br>(Diameter) mm                                                                                            | Above 35     | Above 25     |         |
| (ii)   | <b>Defects by count (Max):-</b><br>(iii) Skin defects- bruising, sunspots, latex burns (up to 4 sq cm of total surface) | 3%           | 5%           | 7%      |
| (iii)  | <b>Range Acceptance</b>                                                                                                 | 95 %         | 90 %         | 85 %    |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                  |              |              |         |
| (iv)   | <b>Color (%)</b><br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual                                      | More than 70 | More than 40 |         |

### 5. Sampling: -

- (iv) 5 % or minimum one container shall be randomly selected for sampling
- (v) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample
- (vi) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 119. Horse gram

Common name: कुल्थी

English name : Horse gram

Botanical name: Macrotyloma uniflorum



### 1. Horse gram shall be:

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| Sr.No.                                     | Tradable Parameters                                                   | Range-1                            | Range-2 | Range-3 |
|--------------------------------------------|-----------------------------------------------------------------------|------------------------------------|---------|---------|
| <b>A. Primary</b>                          |                                                                       |                                    |         |         |
| (i)                                        | Moisture<br>(% by wt) Max                                             | 10.0                               | 12.0    | 14.0    |
| (ii)                                       | Foreign matter<br>(% by wt) Max                                       | 1.0                                | 2.0     | 3.0     |
| (iii)                                      | Admixture/Other edible grains<br>(% by wt) Max                        | 2.0                                | 3.0     | 4.0     |
| (iv)                                       | Damaged grains<br>(% by wt) Max                                       | 4.0                                | 5.0     | 7.0     |
| (v)                                        | Weevilled Grains<br>(% by count) Max                                  | 3.0                                | 6.0     | 10.0    |
| (vi)                                       | Immature & Shrivelled<br>(% by wt) Max                                | 1.0                                | 2.0     | 3.0     |
| <b>B. Optional (Industrial/ Processor)</b> |                                                                       |                                    |         |         |
| (i)                                        | Uric Acid                                                             | Not more than 100 mg/kg            |         |         |
| (ii)                                       | Aflatoxin                                                             | Not more than 30 microgram/kg      |         |         |
| (iii)                                      | Total of foreign matter, other<br>edible grains and damaged<br>grains | Not more than 12 percent by weight |         |         |

### 5. Sampling:-

- 5% or minimum one container shall be randomly selected for sampling.
- 100 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- Weevilled grains means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 120. DRY GINGER

Common name: सोंठ

English name: Dry Ginger

Botanical name: *Zingiber officinale*



### 1. Dry ginger shall be-

- (i) wholesome, firm, clean, same variety; and
- (ii) free from visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                           | Range-1                             | Range-2 | Range-3 |
|-----------|-----------------------------------------------|-------------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                |                                     |         |         |
| (i)       | Moisture (% by Weight) max                    | 10.0                                |         | 12.0    |
| (ii)      | Extraneous matter % (max)                     | 1.0                                 |         | 2.0     |
| (iii)     | Insect Damaged % (max)                        | 1.0                                 |         |         |
| (iv)      | Range Acceptance                              | 95 %                                | 90 %    | 85 %    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>        |                                     |         |         |
| (v)       | Volatile Oil on dry basis                     | Not less than 1.5 %                 |         |         |
| (vi)      | Total ash on dry basis                        | 8.0 ( unbleached)<br>12 ( bleached) |         |         |
| (vii)     | Calcium as Calcium oxide on dry basis % (max) | 1.1 (Unbleached)                    |         |         |
|           |                                               | 2.5 (Bleached)                      |         |         |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 50 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- (iii) 100 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 121. Lobia Pods

Common name: लोबिया फली / चावली (मराठी) English  
name: Green long Cowpea Pods  
Botanical name: *Vigna sinensis*



### 1. Lobia Pods shall be-

- intact, firm, clean, properly drained if washed and
- free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                        | Range-1         | Range-2           | Range-3         |
|-------|----------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|-----------------|
| A.    | <b>Primary</b>                                                                                                             |                 |                   |                 |
| (i)   | <b>Defects<br/>% by weight (Max):-</b><br>(i) Discolouration                                                               | 3%              | 5%                | 7%              |
|       | (ii) Crushed & Split                                                                                                       | 2%              | 4%                | 6%              |
| (ii)  | <b>Range Acceptance</b>                                                                                                    | 95 %            | 90 %              | 85 %            |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                     |                 |                   |                 |
| (iii) | <b>Color (%)</b><br>(Characteristic true to the<br>variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than<br>70 | Between 40-<br>70 | Less than<br>40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 200 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- 200 g shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement

## 122. DOLICHOS BEAN

Common name: सेम फली / घेवड़ा (मराठी)

English name: Dolichos bean, Hyacinth bean or Field bean

Botanical name: *Lablab purpureus* L. (Sweet)



### 1. Green Lobia Beans shall be-

- intact, firm, clean, properly drained if washed and
- free from, visible foreign matter, abnormal external moisture, foreign smell

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                     | Range-1         | Range-2           | Range-3         |
|-------|-------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|-----------------|
| A.    | <b>Essential</b>                                                                                                        |                 |                   |                 |
| (i)   | <b>Defects<br/>% by weight (Max):-</b><br>(iii) Discolouration                                                          | 3%              | 5%                | 7%              |
|       | (iv) Crushed & Split                                                                                                    | 2%              | 4%                | 6%              |
| (ii)  | <b>Range Acceptance</b>                                                                                                 | 95 %            | 90 %              | 85 %            |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                  |                 |                   |                 |
| (iii) | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than<br>70 | Between 40-<br>70 | Less than<br>40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 200 g shall be drawn from each selected container and shall be mixed homogenously called as primary sample.
- 200 g shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 123. CARNATIONS

Common name: गुलनार  
English name: Carnation  
Botanical name: Dianthus caryophyllus



1. Carnation shall be –
  - (i) not in dry condition, Clean free from any visible foreign matter, pest, insect cut and damage caused by them or diseases or mechanical damage due to handling, loading, unloading and transportation affecting the general appearance of the produce, reasonably uniform size & shape, color
  - (ii) free from artificial colouring
2. Variety .....(to be filled by assayer)
3. Actual Picture of the Lot.....
4. **Criteria for Range designation:**

| Sr.No.   | Tradable Parameters                                                                                                        | Range-1                                      | Range-2                                         | Range-3                                           |
|----------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------|---------------------------------------------------|
| <b>A</b> | <b>Primary</b>                                                                                                             |                                              |                                                 |                                                   |
| (i)      | Blooming and Shape                                                                                                         | Perfect bloomed and shaped<br>More than 95 % | Slightly bloomed and in shape<br>More than 90 % | Reasonable bloomed and in shape<br>More than 85 % |
| (ii)     | <b>Minimum length (in cms)</b>                                                                                             | 60 cm to 38 cm                               |                                                 |                                                   |
| <b>B</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |                                              |                                                 |                                                   |
| (iii)    | <b>Color (%)</b><br>(Characteristic true to the variety) Scale<br>(1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70                                 | Between 40-70                                   | Less than 40                                      |
| (iv)     | <b>Minimum Flower Diameter</b>                                                                                             | 6.9 cms to 5 cm                              |                                                 |                                                   |

Note: Due to verietal differences and climatic conditions in the various growing areas, flower dimmensions may vary.

### 5. Sampling:

- (i) 5 % or minimum one bunch shall be randomly selected for sampling from each bunch.
- (ii) 5 Nos spikes shall be drawn from each selected bunch and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. spikes shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 124. GLADIOLUS

Common name: ग्लैडियस

English name: Gladiolus

Botanical name: *Gladiolus*



1. *Gladiolus* shall be –
  - (i) not in dry condition, Clean free from any visible foreign matter, pest, insect cut and damage caused by them or diseases or mechanical damage due to handling, loading, unloading and transportation affecting the general appearance of the produce, reasonably uniform size & shape, color
  - (ii) free from artificial colouring
2. Variety .....(to be filled by assayer)
3. Actual Picture of the Lot.....
4. **Criteria for Range designation:**

| Sr.No. | Tradable Parameters                                                                                              | Range-1                                         | Range-2                                          | Range-3                                                    |
|--------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------|------------------------------------------------------------|
| A      | Primary                                                                                                          |                                                 |                                                  |                                                            |
| (i)    | Blooming and Shape                                                                                               | Perfect bloomed and with stem<br>More than 95 % | Slightly bloomed and with stem<br>More than 90 % | Reasonable bloomed and with/without stem<br>More than 85 % |
| (ii)   | Minimum length (in cms)                                                                                          | 115 cm to 80 cm                                 |                                                  |                                                            |
| B      | Optional (Industrial/Processor)                                                                                  |                                                 |                                                  |                                                            |
| (iii)  | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70                                    | Between 40-70                                    | Less than 40                                               |
| (iv)   | Minimum Flower Diameter                                                                                          | 8.75 cms to 6.25 cm                             |                                                  |                                                            |

Note: Due to verietal differences and climatic conditions in the various growing areas, flower dimensions may vary.

### 5. Sampling:

- (i) 5 % or minimum one bunch shall be randomly selected for sampling from each bunch.
- (ii) 5 Nos spikes shall be drawn from each selected bunch and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. spikes shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 125. APRICOT

Common name: Khubani/खुमानी /Khurmaani / Jardaalu (Kannada)

English name: Apricot

Botanical name: [P. armeniaca](#)



### 6. Apricot shall be.-

- iv. Fresh , clean , mature, firm, of reasonably uniform size & shape, color characteristics to the variety, with/without stalk
- v. Reasonably free from visible foreign matter, abnormal external moisture, foreign smell, mould infestation and bore.
- 7. Variety..... (to be filled by assayer).
- 8. Actual Picture of the lot ..... ( to be uploaded by assayer).
- 9. **Criteria for Range designation:**

| S. No. | Tradable Parameters                                                                      | Range-1      | Range-2      | Range-3  |
|--------|------------------------------------------------------------------------------------------|--------------|--------------|----------|
| A.     | <b>Primary</b>                                                                           |              |              |          |
| (vi)   | Size in Diameter )mm(                                                                    | Above 35     | Below 35     |          |
| (vii)  | <b>Defects by count:-</b><br>Bruised marks / slight bruising/soft patches upto 1 sq. cm. | Upto10%      | Upto 15%     | Upto 20% |
| (viii) | Range Acceptance                                                                         | 95 %         | 90 %         | 85 %     |
| B.     | <b>Optional (Industrial/Processor)</b>                                                   |              |              |          |
| (ix)   | Color (%)<br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual              | More than 70 | More than 40 |          |

### 10. Sampling: -

- (vii) 5 % or minimum one container shall be randomly selected for sampling.
- (viii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (ix) 5 Nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 126. ARHAR DAL SPLIT

Common name: ARHAR DAL SPLIT, अरहर/तूर दाल

English name : Red gram-split

Botanical name: *Cajanus cajan*

### 1. Arhar Dal whole shall be.-

- Clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety, with or without husk.
- Free from any obnoxious smell and fungus infestation.

2. Variety ..... (to be filled by assayer).

3. Actual Picture of the Lot..... ( to be uploaded by assayer) .



### 4. Criteria for Range designation :

| Sr.No.   | Tradable Parameters                     | Range-1                       | Range-2 | Range-3 |
|----------|-----------------------------------------|-------------------------------|---------|---------|
| <b>A</b> | <b>Primary</b>                          |                               |         |         |
| (i)      | Moisture, (% by wt), Max.               | 11.0                          | 12.0    | 14.0    |
| (ii)     | Foreign matter, (% by wt) Max.          | 1.0                           | 1.5     | 2.0     |
| (iii)    | Other edible grains, (% by wt) Max.     | 1.0                           | 2.0     | 3.0     |
| (iv)     | Damaged grains, (% by wt) Max.          | 2.5                           | 3.5     | 5.0     |
| (v)      | Weevilled grains, (% by count) Max.     | 3.0                           | 5.0     | 10.0    |
| (vi)     | Admixture, (% by wt) Max.               | 1.0                           | 2.0     | 3.0     |
| (vii)    | Broken fragments, (% by wt) Max.        | 1.0                           | 2.0     | 3.0     |
| <b>B</b> | <b>Optional (Processor/ Industrial)</b> |                               |         |         |
| (viii)   | Uric Acid                               | Not more than 100mg/kg        |         |         |
| (ix)     | Aflatoxin                               | Not more than 30 microgram/kg |         |         |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter: includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains: means any edible grains (including oil seeds) other than the one which is under consideration.
- Damaged Grains: means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses.
- Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 127. SAFED PETHA

Common name: Safed petha/ सफ़ेद पेठा / कद्दू

English name: Ash gourd

Botanical name: *Benincasa hispida*



### 1. Safed Petha shall be.–

- fresh, clean, wholesome, firm (thick core), reasonably uniform size & shape, of same variety.
- free from, visible foreign matter, abnormal external moisture, foreign smell.

2. Variety..... (to be filled by assayer).

3. Actual Picture of the lot..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameter                                                                                                         | Range-1      | Range-2           | Range-3      |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|-------------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                             |              |                   |              |
| (i)       | Weight / Piece<br>( Kg)                                                                                                    | Upto 12      | Upto 20           | Above 20     |
| (ii)      | <b>Defects<br/>% by count (Max):-</b><br>Skin defects<br>(cracks/cut/rotten spots)<br>(1-9 sq cm of total<br>surface area) | 10%          | 15%               | 20%          |
| (iii)     | Range Acceptance                                                                                                           | 90 %         | 85 %              | 80 %         |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |              |                   |              |
| (iv)      | Color (%)<br>(Characteristic true to the<br>variety)<br>Scale 1 to 100 /Visual                                             | More than 70 | Between 40-<br>70 | Less than 40 |

### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 5 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 128. BER

Common name: Ber/बेर

English name: Indian jujube

Botanical name: *Ziziphus mauritiana*



### 1. Ber shall be.–

- firm, clean, of reasonably uniform size, & shape, color characteristics to the variety, with or without stalk.
- free from, visible foreign matter, abnormal external moisture, foreign smell, mould infestation and bore.

2. Variety ..... (to be filled by assayer) .

3. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                                                    | Range-1         | Range-2       | Range-3      |
|-----------|------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                         |                 |               |              |
| (i)       | Size (mm)<br>Length & Diameter                                                                                         | Upto 50         | Upto 35       | Below 35     |
| (ii)      | <b>Defects</b><br><b>% by count (Max):-</b><br>Slight bruising/ soft patches<br>( upto 1 sq. cm of total surface area) | 3%              | 5%            | 7%           |
| (iii)     | Range Acceptance                                                                                                       | 95 %            | 90 %          | 85%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                 |                 |               |              |
| (iv)      | Color (%)<br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual                                            | More than<br>70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 10 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 10 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 129. BLACK PEPPER WHOLE

Common name: काली मिर्च (सबुत)

English name : Black Pepper (Whole )

Botanical name: *Piper nigrum*



### 1. Black Pepper-Whole shall be.–

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety;
- free from obnoxious smell and fungus infestation.
- free from artificial color , coating of mineral oil.

2. Variety..... (to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                             | Range-1     | Range-2   | Range-3    |
|-----------|-------------------------------------------------|-------------|-----------|------------|
| <b>A.</b> | <b>Primary</b>                                  |             |           |            |
| (i)       | Moisture, %(m/m), Max.                          | 10.0        | 11.0      | 12.0       |
| (ii)      | Extraneous matter, % (m/m)                      | Up to 1%    | Up to 2%  |            |
| (iii)     | Pinheads & broken berries, %(m/m),              | Up to 4.0%  | Up to 6%  | Up to 8%   |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>          |             |           |            |
| (iv)      | Bulk Density g/L,                               | 490 & above | Below 490 |            |
| (v)       | Total ash %(m/m) on dry basis Max.              | 6.0         | 6.5       | 7.0        |
| (vi)      | Light berries %(m/m),                           | Up to 5.0   | Up to 7.5 | Up to 10.0 |
| (vii)     | Piperine content, %(m/m) on dry basis, Min.     | 4.0         | 3.0       | 2.0        |
| (viii)    | Volatile oil, % (ml. 100 gm) on dry basis,      | Up to 2.5   | Up to 2.0 | Above 1.0  |
| (ix)      | Non-volatile ether extract %(m/m) on dry basis, | Up to 6.0   | Up to 5.5 | Up to 5.0  |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Broken berries means berry that has been separated into two or more pieces;
- Extraneous matter includes dust, stones, earth or any other inorganic foreign matter;
- Light berries means berry that has reached an apparently normal stage of development but the kernel does not exist;
- Extraneous matter means all materials other than black pepper berries, irrespective of whether they are of vegetable origin;
- Pinhead means berry of very small size that has not developed;

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



### 130. MOONG DAL SPLIT

Common name: Moong Dal /मूंग दाल

English name : Green gram-Split.

Botanical name: *Vigna radiata* (L).



#### 1. Moong Dal shall be.–

- With or without husk split,
- Wholesome, reasonably uniform in size, shape and color characteristic to the variety.
- free from rodent hair and excreta, moulds, living and dead insects, insects fragments, obnoxious smell and fungus infestation, discoloration.

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot ..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| Sr.No.    | Tradable Parameters                    | Range-1                        | Range-2 | Range-3 |
|-----------|----------------------------------------|--------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                         |                                |         |         |
| (i)       | Moisture, (% by wt), Max               | 11.0                           | 12.0    | 14.0    |
| (ii)      | Foreign matter, (% by wt), Max         | 1.0                            | 1.5     | 2.0     |
| (iii)     | Other edible grains, (% by wt), Max    | 1.0                            | 2.0     | 3.0     |
| (iv)      | Damaged grains, (% by wt) , Max        | 2.5                            | 3.5     | 5.0     |
| (v)       | Weevilled grains, (% by count), Max.   | 3.0                            | 5.0     | 7.0     |
| (vi)      | Admixture, (% by wt), Max.             | 1.0                            | 2.0     | 3.0     |
| (vii)     | Broken fragments, (% by wt), Max.      | 1.0                            | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                                |         |         |
| (viii)    | Uric Acid                              | Not more than 100mg/kg.        |         |         |
| (ix)      | Aflatoxin                              | Not more than 30 microgram/kg. |         |         |

#### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

#### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 131. CHANA DAL SPLIT

Common name: Chana dal/चना दाल

English name : Chana Dal Split, Bengal Gram-Split

Botanical name: *Cicer arietinum*.



#### 1. Chana Dal shall be.–

- With or without husk split.
- clean, wholesome, reasonably uniform in size, shape and color characteristic to the variety.
- free from rodent hair and excreta, moulds, living and dead insects, insects fragments, obnoxious smell and fungus infestation, discoloration.

2. Variety..... (to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

#### 4. Criteria for Range designation:

| Sr.No.    | Tradable Parameters                    | Range-1                        | Range-2 | Range-3 |
|-----------|----------------------------------------|--------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                         |                                |         |         |
| (i)       | Moisture, (% by wt), Max.              | 11.0                           | 12.0    | 14.0    |
| (ii)      | Foreign matter, (% by wt), Max.        | 1.0                            | 1.5     | 2.0     |
| (iii)     | Other edible grains,(% by wt), Max.    | 1.0                            | 2.0     | 3.0     |
| (iv)      | Damaged grains, (% by wt) Max.         | 2.5                            | 3.5     | 5.0     |
| (v)       | Weevilled grains, (% by count), Max.   | 3.0                            | 5.0     | 7.0     |
| (vi)      | Admixture (% by wt) Max                | 1.0                            | 2.0     | 3.0     |
| (vii)     | Broken fragments, (% by wt) Max.       | 1.0                            | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                                |         |         |
| (viii)    | Uric Acid                              | Not more than 100mg/kg.        |         |         |
| (ix)      | Aflatoxin                              | Not more than 30 microgram/kg. |         |         |

#### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

#### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 132. URD DAL SPLIT

Common name: Urd dal/उड़द दाल

English name : Urd Dal/Black Gram-split.

Botanical name: *Vigna mungo* L.



#### 1. Urd Dal shall be.–

- With or without husk split, uniform in size, shape and in color
- Clean, wholesome, reasonably uniform in size, shape and color characteristic to the variety.
- Free from rodent hair and excreta, moulds, living and dead insects, insects fragments, obnoxious smell and fungus infestation, discoloration.

2. Variety ..... (to be filled by assayer).

3. Actual Picture of the lot ..... ( to be uploaded by assayer).

#### 4. Criteria for Range designation:

| Sr.No.    | Tradable Parameters                    | Range-1                        | Range-2 | Range-3 |
|-----------|----------------------------------------|--------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                         |                                |         |         |
| (i)       | Moisture, (% by wt), Max.              | 11.0                           | 12.0    | 14.0    |
| (ii)      | Foreign matter, (% by wt), Max.        | 1.0                            | 1.5     | 2.0     |
| (iii)     | Other edible grains,(% by wt), Max.    | 1.0                            | 2.0     | 3.0     |
| (iv)      | Damaged grains, (% by wt), Max.        | 2.5                            | 3.5     | 5.0     |
| (v)       | Weevilled grains, (% by count), Max.   | 3.0                            | 5.0     | 7.0     |
| (vi)      | Admixture, (% by wt), Max.             | 1.0                            | 2.0     | 3.0     |
| (vii)     | Broken fragments, (% by wt), Max.      | 1.0                            | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                                |         |         |
| (viii)    | Uric Acid                              | Not more than 100mg/kg.        |         |         |
| (ix)      | Aflatoxin                              | Not more than 30 microgram/kg. |         |         |

#### 5. Sampling:

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

#### 6. Definitions:

- Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 133. KABULI CHANA WHOLE

Common name: काबुली चना

English name : Kabuli Chana Whole

Botanical name: *Cicer arietinum*



**1. Kabuli Chana shall be.–**

- i. wholesome, reasonably uniform in size, shape and color characteristic to the variety.
  - ii. free from rodent hair and excreta, moulds, living and dead insects, insects fragments, obnoxious smell and fungus infestation, discoloration.
2. Variety ..... (to be filled by assayer).
3. Actual Picture of the lot ..... ( to be uploaded by assayer).

**4. Criteria for Range designation:**

| Sr.No.    | Tradable Parameters                    | Range-1                        | Range-2 | Range-3 |
|-----------|----------------------------------------|--------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                         |                                |         |         |
| (i)       | Moisture, (% by wt), Max.              | 11.0                           | 12.0    | 14.0    |
| (ii)      | Foreign matter, (% by wt), Max.        | 1.0                            | 1.5     | 2.0     |
| (iii)     | Other edible grains,(% by wt), Max.    | 1.0                            | 2.0     | 3.0     |
| (iv)      | Damaged grains, (% by wt), Max.        | 2.5                            | 3.5     | 5.0     |
| (v)       | Weevilled grains, (% by count), Max.   | 3.0                            | 4.0     | 5.0     |
| (vi)      | Admixture, (% by wt), Max.             | 1.0                            | 2.0     | 3.0     |
| (vii)     | Broken fragments (% by wt), Max.       | 1.0                            | 2.0     | 3.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b> |                                |         |         |
| (viii)    | Uric Acid                              | Not more than 100mg/kg.        |         |         |
| (ix)      | Aflatoxin                              | Not more than 30 microgram/kg. |         |         |

**5. Sampling:**

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample.

**6. Definitions:**

- (i) Foreign matter includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Other Edible Grains means any edible grains (including oil seeds) other than the one which is under consideration;
- (iii) Damaged Grains means grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains in case of whole pulses ;
- (iv) Weevilled grains: means grains that are partially or wholly bored by insects injurious to grains but does not include germ eaten grains and egg spotted grains.
- (v) Immature & Shrivelled Grains means kernels or pieces of kernels that are not fully developed.
- (vi) Admixture means the presence of a variety of the same grain other than the variety in consideration.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 134. TINDA

Common name: Tinda/ टिंडा

English name: Indian round gourd/ Tinda.

Botanical name: *P. fistulosus*.



### 1. Tinda shall be.–

- i. fresh, clean, wholesome, tender, firm and blemish free surface, of reasonably uniform size, of same variety and preferably hairy surface.
- ii. free from visible foreign matter, soil, abnormal external moisture, foreign smell.
- iii. free from artificial color, coating of oil or application of any harmful substances.

### 2. Variety .....(to be filled by assayer)

### 3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation :

| S.No. | Tradable Parameters                                                                                                                           | Range-1      | Range-2       | Range-3             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------------|
| A.    | <b>Primary</b>                                                                                                                                |              |               |                     |
| (i)   | Size (mm)                                                                                                                                     | 30 to 50     |               | Below 30 & Above 50 |
| (ii)  | <b>Defects<br/>% by count :</b><br>Skin defects:<br>(Bruising/scratches/superficial healed cut/blemishes( upto 3 sq cm of total surface area) | Upto 5%      | Upto 10%      | Upto 15%            |
| (iii) | Range Acceptance                                                                                                                              | 95 %         | 90 %          | 85 %                |
| B.    | <b>Optional (Industrial/Processor)</b>                                                                                                        |              |               |                     |
| (iv)  | Color (%)<br>(Characteristic true to the variety)<br>Scale (1 to 100) /Visual                                                                 | More than 70 | Between 40-70 | Less than 40        |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 10 nos. shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 135. JAMUN

Common name: जामुन

English name: Java Plum.

Botanical name: *S. cumini*.



**1. BBBB shall be.–**

- i. intact, clean, of reasonably uniform size & shape, color characteristics to the variety, with or without stalk ;
- ii. free from, visible foreign matter, abnormal external moisture, foreign smell , mould infestation and bore.
- iii. free from artificial color, coating of oil or any harmful substances.

**2. Variety..... (to be filled by assayer).**

**3. Actual Picture of the lot ..... ( to be uploaded by assayer).**

**4. Criteria for Range designation:**

| S.No.     | Tradable Parameters                                                                                               | Range-1         | Range-2       | Range-3      |
|-----------|-------------------------------------------------------------------------------------------------------------------|-----------------|---------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                    |                 |               |              |
| (i)       | Size (mm)<br>Diameter                                                                                             | Above 30        | Above 25      | Upto 25      |
| (ii)      | <b>Defects</b><br><b>% by count (Max):-</b><br>Bruising/ soft patches<br>(upto 1 sq. cm of total<br>surface area) | 4%              | 7%            | 10%          |
| (iii)     | <b>Range Acceptance</b>                                                                                           | 95 %            | 90 %          | 85%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                            |                 |               |              |
| (iv)      | Color (%)<br>(Characteristic true to the variety)<br>Scale (1 to 100) /Visual                                     | More than<br>70 | Between 40-70 | Less than 40 |

**5. Sampling: -**

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 10 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 10 Nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



### 136. CARDAMOMS WHOLE

Common name: Choti elaychi/ छोटी इलायची

English name : Cardamoms Whole capsules

Botanical name: *Elettaria cardamomum*



#### 1. CARDAMOMS capsules shall be.-

- Wholesome, clean, dried, nearly ripened, reasonably uniform in size, oblong shape with a rounded part or three cornered and having ribbed appearance and color characteristic to the variety;
- well formed capsule contains sound seeds;
- free from obnoxious odor or flavor including those of rancidity, mustiness and fungus infestation.

2. Variety ..... (to be filled by assayer)

3. Actual Picture of the lot..... ( to be uploaded by assayer)

#### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                         | Range-1           | Range-2   | Range-3   |
|-----------|---------------------------------------------|-------------------|-----------|-----------|
| <b>A.</b> | <b>Primary</b>                              |                   |           |           |
| (i)       | Moisture, %(m/m), Max.                      | 11.0              | 12.0      | 13.0      |
| (ii)      | Extraneous matter ,<br>%, (m/m), Max.       | 1.0               | 2.5       | 5.0       |
| (iii)     | Empty and malformed<br>capsules % by count  | 3.0               | 5.0       | 7.0       |
| (iv)      | Immature and shriveled<br>capsules (Max.)   | 3.0               | 5.0       | 7.0       |
| (v)       | Black and split percent by<br>count(Max.)   | 5.0               | 10.0      | 15.0      |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>      |                   |           |           |
| (vi)      | Bulk Density g/L                            | Upto 435          | Above 400 | Up to 400 |
| (vii)     | Total Ash, % (m/m),                         | Not more than 9.5 |           |           |
| (viii)    | Volatile oil % (ml/100 gm)<br>on dry basis, | Not less than 3.5 |           |           |

#### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

#### 6. Definitions:

- Extraneous matter includes stones, dust, earth or any other inorganic foreign matter it also includes pieces of Calyx, stalk and foreign matter of plant origin.
- Empty and malformed capsules means which have no seeds or are scantily filled with seeds.
- Immature and Shriveled capsules means which are not fully developed.
- Blacks include capsules which are blackish to black in colour.
- Splits include capsules which are open at corners for more than half the length.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



### 137. CLOVES WHOLE

Common name: लौंग

English name : Clove

Botanical name: *Syzygium aromaticum*



**1. Clove shall be. –**

- i. Wholesome, clean, dried, fully grown but unopened floral buds, reasonably uniform in size, shape and colour characteristic to the variety;
- ii. well formed and contains sound seeds.
- iii. free from obnoxious odour or flavor including those of rancidity and mustiness and fungus infestation.

**2.** Variety..... (to be filled by assayer).

**3.** Actual Picture of the lot..... (to be uploaded by assayer).

**4. Criteria for Range designation:**

| S.No.     | Tradable Parameters                                | Range-1  | Range-2  | Range-3   |
|-----------|----------------------------------------------------|----------|----------|-----------|
| <b>A.</b> | <b>Primary</b>                                     |          |          |           |
| (i)       | Moisture, %(m/m) ( Max.)                           | 11.0     |          | 12.0      |
| (ii)      | Extraneous matter<br>% (m/m) (Max.)                | 1.0      | 2.0      | 5.0       |
| (iii)     | Headless cloves percent by mass                    | Upto 3.0 | Upto 5.0 | Upto 10.0 |
| (iv)      | Immature and Khoker cloves, percent by mass (Max.) | 2.0      | 3.0      | 5.0       |
| (v)       | Insect damage,% (Max)                              | 2.0      | 3.0      | 5.0       |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>             |          |          |           |
| (vi)      | Volatile oil % (ml/ 100 gm) on dry basis (Min.)    | 17.0     | 15.0     | 14.0      |

**5. Sampling: -**

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample.

**6. Definitions:**

- (i) Extraneous matter includes dust, dirt, stones, and lumps of earth, also vegetative part of cloves plant, other than cloves, such as tendrils, peduncles, cloves stems, mother cloves etc.
- (ii) Headless Cloves: A Clove consisting of only the receptacle and sepals and which has lost the domed shaped head.
- (iii) Khoker Cloves: A Clove which has undergone fermentation as a result of incomplete drying as evidenced by its pale brown.
- (iv) Insect damages cloves means the cloves that the partially or wholly bored by the insects.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 138. NUTMEG WHOLE

Common name:जायफल

English name : **Nutmeg**

Botanical name: Myristica  
fragrans



### 1. Nutmeg shall be. –

- i. wholesome, clean, dried seeds, fully grown, reasonably uniform in size, shape and color characteristic to the variety,
- ii. free from obnoxious odor or flavor including those of rancidity, mustiness and fungus infestation,
- iii. spherical or slightly ovoid in shape with grayish brown color and surface marked with numerous furrows,
- iv. free from any extraneous color or oil coating.

2. Variety..... (to be filled by assayer).

3. Actual Picture of the lot..... (to be uploaded by assayer).

| Criteria for Range designation: S.N O. | Tradable Parameters                             | Range-1            | Range-2   | Range-3   |
|----------------------------------------|-------------------------------------------------|--------------------|-----------|-----------|
| <b>A.</b>                              | <b>Primary</b>                                  |                    |           |           |
| (i)                                    | Moisture, %(m/m),                               | Upto 10.00         | Upto 12.0 | Upto 14.0 |
| (ii)                                   | Mace in Nutmeg %(m/m)                           | Upto 3.0           |           |           |
| (iii)                                  | Extraneous matter                               | absent             |           |           |
| (iv)                                   | Insect damaged, broken kernels, percent by mass | 3.0                | 4.0       | 5.0       |
| <b>B.</b>                              | <b>Optional (Industrial/Processor)</b>          |                    |           |           |
| (v)                                    | Volatile oil % content dry basis                | Not less than 6.5% |           |           |

### 4. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250 gm shall be drawn from primary sample called as Laboratory sample.

### 5. Definitions:

- (i) Insect damages kernels means the kernels that partially or wholly bored by the insects.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 139 COCONUT WITH HUSK

Common name: Naariyal / नारियल

English name: Coconut

Botanical name: *Cocos nucifera*



### 1. Coconut shall be.–

- i. With water or without water inside kernel, fibered or
- ii. de- fibered , dry skin of same variety
- iii. Intact ,firm, free from visible defects, foreign smell
- iv. If de-fibered from three sides, but not more than 15 % of fruit weight

2. Variety .....(to be filled by assayer)

3. Actual Picture of the lot.....(to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                             | Range-1   | Range-2    | Range-3   |
|--------|-------------------------------------------------|-----------|------------|-----------|
| A.     | Primary                                         |           |            |           |
| (i)    | Weight (gms.)                                   | Above 500 | 400 to 500 | Below 400 |
| (ii)   | Defects<br>% by count :-<br>Cracks( de-fibered) | absent    | Upto 10%   | Upto 15%  |
| (iii)  | Range Acceptance                                | 95 %      | 90 %       | 85 %      |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 5 Nos. shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 140. OATS RAW

Common name: Jaye / जई  
English name : Oats (Raw)  
Botanical name: *Avena sativa*



### 1. Oats Raw shall be :-

- clean, wholesome, reasonably uniform in size, shape and colour characteristic to the variety.
- free from abnormal flavours, obnoxious smell and fungus infestation.

### 2. Variety..... (to be filled by assayer).

### 3. Actual Picture of the lot..... (to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                           | Range-1                       | Range-2 | Range-3 |
|-----------|-----------------------------------------------|-------------------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                |                               |         |         |
| (i)       | Moisture % by mass, Max.                      | 10.0                          | 12.0    | 15.0    |
| (ii)      | Foreign matter, % by mass, Max.               | 2.0                           | 3.0     | 5.0     |
| (iii)     | Admixture/Other edible grains (% by wt), Max. | 3.0                           | 5.0     | 7.0     |
| (iv)      | Damaged/Weeviled grains % by mass             | 2.0                           | 3.0     | 5.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>        |                               |         |         |
| (v)       | Uric Acid, (Max.)                             | Not more than 100mg/kg        |         |         |
| (vi)      | Aflatoxin, (Max.)                             | Not more than 30 microgram/kg |         |         |

### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 250 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 250 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- Foreign matter: includes husk, straws, weed seeds, other inedible grains, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- Other Edible Grains: means any edible grains (including oil seeds) other than the one which is under consideration;
- Damaged & Weevilled Grains: grains that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured grains.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 141. CHERRY RED / BLACK

Common name: Cherry / चेरी

English name: Cherry

Botanical name: *Prunus avium* , *Prunus cerasus*



### 1. Cherry shall be.–

- i. intact, firm, green un-withered stalk attached to each fruit.
  - ii. clean, of reasonably uniform size, & shape, color characteristics to the variety.
  - iii. reasonably free from visible foreign matter, abnormal external moisture, foreign smell, moulds infestation.
2. Variety..... (to be filled by assayer).
3. Actual Picture of the lot ..... ( to be uploaded by assayer).
4. **Criteria for Range designation:**

| S. No.    | Tradable Parameters                                                         | Range-1      | Range-2      | Range-3 |
|-----------|-----------------------------------------------------------------------------|--------------|--------------|---------|
| <b>A.</b> | <b>Primary</b>                                                              |              |              |         |
| (i)       | Size in diameter (maximum of the equatorial section of the fruit in mm)     | Above 20     | Upto 20      |         |
| (ii)      | <b>Defects by count (Max):-</b><br>(iv) Hail Damage                         | Absent       | 20%          | 40%     |
|           | (v) Bruised / Blemishes (up to 10% of surface area of fruit)                | 5%           | 10%          | 20%     |
| (iii)     | Range Acceptance                                                            | 95 %         | 90 %         | 85 %    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                      |              |              |         |
| (iv)      | Color (%)<br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual | More than 70 | More than 40 |         |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 10 Nos. shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 142. BANANA RAW

Common name: Kaccha Kela / कच्चा केला

English name: Banana (Raw)

Botanical name: *Musa acuminata*



### 1. Green Banana shall be. –

- i. green, firm, white flesh, color characteristics to the variety.
  - ii. reasonably free from visible and other visible defects, foreign smell, mould/pest infestation.
  - iii. with calyx, short green un-withered stalk attached and should be in bunches.
2. The stalk is not torn but clearly cut, without damage to the neighboring fingers.
3. Variety..... (to be filled by assayer).
4. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 5. Criteria for Range designation:

| S.No. | Tradable Parameters                                                                  | Range-1         | Range-2  | Range-3   |
|-------|--------------------------------------------------------------------------------------|-----------------|----------|-----------|
| A.    | <b>Primary</b>                                                                       |                 |          |           |
| (i)   | <b>Defects</b> (% by count)<br>(i) Cuts/Bruising/black spots( upto 20 mm on surface) | Upto 10%        | Upto 15% | Above 15% |
| (ii)  | <b>Range Acceptance</b>                                                              | 95 %            | 90 %     | 85 %      |
| B.    | <b>Optional (Industrial/Processor)</b>                                               |                 |          |           |
| (iii) | Diameter(mm)                                                                         | 30.00 and above |          | Below 30  |
| (iv)  | Length (mm)                                                                          | 110 and above   |          | Below 110 |

### 6. Sampling: -

- (i) 5 % or minimum one bunch shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected bunches and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 143. PAPAYA RAW

Common name: Kaccha Papita/ पपीता (कच्चा)

English name: Green-papaya

Botanical name: Carica papaya



#### 1. Papaya shall be.–

- green, firm, clean, uniform size & shape, color characteristics to the variety.
- reasonably free from visible foreign matter, foreign smell, mould/pestinfestation.
- with short green un-withered stalk attached with the fruit.

2. Variety.....(to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

#### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                   | Range-1        | Range-2        | Range-3                  |
|--------|-----------------------------------------------------------------------------------------------------------------------|----------------|----------------|--------------------------|
| A.     | <b>Primary</b>                                                                                                        |                |                |                          |
| (i)    | <b>Defects % by count :-</b><br>Cuts, cracks, bruises, black spots and soft patches ( upto 3 sq. cm of surface area ) | <b>Upto 10</b> | <b>Upto 15</b> | <b>Above 15</b>          |
| (ii)   | Range Acceptance                                                                                                      | 95 %           | 90 %           | 85 %                     |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                |                |                |                          |
| (iii)  | Weight of individual fruit (gm.)                                                                                      | 250- 1000      |                | Below 250 and above 1000 |

#### 5. Sampling: -

- 5 % or minimum one container shall be randomly selected for sampling.
- 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- 5 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 144. NEEM SEEDS

Common name: Nimbori / निंबोरी

English name: Neem seed

Botanical name: *Azadirachta indica* var. *minor* Valetton



### 1. Neem seed shall be.–

- (i) naturally dried, wholesome, reasonably uniform size, & shape, color characteristics to the variety.
  - (ii) free from visible foreign matter, abnoxious smell, fungus infestation and soil clumps or particles.
  - (iii) hard and without pulp and outer seed coat.
2. Variety..... (to be filled by assayer).
3. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                    | Range-1       | Range-2    | Range-3       |
|--------|----------------------------------------|---------------|------------|---------------|
| A.     | <b>Primary</b>                         |               |            |               |
| (i)    | Foreign matter (% by wt.),             | Upto 2.0      | Upto 4.0   | Upto 6.0      |
| (ii)   | Damaged seeds (% by wt.),              | Upto 4.0      | Upto 5.0   | Upto 6.0      |
| B.     | <b>Optional (Industrial/Processor)</b> |               |            |               |
| (iii)  | No. of seeds /kg                       | Less than 300 | 300 to 325 | More than 325 |
| (iv)   | Moisture (% by wt.),                   | Upto 4.0      | Upto 8.0   | Above 8       |
| (v)    | Oil content (% by wt.)                 | Above 20      |            |               |

### 5. Sampling-

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions –

- (i) Foreign matter: includes husk, straws, weed seeds, other inedible items, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged seeds: seeds that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured seeds.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 145. JUTE SEEDS

Common name: Jute seeds / जूट बीज

English name: Jute seeds

Botanical name: white jute (Corchorus copluris) , tossa jute ( C. olitorius)



### 1. Jute seeds shall be.–

- (i) naturally dried, clean, of reasonably uniform size & shape, color characteristics to the variety.
  - (ii) free from visible foreign matter, abnoxious smell, external moisture, soil clumps or particles.
  - (iii) hard, detached completely from the spike, free from each other.
2. Variety..... (to be filled by assayer).
3. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                    | Range-1  | Range-2  | Range-3   |
|--------|----------------------------------------|----------|----------|-----------|
| A.     | <b>Primary</b>                         |          |          |           |
| (i)    | Foreign matter (% by wt.)              | Upto 3.0 | Upto 5.0 | Upto 8.0  |
| (ii)   | Damaged seeds (% by wt.)               | Upto 4.0 | Upto 5.0 | Upto 6.0  |
| B.     | <b>Optional (Industrial/Processor)</b> |          |          |           |
| (iii)  | Moisture (% by wt.)                    | Upto 4.0 | Upto 8.0 | Above 8.0 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 250gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 250gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions –

- (i) Foreign matter: includes husk, straws, weed seeds, other inedible items, metallic pieces, sand, gravel, dirt, pebbles, stones, lumps of earth, clay, mud and animal filth etc.
- (ii) Damaged seeds: seeds that are sprouted or internally damaged as a result of heat, microbe, moisture or weather and mechanical injured seeds.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 146. TAPIOCA

Common name: **Tapioca** / टैपिओका

English name: **Tapioca**

Botanical name: *Manihot esculanta*



### 1. Tapioca shall be.–

- i. firm and straight, with small portion of root firmly cut, characteristics to the variety.
  - ii. free from insects and pest
2. Variety..... (to be filled by assayer).
3. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                           | Range-1  | Range-2  | Range-3  |
|-----------|-----------------------------------------------|----------|----------|----------|
| <b>A.</b> | <b>Primary</b>                                |          |          |          |
| (i)       | Tuber Length(cm)                              | Above 30 | 20 to 30 | Below 20 |
| (ii)      | <b>Defects by count :-</b><br>(cut/blemishes) | Upto 4.0 | Upto 6.0 | Upto 8.0 |
| (iii)     | Range Acceptance                              | 95 %     | 90 %     | 85 %     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>        |          |          |          |
| (iv)      | Woody %                                       | absent   | Upto 5   | Upto 10  |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 5 Nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 147. ROSE CUT FLOWER

Common name: Gulab/गुलाब

English name: Rose

Botanical name: *Rosa*



### 6. Rose shall be –.–

- i. fresh, clean, not in dry condition, free from any visible foreign matter, pest, insect, cut and damage caused by them or diseases or mechanical damage due to handling, loading, unloading and transportation affecting the general appearance of the produce,
- ii. free from artificial color and flavor.

7. Variety .....(to be filled by assayer)

8. Actual Picture of the Lot..... ( to be uploaded by assayer).

### 9. Criteria for Range designation:

| Sr.No.   | Tradable Parameters                           | Range-1                                                   | Range-2                                                       | Range-3                                                          |
|----------|-----------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------|
| <b>A</b> | <b>Primary</b>                                |                                                           |                                                               |                                                                  |
| (v)      | Color<br>(Characteristic true to the variety) | More than 70                                              | Between 40 to 70                                              | Less than 40                                                     |
| (vi)     | Shape & Blooming                              | Perfect bloomed and straight, strong stems more than 95 % | Slightly bloomed and practically straight stem more than 90 % | Reasonable bloomed, sufficiently strong, straight, more than 80% |
| (vii)    | Range Acceptance                              | 95%                                                       | 90 %                                                          | 85%                                                              |
| <b>B</b> | <b>Optional (Industrial/Processor)</b>        |                                                           |                                                               |                                                                  |
| (viii)   | Stems length in cms                           | Above 50                                                  | 30 to 50                                                      | Below 30                                                         |

### 10. Sampling:

- (iv) 5 % or minimum one bunch shall be randomly selected for sampling from each bunch.
- (v) 5 Nos spikes shall be drawn from each selected bunch and shall be mixed homogenously, called as primary sample.
- (vi) 5 nos. spikes shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 148. CHHAPPAN KADDU

Common name: Chhappan kaddu/छप्पन कद्दू

English name: Summer squash/vegetable marrow

Botanical name: *Cucurbita pepo*



### 1. Chhappan kaddu shall be.–

- i. fresh, firm, clean, characteristics to the variety, with stalk intact
- ii. free from visible foreign matter, abnormal external moisture, foreign smell, insect infestation.

### 2. Variety ..... (to be filled by assayer)

### 3. Actual Picture of the lot ..... ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                                                  | Range-1         | Range-2       | Range-3      |
|-----------|----------------------------------------------------------------------------------------------------------------------|-----------------|---------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                       |                 |               |              |
| (i)       | <b>Weight/piece(gm)</b>                                                                                              | Upto 100        | above 100     | above 150    |
| (ii)      | <b>Defects % by count (Max):</b><br>Skin defects<br>(cracks/cut/rotten spots)<br>(1-5)sq. cm. of total surface area. | 10%             | 15%           | 20%          |
| (iii)     | <b>Range Acceptance</b>                                                                                              | 90 %            | 85 %          | 80 %         |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                               |                 |               |              |
| (iv)      | Color (%)<br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual                                          | More than<br>70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 149. PERSIMMON

Common name: Persimon/जापानी फल

English name: Persimmon

Botanical name: Diospyros kaki



### 1. Persimmon shall be.–

- i. intact, with the calyx attached, which may be with or without peduncle and dry and brown, clean, reasonable uniform in size, and characteristics to the variety,
- ii. free from visible foreign matter, abnormal external moisture, foreign smell, insect infestation.

2. Variety ..... (to be filled by assayer).

3. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                                                               | Range-1      | Range-2       | Range-3      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                                    |              |               |              |
| (i)       | <b>Defects</b> % by count:<br>Skin defects (sun defects/ bruising/ cracks/cut/rotten spots) upto 3 sq. cm. of total surface area. | Upto10%      | Upto 15%      | Upto 20%     |
| (ii)      | <b>Range Acceptance</b>                                                                                                           | 90 %         | 85 %          | 80 %         |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                            |              |               |              |
| (iii)     | Color (%)<br>(Characteristic true to the variety)<br>Scale 1 to 100 /Visual                                                       | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling.
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 150. WALNUTS INSHELL

Common name: Akharot- in shell/ अखरोट खोल सहित

English name: Walnut-in shell

Botanical name: *Juglans regia*



### 1. Walnut in shell shall be.–

- i. dry, intact, matured, of reasonably uniform size, shape and color characteristics to the variety.
  - ii. free from visible foreign matter, abnormal, external moisture, foreign smell, mould/insect infestation visible to the naked eye.
  - iii. of current year crop
2. Variety ..... (to be filled by assayer).
3. Actual Picture of the lot ..... ( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                                                                                                    | Range-1      | Range-2      | Range-3  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|----------|
| <b>A.</b> | <b>Primary</b>                                                                                                                                                         |              |              |          |
| (i)       | Moisture, (% by wt.)                                                                                                                                                   | Upto 10%     | Upto 11%     | Upto 12% |
| (ii)      | <b>Defects by count :</b><br>affecting the external appearances of the shell such as shells with adhering husk/hull, dirt and blemishes, open, broken or damaged shell | Upto 7%      | Upto 10%     | Upto 15% |
| (iii)     | Range Acceptance                                                                                                                                                       | 95 %         | 90 %         | 85 %     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                                                                 |              |              |          |
| (iv)      | Size in diameter (mm)                                                                                                                                                  | 26 and above | 24 and above | Below 24 |
| (v)       | Empty and stony shell<br>% in wt.                                                                                                                                      | Absent       | 1%           | 2%       |

### 5. Sampling: -

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 151.Chrysanthemum

Common name: गुलदाउदी

English name: Chrysanthemum

Botanical name: *Dendranthema grandiflora*



### 1. Chrysanthemum shall be. -

- (i) clean and fresh;
- (ii) Reasonably free from any foreign matter, damage caused by insects & diseases;
- (iii) uniform in size/color and stage of openness within each bunch;
- (iv) free from shattering, burning, diseases and chemical residue;
- (v) having two third of stem with foliage.

2. Variety/colour.....(to be filled by assayer).

3. Actual Picture of the lot.....(to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No. | Tradable Parameters                                   | Range-1                                                              | Range-2                                                                     | Range-3                                                                         |
|-------|-------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| A.    | Primary                                               |                                                                      |                                                                             |                                                                                 |
| (i)   | Shape and Blooming                                    | Perfect stage of openness, flower should be uniform<br>More than 95% | Slight Perfect stage of openness, flower should be uniform<br>More than 90% | Reasonably Perfect stage of openness, flower should be uniform<br>More than 85% |
| (ii)  | Flower diameter (in cm.)                              | Above 12                                                             | 10-12                                                                       | below10                                                                         |
| (iii) | length of stem (in cm.)                               | Above 60                                                             | below 60                                                                    |                                                                                 |
| (iv)  | Min. flower per stem                                  | 1                                                                    |                                                                             |                                                                                 |
| (v)   | Range Acceptance                                      | 95 %                                                                 | 90 %                                                                        | 85 %                                                                            |
| B.    | Optional (Industrial/Processor)                       |                                                                      |                                                                             |                                                                                 |
| (vi)  | Color (in % )<br>(Characteristic true to the variety) | More than 70                                                         | Between 40 to70                                                             | Less than 40                                                                    |

### 5. Special Note

- (i) Removal of all flowers buds except the terminal one to allow one flower to develop per stem.
- (ii) Recommended storage temperature range is 20-22 degree centigrade, relative humidity 90%.
- (iii) Flowers may be packed with a net covering.

### 6. Sampling

- (i) 5 % or minimum one picking unit shall be randomly selected for sampling.
- (ii) 5 nos. picking unit shall be drawn from each selected bunch and shall be mixed homogenously called as primary sample.
- (iii) 5 nos. picking unit shall be drawn from primary sample called as laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 152. Gerbera

Common name: जरबेरा

English name: Gerbera

Botanical name: *Gerbera jamesonii*



### 1. Gerbera shall be.-

- (i) clean and fresh;
- (ii) reasonably free from any visible foreign matter;
- (iii) at the stage of openness and free from shattering;
- (iv) free from shattering, discoloration, burning, and damage caused by insects, diseases, and chemical residues.

2. Variety/colour.....(to be filled by assayer).

3. Actual Picture of the lot.....(to be uploaded by assayer).

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                    | Range-1                                                              | Range-2                                                                     | Range-3                                                                         |
|--------|--------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| A.     | <b>Primary</b>                                         |                                                                      |                                                                             |                                                                                 |
| (i)    | Stem length (in cm.)                                   | More than 60                                                         | 50-60                                                                       | below 50                                                                        |
| (ii)   | Flower diameter (in cm.)                               | More than 10                                                         | 8-10                                                                        | Below 8                                                                         |
| (iii)  | Shape and Blooming                                     | Perfect stage of openness, flower should be uniform<br>More than 95% | Slight Perfect stage of openness, flower should be uniform<br>More than 90% | Reasonably Perfect stage of openness, flower should be uniform<br>More than 85% |
| (iv)   | Range Acceptance                                       | 95 %                                                                 | 90 %                                                                        | 85 %                                                                            |
| B.     | <b>Optional (Industrial/Processor)</b>                 |                                                                      |                                                                             |                                                                                 |
| (v)    | Min. Flowers per stem                                  | 1                                                                    |                                                                             |                                                                                 |
| (vi)   | Color ( in % )<br>(Characteristic true to the variety) | More than 70                                                         | Between 40 to 70                                                            | Less than 40                                                                    |

### 5. Special Note

- (i) Foliage and stems should be free from physical damage (bruises and breakage) including spots, holes, rotted tissue, wilting, burning or discoloration.
- (ii) The bunch/individual bud may be wrapped in corrugated paper to protect them from mechanical injury.
- (iii) Flowers uniform in size, not less than 8 cm diameter with straight, strong stem minimum length of 40 cm, if packed loose and mixed .
- (iv) Recommended storage temperature range is 20-22 degree centigrade, relative humidity 90%.

### 6. Sampling:

- (i) 5 % or minimum one picking unit shall be randomly selected for sampling.
- (ii) 5 nos of picking unit shall be drawn from each selected bunch and shall be mixed homogenously , called as primary sample.
- (iii) 5 nos. picking unit shall be drawn from primary sample called as laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 153. Anthurium

Common Name: एन्थुरीअम - पुष्प

English name: Anthurium

Botanical name: *Anthurium andraeanum*



1. **Anthurium shall be. -**

- (i) clean and fresh;
- (ii) reasonably free from any visible foreign matter, damage caused by insects & diseases;
- (iii) uniform in size/color and stage of openness within each bunch;
- (iv) free from physical damage (bruises and breakage) including spots, holes, rotted tissue wilting, burning, or discoloration;
- (v) called as anthurium, tailflower, flamingo flower, laceleaf etc.

2. Variety/colour.....(to be filled by assayer).

3. Actual Picture of the lot.....(to be uploaded by assayer).

4. **Criteria for Range designation:**

| S. No. | Tradable Parameters                                                                | Range-1                    | Range-2                                   | Range-3                 |
|--------|------------------------------------------------------------------------------------|----------------------------|-------------------------------------------|-------------------------|
| A.     | <b>Primary</b>                                                                     |                            |                                           |                         |
| (i)    | Shape (colorful heart shaped structure, generally referred to as flower)           | Perfect heart shape Spathe | 90 % heart shape Spathe (small to medium) | 85 % heart shape Spathe |
| (ii)   | Spathe diameter (in cm.)                                                           | 15 above                   | 10 -15                                    | Less than 10            |
| (iii)  | Length of spathe (overall length from tip of spathe to base of the stem). (in cm.) | 50 above                   | 25-50                                     | Less than 25            |
| (iv)   | Min. flower per stem                                                               | 1                          |                                           |                         |
| (v)    | Range Acceptance                                                                   | 95 %                       | 90 %                                      | 85 %                    |
| B.     | <b>Optional (Industrial/Processor)</b>                                             |                            |                                           |                         |
| (vi)   | Color ( in % )<br>(Characteristic true to the variety)                             | More than 70               | Between 40 to70                           | Less than 40            |

5. **Special Note**

- (i) Recommended storage temperature range is 20-22 degree centigrade, relative humidity 90%.
- (ii) Flowers may be packed with a net covering.
- (iii) Stem should be free from discoloration, burning and physical damage caused by insects, diseases and chemical residue.
- (iv) Spathe has a waxy appearance and shapes varies depending on the variety.
- (v) Overall length equals to tip of spathe to base plus stem.

6. **Sampling: -**

- (i) 5 % or minimum one picking unit shall be randomly selected for sampling.
- (ii) 5 nos. spathe shall be drawn from each selected bunch and shall be mixed homogenously called as primary sample.
- (iii) 5 nos. spathe shall be drawn from primary sample called as laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 154. TULIP

Common name: ट्यूलिप

English name: Tulips

Botanical name : *Tulipa hybrid*, *T.orphanidea*, *T.clusiana*



### 1. Tulips shall be.-

- (i) clean and fresh;
- (ii) reasonably free from soil, dust, damage caused by insects & diseases;
- (iii) large, showy, many colors, bi-colors and of various petal shapes;
- (iv) single, double, ruffled, fringed or lily-shaped depending on the variety;
- (v) free from shattered petals, burning, wilting, discoloration, black spots & chemical residues.

2. Variety & color.....( to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                  | Range-1                                                                | Range-2                                                                | Range-3                                                                |
|-----------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>A.</b> | <b>Primary</b>                                                                       |                                                                        |                                                                        |                                                                        |
| (i)       | Shape and blooming (may have incredible height and shapes characteristic of variety) | Perfect stage of openness, flowers should be uniform.<br>More than 95% | Perfect stage of openness, flowers should be uniform.<br>More than 90% | Perfect stage of openness, flowers should be uniform.<br>More than 85% |
| (ii)      | Min. flower per stem                                                                 | 1                                                                      |                                                                        |                                                                        |
| (iii)     | Stem length (in cm.)                                                                 | 35 above                                                               | 35-30                                                                  | below 30                                                               |
| (iv)      | Range Acceptance                                                                     | 95 %                                                                   | 90 %                                                                   | 85 %                                                                   |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                               |                                                                        |                                                                        |                                                                        |
| (v)       | Stem deviation/ Curvature(max.)                                                      | 2cm                                                                    | 2.2cm                                                                  | 2.5cm                                                                  |
| (vi)      | Color ( in % )<br>(Characteristic true to the variety)                               | More than 70                                                           | Between 40 to70                                                        | Less than 40                                                           |

### 5. Special Note

- (i) Recommended storage temperature range is 22-20 degree centigrade, relative Humidity 90%.
- (ii) Packing of bunch (5-10 stems/bunch) with uniform flowers in stage of maturity & color.
- (iii) Stem strength and deviation should be measured after hydration.

### 6. Sampling: -

- (i) 5 % or minimum one picking unit shall be randomly selected for sampling
- (ii) 5 nos shall be drawn from each selected bunch and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 155. Spray Chrysanthemum

Common name: स्प्रै गुलदाउदी

English name : Spray Chrysanthemum

Botanical name: *Chrysanthemum indicum*, *C.japonense*,  
*C.lavandulifolium*



### 1. Spray Chrysanthemum shall be.-

- (i) clean and fresh;
- (ii) free from any visible foreign matter, damage caused by insects, diseases;
- (iii) cushion, daisy, spider, button, anemone or micro type shape;
- (iv) having each stem bearing many lateral flowering stems with similar flowers;
- (v) free from shattered petals, discoloration, burning, bruises, breakage, soil, dust.

2. Variety & color.....( to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                         | Range-1                                                               | Range-2                                                              | Range-3                                                               |
|-----------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>A</b>  | <b>Primary</b>                                                                              |                                                                       |                                                                      |                                                                       |
| (i)       | Shape and blooming (may be cushion, daisy, spider, button, anemone or micro shaped flowers) | Perfect stage of openness, flowers should be uniform<br>More than 95% | Perfect stage of openness, flower should be uniform<br>More than 90% | Perfect stage of openness, flowers should be uniform<br>More than 85% |
| (ii)      | length of stem (in cms)                                                                     | Above 75                                                              | 75-70                                                                | Below 70                                                              |
| (iii)     | Min.bloom per stem                                                                          | 5-6                                                                   | 4-5                                                                  | 3-4                                                                   |
| (iv)      | Range Acceptance                                                                            | 95 %                                                                  | 90 %                                                                 | 85 %                                                                  |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                      |                                                                       |                                                                      |                                                                       |
| (v)       | Color ( in % )<br>(Characteristic true to the variety)                                      | More than 70                                                          | Between 40 to 70                                                     | Less than 40                                                          |

### 5. Special Note

- (i) Recommended storage temperature range is 20-22 degree centigrade, Relative humidity 90%.
- (ii) Bunch may be packed with a net covering.

### 6. Sampling: -

- (i) 5 % or minimum one picking unit shall be randomly selected for sampling.
- (ii) 5 nos. bloom shall be drawn from each selected bunch and shall be mixed homogenously, called as primary sample.
- (iii) 5 nos. bloom shall be drawn from primary sample called as laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 156. Lily

Common Name: लिली

English name: Lily

Botanical name: *Lilium* sps.



### 1. Lilies shall be.-

- (i) clean and fresh;
- (ii) free from any visible foreign matter, damage caused by insects and diseases;
- (iii) uniform in size/color and stage of openness within each bunch;
- (iv) having foliage, stem free from physical damage (bruises and breakage) including spots, holes, rotted tissue, wilting, burning, or discoloration.

2. Lily is leafy star, bowl, funnel and recurved, trumpet shaped flowers at the top of each stem.

3. Variety/colour.....(to be filled by assayer).

4. Actual Picture of the lot.....(to be uploaded by assayer).

### 5. Criteria for Range designation:

| S. No. | Tradable Parameters                    | Range-1                                 | Range-2                                  | Range-3                                        |
|--------|----------------------------------------|-----------------------------------------|------------------------------------------|------------------------------------------------|
| A.     | <b>Primary</b>                         |                                         |                                          |                                                |
| (i)    | Color (characteristics of the variety) | Super bright (available in many colors) | Slight bright (available in many colors) | Reasonable brighter (available in many colors) |
| (ii)   | Min. flower per stem                   | 6                                       | 3                                        | 1                                              |
| (iii)  | Range Acceptance                       | 95 %                                    | 90 %                                     | 85 %                                           |
| B.     | <b>Optional (Industrial/Processor)</b> |                                         |                                          |                                                |
| (iv)   | Flower diameter (in cm.)               | Above 10                                | 10-7                                     | Below 7                                        |
| (v)    | Length of stem (in cm.)                | Above 60                                | 60-50                                    | Below 50                                       |

### 6. Special Note.-

- (i) Recommended storage temperature range is 20-22 degree centigrade, relative humidity 90%.
- (ii) Flowers may be packed with a net covering.
- (iii) Stem should be free from discoloration, burning and physical damage caused by insects, diseases and chemical residue.

### 7. Sampling: -

- (i) 5 % or minimum one picking unit shall be randomly selected for sampling.
- (ii) 5 nos. of lilies bunch shall be drawn from each selected sample and shall be mixed homogeneously called as primary sample.
- (iii) 5 nos. of bunch shall be drawn from primary sample called as laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 157. CHAKHAO or BLACK RICE

Common name: चाखौं

English name: Chakhao, Black Rice

Botanical name: *Oryza sativa* Linn,



colour

### 1. Raw milled Black Scented Rice shall be.-

- (i) clean, wholesome, long, slender, reasonably uniform in size, shape and black to purple in with marked degree of natural fragrance characteristic to the variety;
- (ii) free from obnoxious smell, fungus infestation;
- (iii) free from rodent hair and excreta, animal filth, moulds, live insects, dead insects, insects fragments.

2. Variety/colour.....(to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

### 4. Criteria for Range designation:

| Sr. No.   | Tradable Parameters                          | Range-1          | Range-2 | Range-3 |
|-----------|----------------------------------------------|------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                               |                  |         |         |
| (i)       | Moisture (% by wt) Max                       | 12.0             | 13.0    | 14.0    |
| (ii)      | Foreign matter (% by wt) Max                 | 0.20             | 0.50    | 0.60    |
| (iii)     | Broken/Damaged/Chalky grains (% by wt) Max   | 5.0              | 10.0    | 17.0    |
| (iv)      | Red /Green grains/Paddy Grains (% by wt) Max | Nil              | 3.0     | 6.0     |
| (v)       | Other varieties of rice (% by count) Max     | 5.0              | 10.0    | 15.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>       |                  |         |         |
| (vi)      | Kernel Breadth(mm)                           | 2.0              | 2.3     | 2.5     |
| (vii)     | Kernel Length(mm)                            | 5.0 mm and above |         |         |
| (viii)    | Length- breadth ratio                        | 2.0 and above    |         |         |
| (ix)      | Alkali Spreading Value                       | 2-10             |         |         |
| (x)       | Amylose content (%)                          | 4 to 10          |         |         |

### 5. Sampling: -

- (i) 5 % or minimum one carrying packaging unit shall be randomly selected for sampling.
- (ii) 200 gms shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 200 gms. shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 158. MACE WHOLE

Common name: जावित्री

English name : Mace

Botanical name: Myristica fragrans



### 1. Mace shall be.–

- (i) clean, dried flattened seed coat or aril of the ripe fruit of nutmeg tree, myristica fragrans Houttuyn, fully grown, color characteristic to the variety.
  - (ii) free from admixture, added coloring matter;
  - (iii) buff of light brown colour, smooth, brittle and slightly warm taste and a fragrance similar to that of nutmeg;
  - (iv) having the characteristic fresh and aromatic odour and bitter acid and warm flavour;
  - (v) free from rancid taste musty odour, off-flavour, mould growth, rodent hair, excreta, filth, moulds, live insects, dead insects insect infestation, rodent contamination.
2. Variety/colour .....(to be filled by assayer).
3. Actual Picture of the lot.....( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                | Range-1            | Range-2 | Range-3 |
|-----------|----------------------------------------------------|--------------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                     |                    |         |         |
| (i)       | Moisture, % by wt. (Maximum)                       | 10.0               | 12.0    |         |
| (ii)      | Extraneous matter % by wt. (Maximum)               | 0.5                | 1.5     | 2.0     |
| (iii)     | Insect damaged, broken kernels, % by wt. (Maximum) | 3.0                | 4.0     | 5.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>             |                    |         |         |
| (iv)      | Volatile oil % content dry basis, (Minimum)        | Not less than 7.5% |         |         |

### 5. Sampling: -

- (i) 5 % or minimum one bag shall be randomly selected for sampling.
- (ii) 100
- (iii) gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iv) 100 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Insect damages seed coat means the arils that partially or wholly bored by the insects.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 159. POPPY SEED

Common name: खसखस

English name : Poppy seed

Botanical name: *Papaver somniferum* L.



### 1. Poppy seed shall be.–

- (i) wholesome, clean, dried seeds, fully grown, reasonably uniform in size, shape and color characteristic to the variety;
- (ii) free from obnoxious odor or flavor including those of rancidity, mustiness, fungus infestation, rodent hair, excreta, filth, moulds, live insects, dead insects, insects fragments and fungus;
- (iii) free from any color or oil.

2. Variety/colour .....(to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                             | Range-1    | Range-2 | Range-3 |
|-----------|-----------------------------------------------------------------|------------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                                  |            |         |         |
| (i)       | Moisture, % by wt. (Maximum)                                    | 10.00      | 11.00   |         |
| (ii)      | Extraneous matter, % by wt. (Maximum)                           | 1.0        | 2.0     | 3.0     |
| (iii)     | Insect damaged, broken kernels, % by wt. (Maximum)              | 3.0        | 5.0     | 10.0    |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                          |            |         |         |
| (iv)      | Non Volatile ether extracts, on % by wt. on dry basis (Minimum) | 50 & above | Upto 45 | Upto 40 |

### 5. Sampling: -

- (i) 5 % or minimum one carrying packaging unit shall be randomly selected for sampling.
- (ii) 50 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 50 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Insect damages seeds means the seeds that are partially or wholly bored by the insects.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 160. CHIRONJI

Common name: चिरोँजी

English name : Cuddapah almond

Botanical name: *Buchanania lanzan* Spreng



### 1. Chironji shall be.–

- (i) mature, wholesome, clean, dried seeds;
  - (ii) reasonably uniform in size, shape and color characteristic to the variety;
  - (iii) free from obnoxious odor or flavor including those of rancidity, mustiness and fungus infestation, any other edible and non-edible seeds;
  - (iv) free from rodent hair, excreta, animal filth, moulds, live insects, dead insects.
2. Variety/colour .....(to be filled by assayer).
  3. Actual Picture of the lot.....( to be uploaded by assayer).
  4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                        | Range-1 | Range-2 | Range-3 |
|-----------|------------------------------------------------------------|---------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                             |         |         |         |
| (i)       | Moisture, % by wt. (Maximum)                               | 5.5     | 6.0     | 6.5     |
| (ii)      | Extraneous matter % by wt. (Maximum)                       | 0.1     | 0.25    | 0.5     |
| (iii)     | Damaged, Discolored and broken kernels, % by wt. (Maximum) | 2.0     | 3.5     | 5.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                     |         |         |         |
| (iv)      | Protein content, % by wt. (Minimum)                        | 24      | 22      | 19      |
| (v)       | Total soluble solids, % by wt. (Minimum)                   | 10.0    | 8.5     | 7.5     |
| (vi)      | Oil content, % by wt. (Minimum)                            | 54.0    | 52.0    | 48.0    |
| (vii)     | Acidity of extracted fat expressed as Oleic acid, Max      | 10.0    | 12.0    | 14.0    |

### 5. Sampling: -

- (i) 5 % or minimum one carrying packaging unit shall be randomly selected for sampling.
- (ii) 50 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 50 gm shall be drawn from primary sample called as Laboratory sample.

### 6. Definitions:

- (i) Insect damages seedss means the seeds that are partially or wholly bored by the insects.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 161. SAFED MUSLI

Common name: सफ़ेद मूसली

English name : Chlorophytum, Indian Spider Plant, White Musli

Botanical name: *Chlorophytum borivillianum* L.



### 1. Safed Musli shall be.–

- (i) Mature, wholesome, clean, dried, fully grown, whitish in colour, reasonably uniform in size, shape and color characteristic to the variety;
- (ii) free from obnoxious odor or flavor including those of rancidity, mustiness and fungus infestation;
- (iii) free from rodent hair, excreta, animal filth, moulds, live insects, dead insects.

2. Variety/colour .....(to be filled by assayer).

3. Actual Picture of the lot.....( to be uploaded by assayer).

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                        | Range-1 | Range-2 | Range-3 |
|-----------|------------------------------------------------------------|---------|---------|---------|
| <b>A.</b> | <b>Primary</b>                                             |         |         |         |
| (i)       | Moisture, % by wt. (Maximum)                               | 9.0     | 10.0    | 11.0    |
| (ii)      | Extraneous matter, % by wt. (Maximum)                      | 0.2     | 0.5     | 0.75    |
| (iii)     | Damaged, Discolored and broken kernels, % by wt. (Maximum) | 2.0     | 3.5     | 5.0     |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                     |         |         |         |
| (iv)      | Crude Fiber, % by wt. (Maximum) (on dry basis)             | 3.0     | 4.0     | 5.0     |
| (v)       | Total Ash, % by wt. (Maximum) (on dry basis)               | 3.5     | 4.5     | 5.5     |
| (vi)      | Acid Insoluble ash, % by wt. (Maximum) (on dry basis)      | 0.5     | 0.7     | 0.9     |
| (vii)     | Saponin content, % by wt. (Maximum) (on dry basis)         | 15.0    | 10.0    | 5.0     |

### 5. Sampling: -

- (i) 5 % or minimum one packaging unit shall be randomly selected for sampling.
- (ii) 100 gm shall be drawn from each selected container and shall be mixed homogenously, called as primary sample.
- (iii) 100 gm shall be drawn from primary sample called as Laboratory sample.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 162. SPONGE GOURD

Common name: गिल्की तुराई

English name: Sponge Gourd

Botanical name: *Luffa aegyptiaca* Mill., syn. *Luffa cylindrica* Roem



**1. Sponge Gourd shall be.-**

- (i) wholesome, firm, fresh, clean, of reasonably uniform size, of same variety;
- (ii) free from any infestation and undesirable defect.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

**4. Criteria for Range designation :**

| S.No. | Tradable Parameters                                                                                                                         | Range-1       | Range-2   | Range-3                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|-----------------------------------------|
| A.    | Primary                                                                                                                                     |               |           |                                         |
| (i)   | Length                                                                                                                                      | 100 to 250 MM |           | All sizes not conforming to Range 1 & 2 |
|       | Diameter                                                                                                                                    | 50-60 MM      |           |                                         |
|       | Skin defects<br>Damage<br>(Bruising/scratches/cut)<br>(1-5 sq cm of total surface area)                                                     | 3%            | 5%        | 10%                                     |
| (ii)  | Range Acceptance                                                                                                                            | 95 %          | 90 %      | 85 %                                    |
| B.    | Optional (Industrial/Processor)                                                                                                             |               |           |                                         |
| (iii) | Color (%) Green with white strips<br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | more than 70% | below 70% |                                         |

**5. Sampling: -**

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 5 nos shall be drawn from each selected packing unit and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

### 163. PASSION FRUIT

Common name: पैशन फ्रूट / कृष्णा फल

English name: Passion fruit

Botanical name: *Annona reticulata* for purple, *Passiflora edulis* f. *flavicarpa* Deg. for yellow



1. **Passion Fruit shall be.-**

- (i) whole, with stem/stalk intact, firm, clean, fresh in appearance & color characteristics to the variety ;
- (ii) free from, visible foreign matter, abnormal external moisture, foreign smell, pests and damage, excluding condensation following removal from cold storage, free of cracking.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

4. **Criteria for Range designation:**

| S.No.     | Tradable Parameters                                                                                                                        | Range-1      | Range-2       | Range-3      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                                             |              |               |              |
| (i)       | <b>Defects<br/>% by count (Max):-</b><br>Blemishes, scars, sunspots, very slight superficial defects (0.5-2 sq. cm. of total surface area) | 3%           | 5%            | 10%          |
| (ii)      | Range Acceptance                                                                                                                           | 95 %         | 90 %          | 85 %         |
|           | <b>Optional (Industrial/Processor)</b>                                                                                                     |              |               |              |
| (iii)     | Size (diameter in mm)                                                                                                                      | 78 & above   | 56 to 77      | Less than 56 |
| (iv)      | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour)                        | More than 70 | Between 40-70 | Less than 40 |

5. **Sampling: -**

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 5 nos shall be drawn from each selected packing unit and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 164. Broccoli/Calabrese

Common name: ब्रोकोली

English name: Broccoli

Botanical name: *Brassica oleracea* L.

Varieties name: Calabrese broccoli, destiny broccoli, and belstar broccoli



### 1. Broccoli flower clusters shall be.-

- (i) fresh in appearance and dark bright green, colour with closed flower buds; clean, free from visible foreign matter, traces of fertilizer or chemicals;
- (ii) intact, sound, free from rotting or deterioration;
- (iii) free of pests affecting the general appearance of the produce, any foreign smell and/or taste;
- (iv) free of damage caused by pests and diseases;
- (v) free of abnormal external moisture, larvae, mites, caterpillars etc, blackening of tips of flower.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot....., colour ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                 | Range-1      | Range-2       | Range-3       |
|-----------|---------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|
| <b>A.</b> | <b>Primary</b>                                                                                                      |              |               |               |
| (i)       | <b>Defects<br/>% by count (Max):-</b>                                                                               |              |               |               |
|           | (i) Bruising<br>(3 sq cm of total surface area)                                                                     | 5 %          | 5 %           | 7 %           |
|           | (ii) Colour<br>(1-3 sq cm of total surface area)                                                                    | 3 %          | 5 %           | 7 %           |
|           | (iii) Wooliness<br>(1-5 sq cm of total surface area)                                                                | 3%           | 5%            | 7%            |
| (ii)      | Range Acceptance                                                                                                    | 95 %         | 90 %          | 85 %          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                              |              |               |               |
| (iii)     | Color (%)<br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 90 | Between 60-90 | Between 60-90 |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 5 nos shall be drawn from each selected packing unit and shall be mixed homogeneously, called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 165. SUGAR SNAP PEAS

Common name: मटर

English name : Sugar Snap peas, Snow pea

Botanical name: *Pisum sativum* (L)



### 1. The Peas pods shall be.–

- (i) intact, clean and sound, fresh in appearance;
- (ii) free from black/brown spots on pods, insect damage, rotten tips, free of foreign matter, external moisture; pest and damage caused by pests, any foreign smell;
- (iii) free from seeds, or if present shall be small and undeveloped.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| Sl.No.    | Tradable Parameters                                                                                              | Range-1      | Range-2       | Range-3      |
|-----------|------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------------|
| <b>A.</b> | <b>Primary</b>                                                                                                   |              |               |              |
| (i)       | <b>Defects (Max)</b>                                                                                             |              |               |              |
|           | (i) Dry skin<br>(covering surface area)                                                                          | 5%           | 10%           | 15 %         |
|           | (ii) Black/ brown spots                                                                                          | Upto 3%      | Upto 7%       | Upto 10%     |
| (ii)      | Size (Length in mm)                                                                                              | Upto 65mm    | Upto 55mm     | Upto 45 mm   |
| (iii)     | Range Acceptance                                                                                                 | 95%          | 90%           | 85%          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                           |              |               |              |
| (iv)      | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40 |

### 5. Sampling.

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling.
- (ii) 200 g shall be drawn from each selected packing unit and shall be mixed homogenously called as primary sample
- (iii) 200 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 166. ALOEVERA

Common name: घृत कुमारी

English name: Aloe vera

Botanical name: *Aloe barbadensis miller*.



### 1. Aloe vera shall be.-

- (i) fresh in appearance and dark green, bright green in colour;
- (ii) intact, firm, clean, sound, produce affected by rotting or deterioration such as to make proper selected produce;
- (iii) free of pests affecting the general appearance of the produce, damage caused by pests and diseases.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot....., colour ( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                 | Range-1      | Range-2       | Range-3       |
|--------|---------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|
| A.     | <b>Primary</b>                                                                                                      |              |               |               |
| (i)    | <b>Defects<br/>% by count (Max):-</b><br>Bruising (1 sq cm of total surface area)                                   | 3 %          | 5 %           | 7 %           |
| (ii)   | Length of stem (in cm)                                                                                              | More than 50 | 40-50         | below 40      |
| (iii)  | Thickness (in cm)                                                                                                   | More than 3  | Upto 2-3      | below 2       |
| (iv)   | Width (in cm)                                                                                                       | More than 6  | Upto 5        | below 5       |
| (v)    | Range Acceptance                                                                                                    | 95 %         | 90 %          | 85 %          |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                              |              |               |               |
| (vi)   | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 90 | Between 60-90 | Between 60-90 |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 5 nos shall be drawn from each selected packing unit and shall be mixed homogenously , called as primary sample
- (iii) 5 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 167. OYSTER MUSHROOM

Common name: ढिंगरी मशरूम (ऑयस्टर मशरूम)

English name: Oyster Mushroom

Botanical name: *Pleurotus species of the family Pleurotaceae*  
excluding King Oyster mushroom (*Pleurotus eryngii species*)



### 1. Oyster Mushroom shall be.-

- (i) whole, cluster or single, fresh - characteristic of the variety;
- (ii) sound, produce affected by rotting or deterioration such as to make it unfit for consumption is excluded;
- (iii) clean, practically free of any visible foreign matter;
- (iv) practically free of pests and damage caused by them affecting the general appearance of the produce;
- (v) free of abnormal external moisture, free of bruises and blemishes.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                                                                                                                                                                                                           | Range-1      | Range-2       | Range-3       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|
| <b>A.</b> | <b>Primary</b>                                                                                                                                                                                                                                |              |               |               |
| (i)       | <b>Defects ( Oyster mushroom in cluster) for bloom- cap % by count (Max):-</b><br>Bruising, rubbing and other superficial defects such as brakage and blemishes, (total surface area )<br>The defects must not, in any case, affect the gill. | 3 %          | 5 %           | 7 %           |
|           | Shape                                                                                                                                                                                                                                         | 5%           | 10%           | 15%           |
| (ii)      | Size<br>(Diameter of the cap in cm)                                                                                                                                                                                                           | 5 & above    | Upto 5        | below 3       |
|           | Range Acceptance                                                                                                                                                                                                                              | 95 %         | 90 %          | 85 %          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                                                                                                                                        |              |               |               |
| (iii)     | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*) /Visual<br>(*100% is true to variety colour)                                                                                                                              | More than 90 | Between 60-90 | Between 60-90 |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 10 nos shall be drawn from each selected packing unit and shall be mixed homogenously called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 168. Button Mushroom

Common name: कुंभी, कुकरमुत्ता

English name: **Mushroom**

Botanical name: *Agaricus*.



### 1. Button Mushroom shall be.-

- whole, cluster or single, free from poisonous variety, fresh - characteristic of the variety;
- sound, produce affected by rotting or deterioration such as to make it unfit for consumption is excluded;
- clean, practically free of any visible foreign matter with the exception of the cultivation substrate adhered to the mushroom base;
- free of abnormal external moisture, free of bruises and blemishes.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

4. The number of stalks shall not exceed the number of caps.

### 5. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                                                                                                                                   |                       | Range-1                       | Range-2                      | Range-3                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|------------------------------|-----------------------------|
| A.     | <b>Primary</b>                                                                                                                                                                                                                        |                       |                               |                              |                             |
| (i)    | Defects ( mushroom in cluster) for bloom- cap % by count (Max):-<br>Bruising, rubbing and other superficial defects such as breakage and blemishes, (total surface area) The defects must not, in any case, affect the gill.<br>Shape |                       | 3 %                           | 5 %                          | 7 %                         |
| (ii)   | Size (Diameter of the cap in cm)                                                                                                                                                                                                      |                       | 5 & above                     | Upto 5                       | Below 3                     |
| (iii)  | <b>Range Acceptance</b>                                                                                                                                                                                                               |                       | 95 %                          | 90 %                         | 85 %                        |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                                                                                                                                |                       |                               |                              |                             |
| (iv)   | Color (%)<br>(Characteristic true to the variety)<br>Scale (1 to 100*) /Visual<br>(*100% is true to variety colour)                                                                                                                   |                       | More than 90                  | Between 60-90                |                             |
| (v)    | Mineral impurities% (m/m)                                                                                                                                                                                                             | Wild growing mushroom | 0.1                           | 0.3                          |                             |
|        |                                                                                                                                                                                                                                       | Cultivated            | 0.25                          | 0.5                          |                             |
| (vi)   | Organic impurities of vegetable origin,% (m/m)                                                                                                                                                                                        | Wild growing mushroom | 0.5                           | 1                            |                             |
|        |                                                                                                                                                                                                                                       | Cultivated            | 3(for uncut)<br>0.5 (for cut) | 5 (for uncut)<br>1 (for cut) | 8(for uncut)<br>1 (for cut) |
| (vii)  | Content of maggot damaged mushroom,% (m/m)                                                                                                                                                                                            | Wild growing mushroom | 2                             | 4                            | 6                           |
|        |                                                                                                                                                                                                                                       | Cultivated            | 0.5                           | 1                            |                             |

### 6. Sampling: -

- 5 % or minimum one packing unit shall be randomly selected for sampling
- 10 nos shall be drawn from each selected packing unit and shall be mixed homogenously called as primary sample
- 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 169. Gherkin

Common name: घर्किन

English name : Gherkin

Botanical name: *Cucumis sativus*,, *C.angura*



### 1. Gherkin shall be.-

- (i) fresh in appearance, well formed, intact;
  - (ii) free from disease, soil, dust, wilting, burning, spots, holes and discoloration;
  - (iii) free of damage caused by high & low temperature, injury caused by scars, bruises ,badly curved/deformed, or shriveled;
  - (iv) uniform in size, shape, color and free from stem or flowers.
2. Variety/colour.....( to be filled by assayer).
  3. Actual Picture of the lot.....( to be uploaded by assayer)
  4. Criteria for Range designation:

| S.No.     | Tradable Parameters                                          | Range-1                                                 | Range-2                                                         | Range-3                                                        |
|-----------|--------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| <b>A.</b> | <b>Primary</b>                                               |                                                         |                                                                 |                                                                |
| (i)       | Defects<br>% by count (max.)                                 | 3%                                                      | 5%                                                              | 7%                                                             |
| (ii)      | Shape and color<br>(not less than of the total surface area) | well developed & green or dark green(not less than 3/4) | well developed & green with slight yellowing(not less than 3/4) | well developed & green with some yellowing( not less than 2/3) |
| (iii)     | Length of the fruit (in cm)                                  | Upto 3-5                                                | Upto 5-7                                                        |                                                                |
| (iv)      | Range Acceptance                                             | 95 %                                                    | 90 %                                                            | 85 %                                                           |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                       |                                                         |                                                                 |                                                                |
| (v)       | Color(%)characteristic to the variety(1-100)/visual          | More than 70                                            | Between 30-70                                                   |                                                                |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 100 gms shall be drawn from each selected packing unit and shall be mixed homogenously , called as primary sample
- (iii) 50 gms. shall be drawn from primary sample called as laboratory sample

**Optional parameters:** are voluntary and may be analyzed as per buyer-seller agreement.

Note: Gherkin is practically straight and not more than very slightly constricted or more than moderately tapered or pointed.

## 170. Grapefruit

Common name: चकोतरा

English name : Grapefruit

Botanical name: *Citrus × paradisi*



### 1. Grapefruit shall be-

- whole, fresh, well formed, intact, firm;
- free from disease, pest, abnormal external moisture, foreign smell or taste;
- uniform in shape, size, color and undamaged skin(peel),dry skin, decay;
- free from dust or any visible foreign matter, black spot, scab, spray burns.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                              | Range-1                                       | Range-2                                       | Range-3                                       |
|-----------|----------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| <b>A.</b> | <b>Primary</b>                                                                   |                                               |                                               |                                               |
| (i)       | Defects<br>%by count(max.)<br>Discoloration (0.5-2 sq. cm of total surface area) | 2%                                            | 5%                                            | 10%                                           |
| (ii)      | Shape(oblate) and color of fruit(orange-yellow or orange-red)                    | uniform color and shape normal to the variety | uniform color and shape normal to the variety | uniform color and shape normal to the variety |
| (iii)     | Size of the fruit(diameter of the equatorial section)                            | Upto1500 mm                                   | Upto1000 mm                                   | Upto 500 mm                                   |
| (iv)      | Range Acceptance                                                                 | 95 %                                          | 90 %                                          | 85 %                                          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                           |                                               |                                               |                                               |
| (v)       | Color(%)characteristic to the variety(1-100)visual                               | More than 85                                  | More than 70                                  | More than 60                                  |
| (vi)      | Juice content % min.<br>Juice wt./fruit wt.                                      | 40%                                           | 38%                                           | 35%                                           |

### 5. Sampling: -

- 5 % or minimum one packing unit shall be randomly selected for sampling
- 5 nos shall be drawn from each selected packing unit and shall be mixed homogenously , called as primary sample
- 10 nos. shall be drawn from primary sample called as laboratory sample

6. **Special Note:** Fruits should not have coarse corrugated skin(peel),shriveled or immature, degreening is allowed provided that it does not affect the produce.

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 171. Rapeseed

Common name: तोरीआ

English name : Taramira, Rapeseed

Botanical name: Brassica rapa, B.napus



### 1. Rapeseed shall be-

- (i) clean, wholesome reasonably uniform in size, shape and color, free from argemone seeds;
  - (ii) free from obnoxious smell and fungus infestation.
2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

4. Criteria for Range designation:

| S. No.    | Tradable Parameters                      | Range-1                             | Range-2   | Range-3    |
|-----------|------------------------------------------|-------------------------------------|-----------|------------|
| <b>A.</b> | <b>Primary</b>                           |                                     |           |            |
| (i)       | Oil content(%by wt) Min.                 | 40.0                                | 35.0-40.0 | Below 35.0 |
| (ii)      | Moisture (%by wt) Max.                   | 8.0                                 | 9.0       | 10.0       |
| (iii)     | Foreign matter (%by wt) Max.             | 3.0                                 | 5.0       |            |
| (iv)      | Damaged and discolored seeds(%by wt)Max. | 3.0                                 | 5.0       |            |
| (v)       | Other edible seeds (%by wt) Max.         | 0.5                                 | 0.8       | 1.0        |
| (vi)      | Range Acceptance                         | 95 %                                | 90 %      | 85 %       |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>   |                                     |           |            |
| (vii)     | Allyl iso thiocynate(m/m) On dry basis   | Not less than 0.7 percent by weight |           |            |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling
- (ii) 100 gm shall be drawn from each selected packing unit and shall be mixed homogenously , called as primary sample
- (iii) 50 gm shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.



## 172. Hilsa

Common name: Hilsa, हिल्सा

Zoological name: *Hilsa shad*, *Tenualosa ilisha*



1. **Hilsa shall be-**

- (i) clean, wholesome finfish, intact, either in raw chilled or frozen condition;
- (ii) free from disease, skin patches, stomach broken and obnoxious smell, or any other added chemical free from visible scar, parasite, grease, soil;
- (iii) free from any scar/cuts, dip in formaline or any chemical, use of added coloring/chemicals.

2. Actual Picture of the lot.....( to be uploaded by assayer)

3. Criteria for Range designation:

| S.No. | Tradable Parameters                      | Range-1          | Range-2      | Range-3      |
|-------|------------------------------------------|------------------|--------------|--------------|
| A.    | <b>Primary</b>                           |                  |              |              |
| (i)   | Size determined by (wt. per piece in gm) | More than 800 gm | 300-800 gms. | 300 gm below |

4. **Quality Index Method(QIM):**

| Quality parameter  | Character               | Score (ice/seawater)                                                          |
|--------------------|-------------------------|-------------------------------------------------------------------------------|
| General appearance | Skin                    | 0 Bright, shining<br>1 Bright<br>2 Dull                                       |
|                    | Bloodspot on gill cover | 0 None<br>1 Small, 10-30%<br>2 Big, 30-50%<br>3 Very big, 50-100%             |
|                    | Stiffness               | 0 Stiff, in <i>rigor mortis</i><br>1 Elastic<br>2 Firm<br>3 Soft              |
|                    | Belly                   | 0 Firm<br>1 Soft<br>2 Belly burst                                             |
|                    | Smell                   | 0 Fresh, seaweed/metallic<br>1 Neutral<br>2 Musty/sour<br>3 Stale meat/rancid |
| Eyes               | Clarity                 | 0 Clear<br>1 Cloudy                                                           |
|                    | Shape                   | 0 Normal<br>1 Plain<br>2 Sunken                                               |
| Gills              | Colour                  | 0 Characteristic, red<br>1 Faded, discoloured                                 |

|               |       |                                                                                                  |
|---------------|-------|--------------------------------------------------------------------------------------------------|
|               | Smell | 0 Fresh, seaweed/metallic<br>1 Neutral<br>2 Sweaty/slightly rancid<br>3 Sour stink/stale, rancid |
| Sum of scores |       | (min. 0 and max. 20)                                                                             |

**Results of QIM:**

| Sum of Scores QIM | Range |
|-------------------|-------|
| 0-9               | 1     |
| 9-15              | 2     |
| 15-20             | 3     |

**5. Other requirements:**

- (i) Chilling can be achieved either by using ice, chilled water or refrigerated water which will not affect the quality of fish.
- (ii) Freezing or any cold chain management process shall be reinforced or any other traditional process by which the quality of the fish product have been maintained at low temperature during transportation, storage, distribution and including the time of final sale.

**6. Sampling: -**

- (i) 5 % or minimum one container shall be randomly selected for sampling
- (ii) 5 Nos shall be drawn from each selected container and shall be mixed homogenously , called as primary sample
- (iii) 10 nos. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 173. MINT LEAVES

Common name: पुदीना पत्ती  
English name: Mint leaves  
Botanical name: *Mentha spicata* L.



### 1. Mint Leaves shall be –

- (i) fresh in appearance; clean, properly drained if washed;
  - (ii) free from visible foreign matter, abnormal external moisture, foreign smell/taste, damage caused by pests and produce affected by rotting or deterioration such as to make it unfit for consumption is excluded;
  - (iii) free from wilting or yellow leaves (dehydrated senescent).
2. Variety/colour.....(to be filled by assayer)
  3. Actual Picture of the lot.....( to be uploaded by assayer)
  4. **Criteria for Range designation:**

| S. No. | Tradable Parameters                                                                                              | Range-1      | Range-2       | Range-3       |
|--------|------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|
| A.     | <b>Essential</b>                                                                                                 |              |               |               |
| (i)    | Defects<br>% by weight (Max):-<br>(i) Discolouration/ black leaves                                               | 3%           | 5%            | 7%            |
|        | (ii) Damaged leaves: cracks, bruises, holes or tears, split leaves                                               | 3%           | 5%            | 7%            |
| (ii)   | Range Acceptance                                                                                                 | 95 %         | 90 %          | 85 %          |
| (iii)  | Bunch stem length (in mm)                                                                                        | 200          | 200-250       | More than 250 |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                           |              |               |               |
| (iv)   | Pre-packed produce stem length (in mm)                                                                           | 75-100       | 100-150       | more than 150 |
| (v)    | Color (%)<br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual (*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40  |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling.
- (ii) 100 g shall be drawn from each selected packing unit and shall be mixed homogeneously called as primary sample.
- (iii) 200 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 174. CURRY LEAVES

Common name: करी पत्ता

English name: Curry leaves

Botanical name: *Murraya Koenigii*(L.)



### 1. Curry Leaves shall be –

- (i) fresh in appearance; intact with stalk, sound, clean, properly drained if washed;
- (ii) free from visible foreign matter, abnormal external moisture, foreign smell/taste, damage caused by pests and produce affected by rotting or deterioration such as to make it unfit for consumption is excluded;
- (iii) free from sun-burn, bud formation.

2. Variety colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No. | Tradable Parameters                                                                                                        | Range-1      | Range-2 | Range-3 |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------|---------|
| A.     | <b>Essential</b>                                                                                                           |              |         |         |
| (i)    | Defects<br>% by weight (Max):-<br>(i) Discolouration/ black leaves                                                         | 1%           | 5%      | 7%      |
|        | (ii) Damaged leaves:<br>cracks, bruises, holes or tears, split leaves                                                      | Absent       | 5%      | 7%      |
| (ii)   | Range Acceptance                                                                                                           | 95 %         | 90 %    | 85 %    |
| B.     | <b>Optional (Industrial/Processor)</b>                                                                                     |              |         |         |
| (iii)  | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 90 | 70-90   | 50-70   |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling.
- (ii) 100 g shall be drawn from each selected packing unit and shall be mixed homogenously called as primary sample.
- (iii) 200 g shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.

## 1 7 5 . SPRING ONION

Common name: हरी प्याज

English name: Spring Onion, Welsh Onion

Botanical name: *Allium fistulosum*



### 1. Spring Onion Leaves shall be –

- (i) fresh in appearance; intact with bulb, mid to dark green leaves with bulb, clean, properly drained if washed;
- (ii) free from, foreign matter, abnormal external moisture, foreign smell/taste, damage caused by pests and produce affected by rotting or deterioration such as to make it unfit for consumption is excluded;
- (iii) free from yellow leaves, mildew, spray deposits, breakdown including rots and decaying leaves.

2. Variety/colour.....(to be filled by assayer)

3. Actual Picture of the lot.....( to be uploaded by assayer)

### 4. Criteria for Range designation:

| S. No.    | Tradable Parameters                                                                                                        | Range-1      | Range-2       | Range-3       |
|-----------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|
| <b>A.</b> | <b>Essential</b>                                                                                                           |              |               |               |
| (i)       | Defects<br>% by count (Max):-<br>(i) Double split                                                                          | 2.0          | 5.0           | Above 8.0     |
|           | (ii) Discolouration/ black leaves                                                                                          | 3.0          | 5.0           | 7.0           |
|           | (iii) Damaged leaves: cracks, bruises, holes or tears, split leaves                                                        | 3%           | 5%            | 7%            |
| (ii)      | Diameter of bulb (mm)                                                                                                      | 10-15        | 15-18         | More than 18  |
| (iii)     | Stem length (mm)                                                                                                           | 250          | 300-350       | More than 350 |
| (iv)      | Range Acceptance                                                                                                           | 95 %         | 90 %          | 85 %          |
| <b>B.</b> | <b>Optional (Industrial/Processor)</b>                                                                                     |              |               |               |
| (v)       | <b>Color (%)</b><br>(Characteristic true to the variety) Scale (1 to 100*)<br>/Visual<br>(*100% is true to variety colour) | More than 70 | Between 40-70 | Less than 40  |

### 5. Sampling: -

- (i) 5 % or minimum one packing unit shall be randomly selected for sampling.
- (ii) 200 no. shall be drawn from each selected packing unit and shall be mixed homogenously called as primary sample.
- (iii) 100 no. shall be drawn from primary sample called as Laboratory sample

**Optional parameters** are voluntary and may be analyzed as per buyer-seller agreement.